

사이클별 지표 패널

M02Ch111

각 패널 = 한 지표의 cycle 궤적 (routine 0.5C).

early/late 점선 = 앞·뒤 5포인트 median.

트렌드 = 선형 slope/100cyc + aging 기대방향 대조.

주의: LAM_PE / *_proxy / pattern_score는 가설·proxy이며 절대 LAM%/LLI%가 아닙니다. R_ohmic은 $\sqrt{t}$ 외삽 proxy (0.1s 단일값 ≠ 순수 음).

트렌드 요약 — M02Ch111

- 유효 지표: 358개 · aging 방향 일치 48 · 반대 13

하락 트렌드 (상위)

- 충전 dQ/dV 피크1 높이: early=80.48 → late=81.33 ($\Delta=+0.8509$, -3433Ah/V/100cyc) → decreasing · context
- 국소 CE(20): early=498.2 → late=301.7 ($\Delta=-196.5$, -63.24%/100cyc) → decreasing · matches_aging
- 충전 에너지: early=249.7 → late=194.6 ($\Delta=-55.06$, -21.6Wh/100cyc) → decreasing · matches_aging
- 방전후 휴지 완화 τ : early=361.7 → late=324.2 ($\Delta=-37.57$, -20.28s/100cyc) → decreasing · context
- 방전후 완화 τ Δ vs기준: early=3.59 → late=-33.98 ($\Delta=-37.57$, -20.28s/100cyc) → decreasing · context

상승 트렌드 (상위)

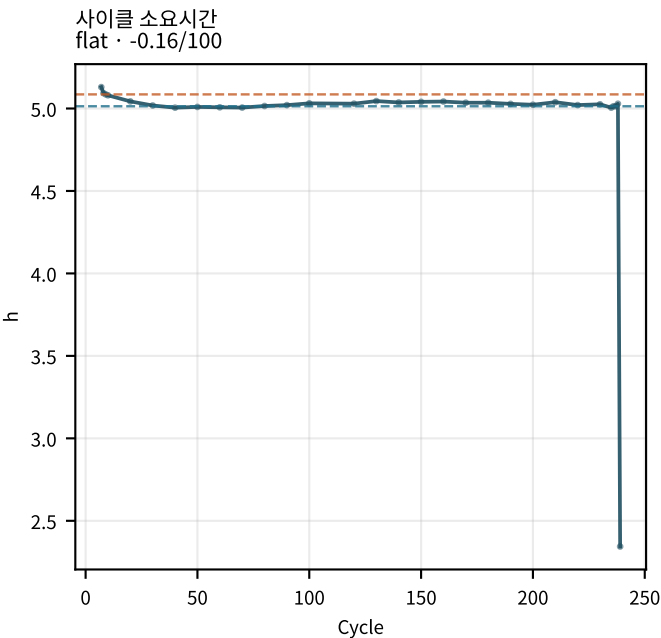
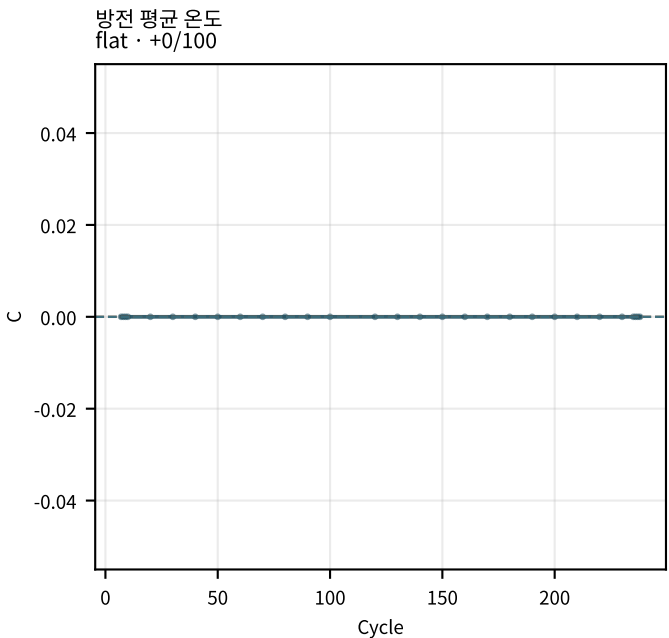
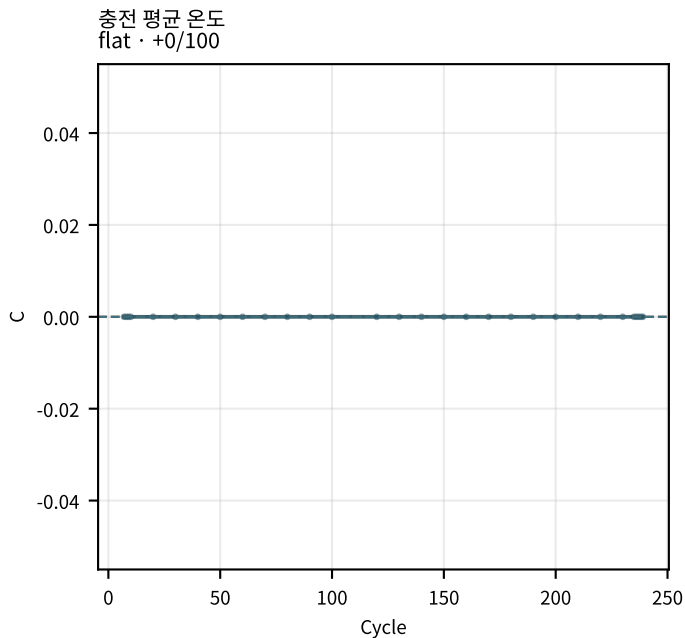
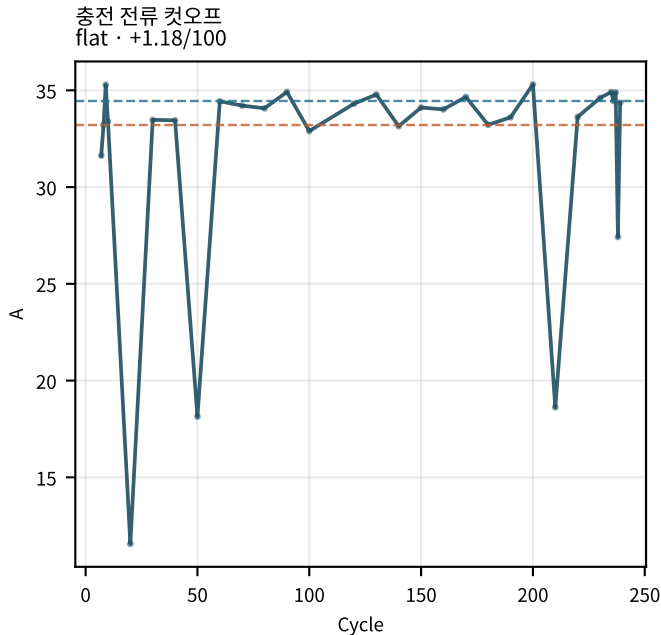
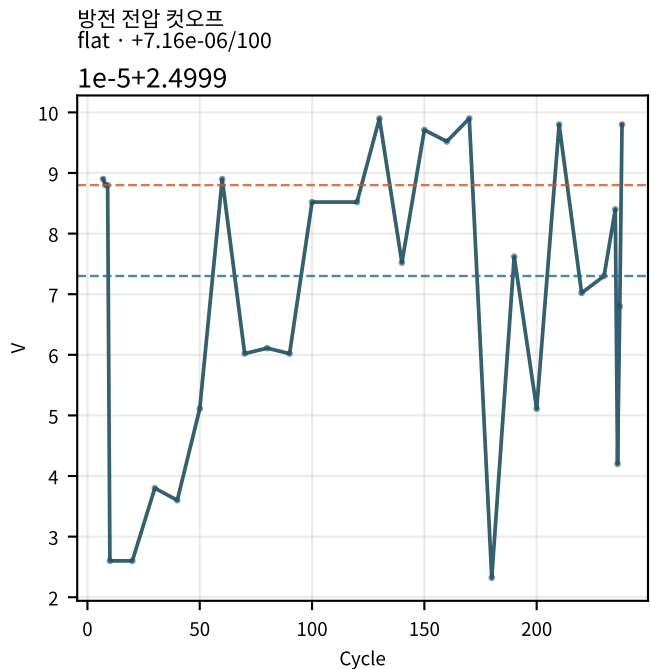
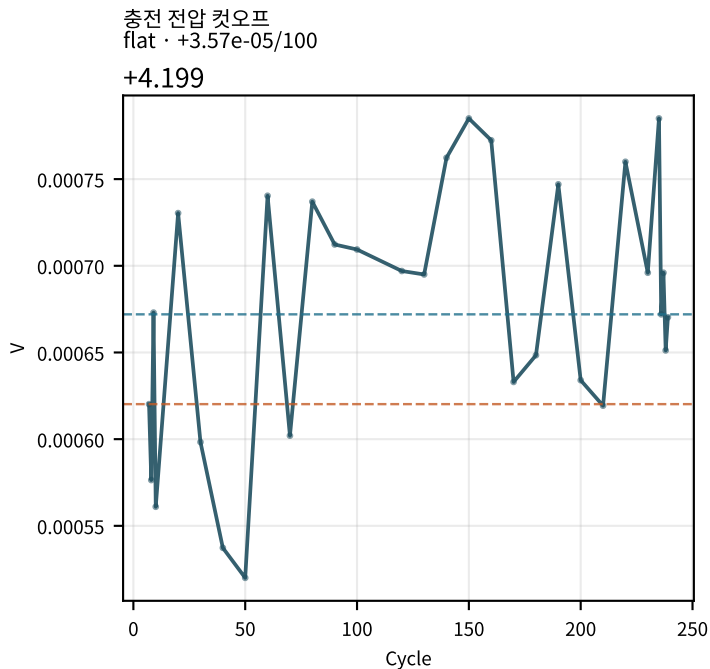
- CV 시정수: early=843 → late=843 ($\Delta=+0$, +80.05s/100cyc) → increasing · matches_aging
- fit 잔차 max: early=34.04 → late=203.3 ($\Delta=+169.3$, +62.24mV/100cyc) → increasing · matches_aging
- 충전 dQ/dV 피크4 높이: early=96.7 → late=96.7 ($\Delta=+0$, +40.62Ah/V/100cyc) → increasing · context
- EoC 방전 10s 증가%: early=24.02 → late=69.66 ($\Delta=+45.64$, +21.57%/100cyc) → increasing · matches_aging
- EoC 방전 30s 증가%: early=18.59 → late=50.31 ($\Delta=+31.72$, +15.2%/100cyc) → increasing · matches_aging

CycleDiag : plot_cycle_metrics_panel.py

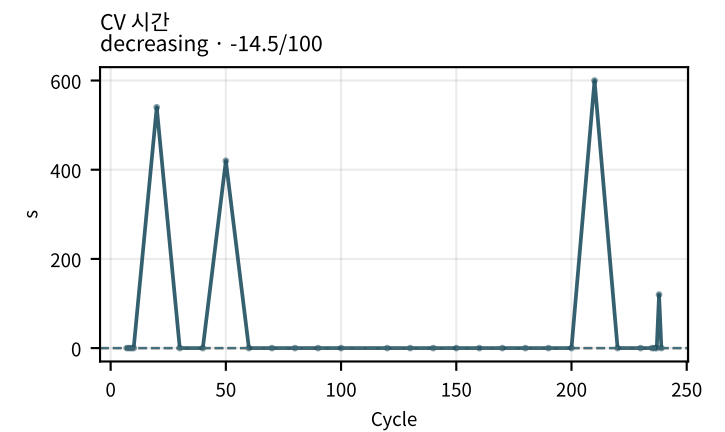
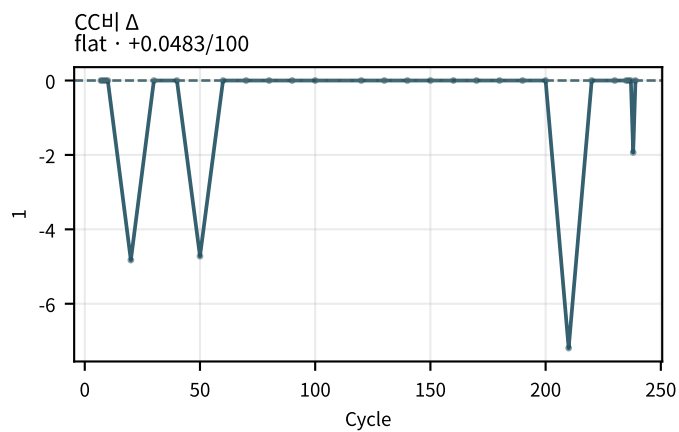
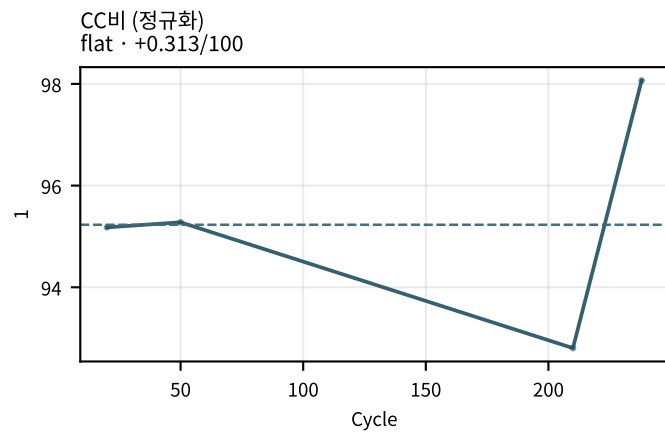
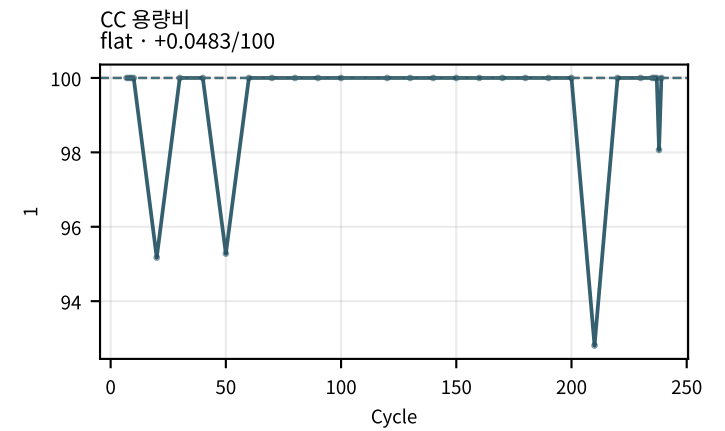
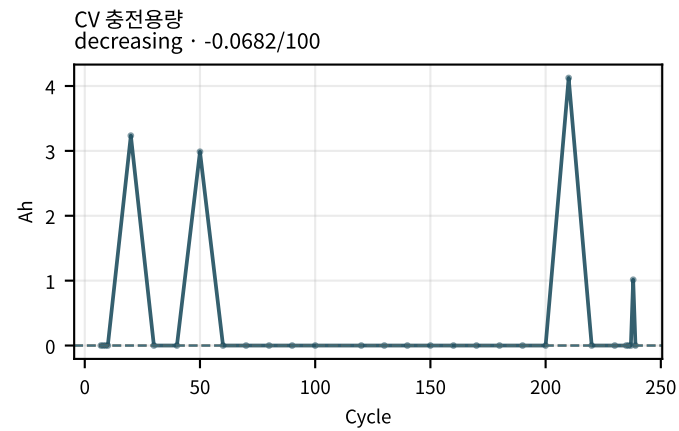
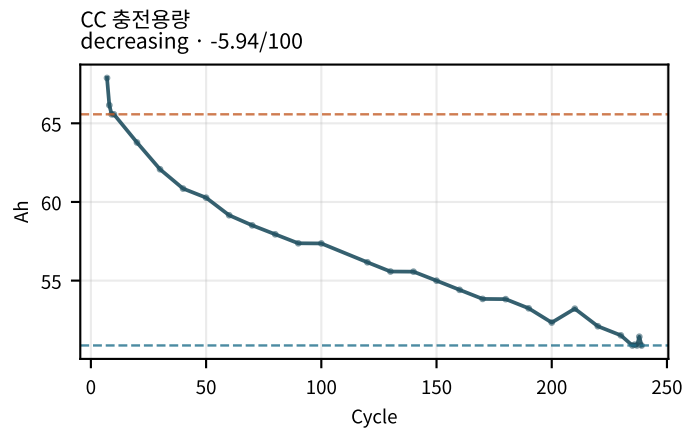
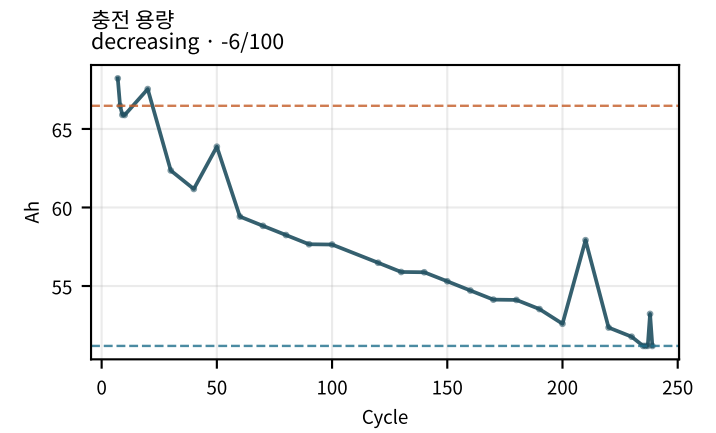
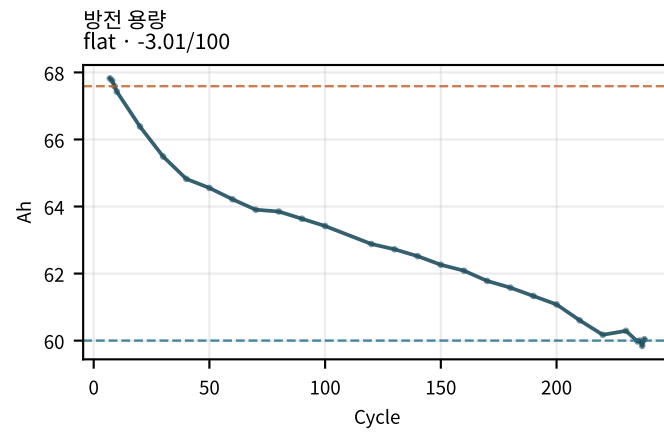
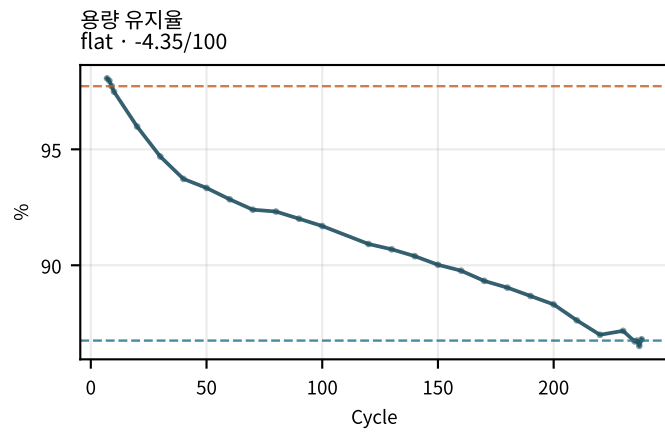
기대aging과 반대

- CV 충전용량: early=0 → late=0 ($\Delta=+0$, -0.06824Ah/100cyc) → decreasing · opposite_aging
- CV 시간: early=0 → late=0 ($\Delta=+0$, -14.51s/100cyc) → decreasing · opposite_aging
- 쿨롱 효율: early=101.9 → late=116.9 ($\Delta=+14.98$, +6.03%/100cyc) → increasing · opposite_aging
- 쿨롱 비효율: early=-1.924 → late=-16.9 ($\Delta=-14.98$, -6.03%/100cyc) → decreasing · opposite_aging

M02Ch111 · 프로토콜 · 온도

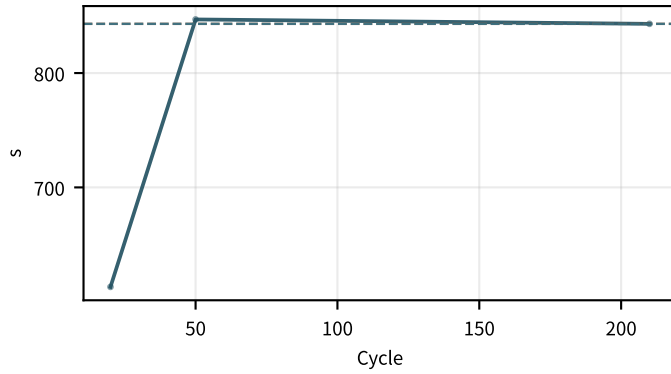


M02Ch111 · 용량 · 효율 · CV

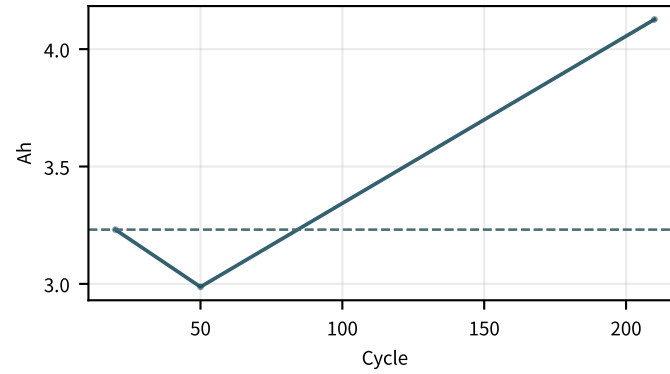


M02Ch111 · 용량 · 효율 · CV

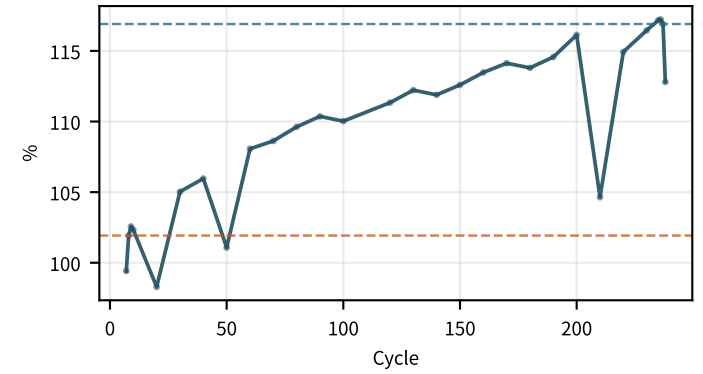
CV 시정수
increasing · +80.1/100



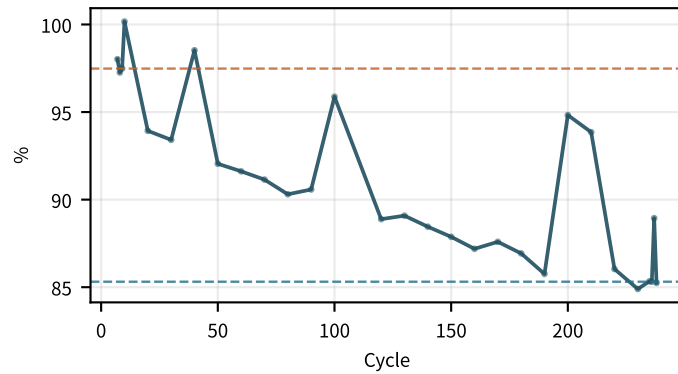
CV Q @Tref
increasing · +0.551/100



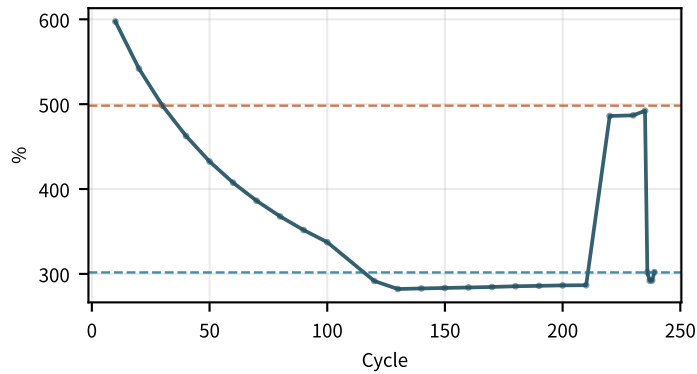
쿨롱 효율
increasing · +6.03/100



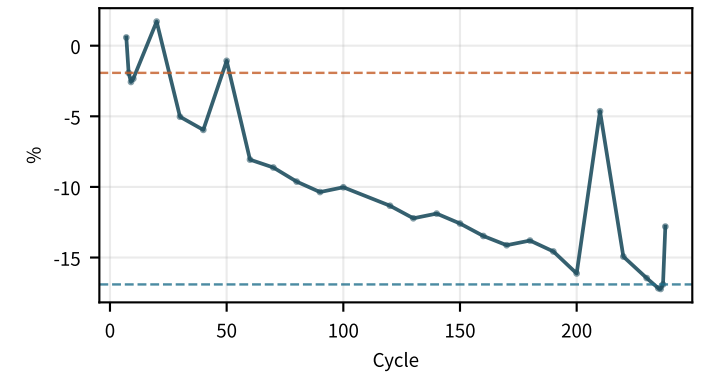
가역 CE
flat · -4.38/100



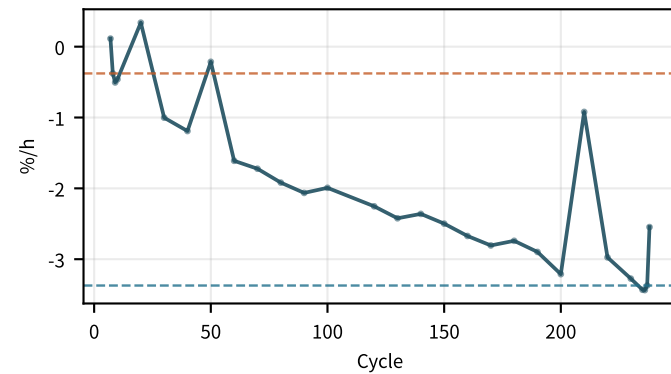
국소 CE(20)
decreasing · -63.2/100



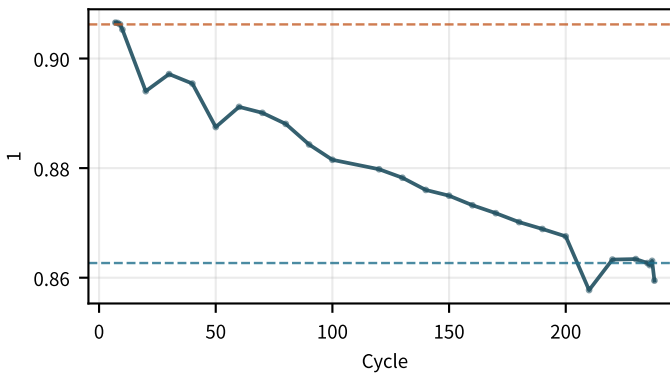
쿨롱 비효율
decreasing · -6.03/100



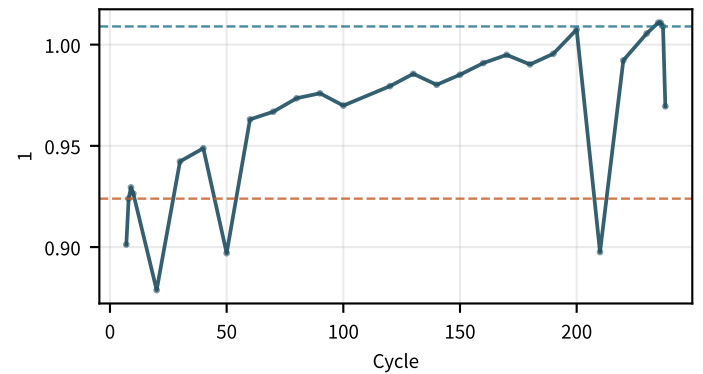
CI/시간
decreasing · -1.2/100



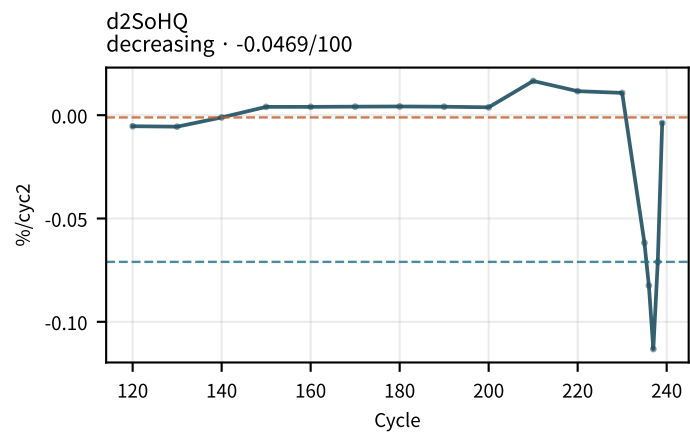
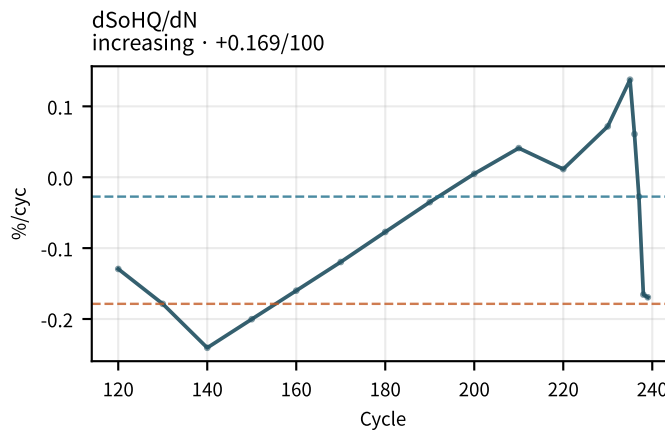
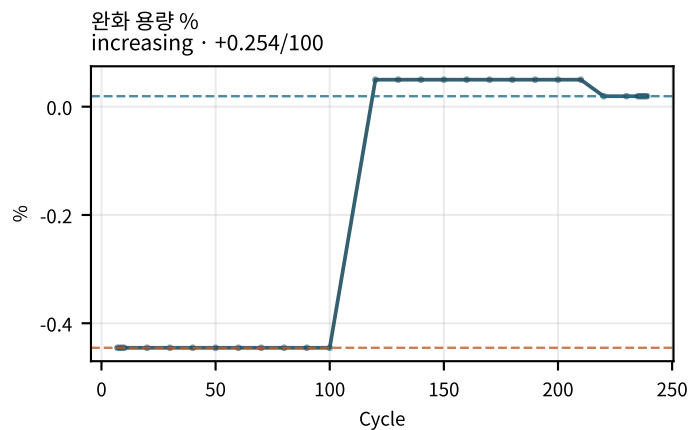
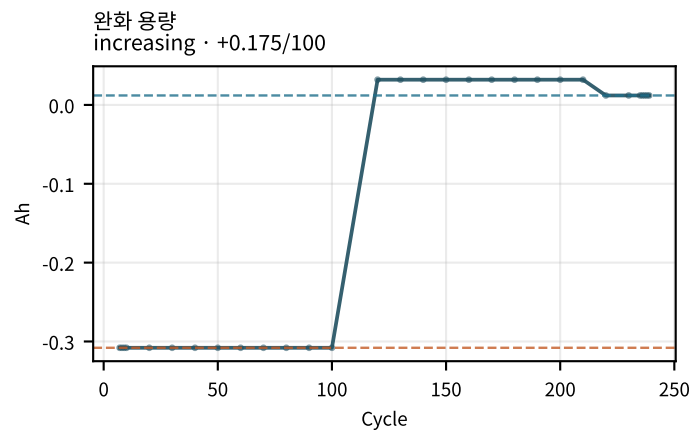
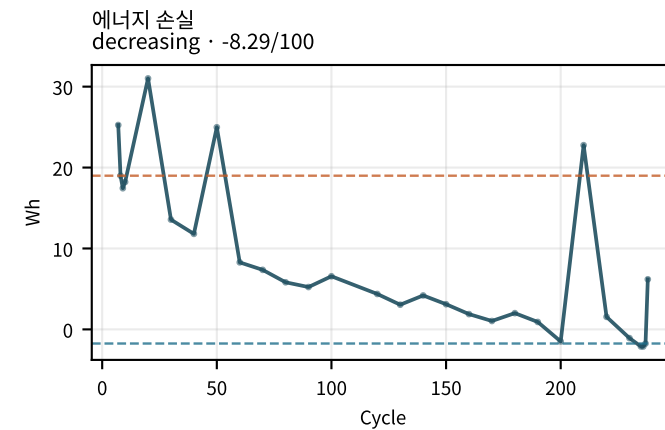
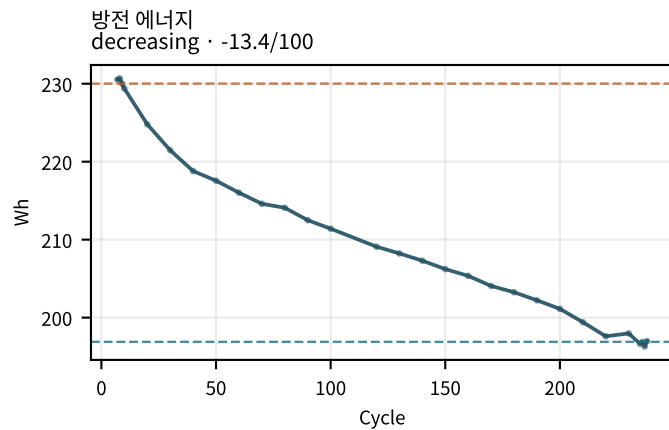
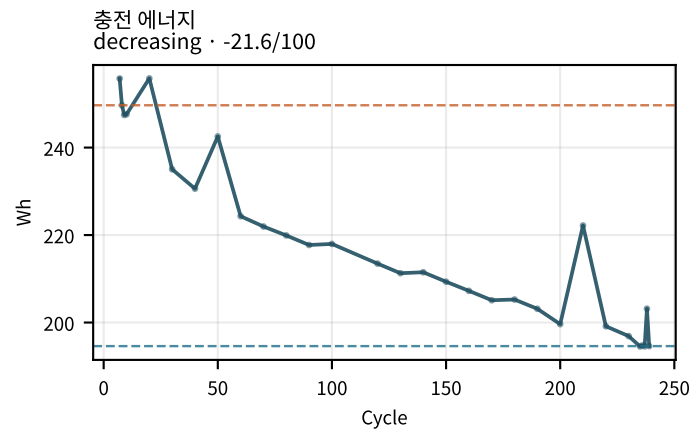
전압 효율
flat · -0.0183/100



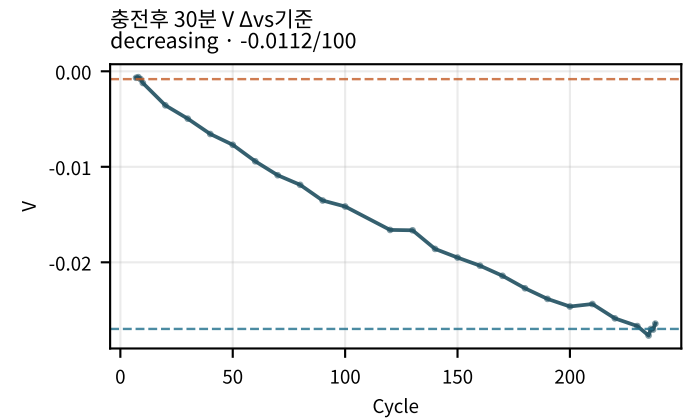
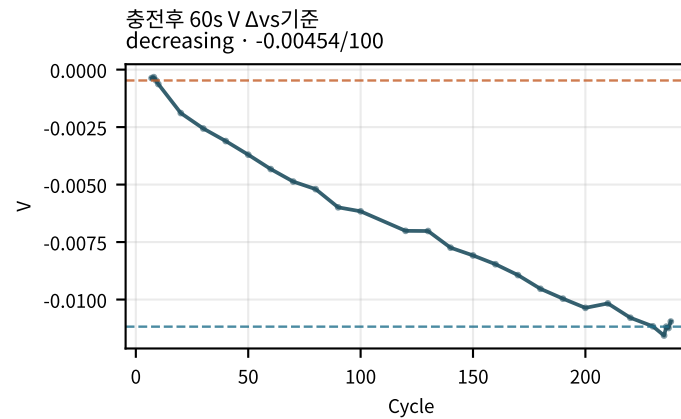
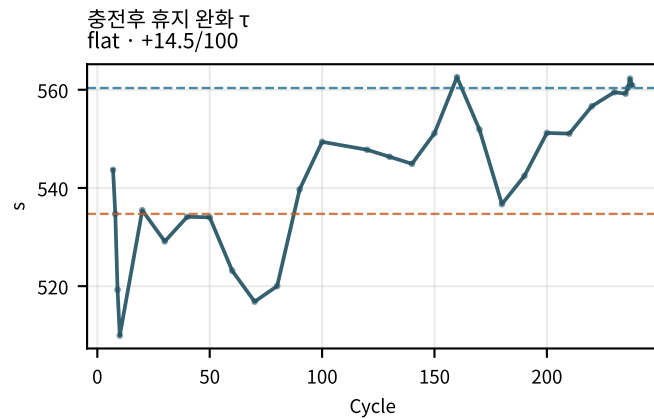
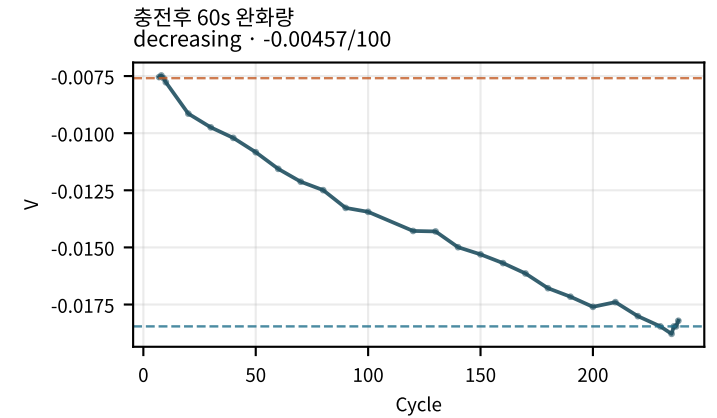
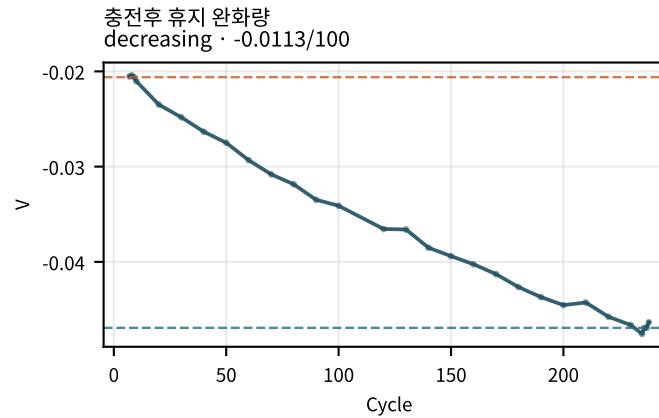
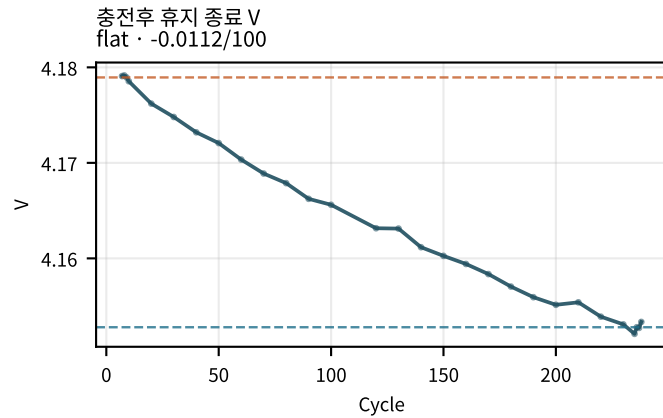
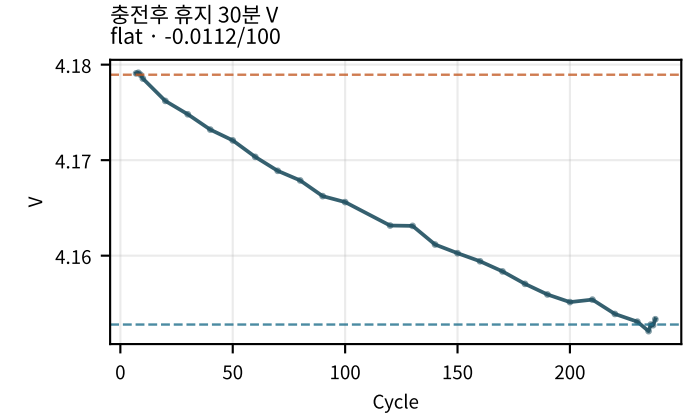
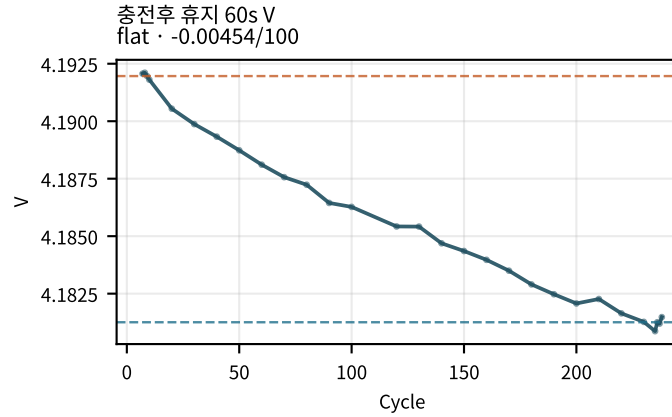
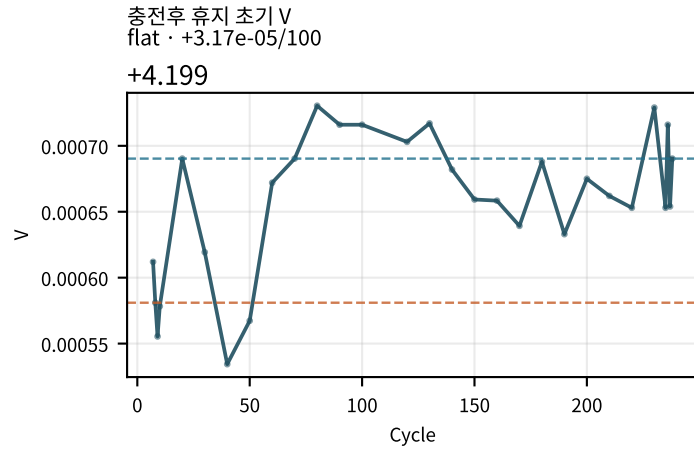
에너지 효율
flat · +0.0332/100



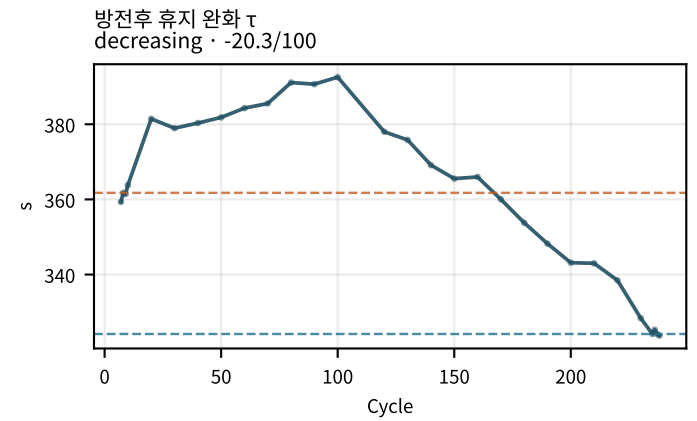
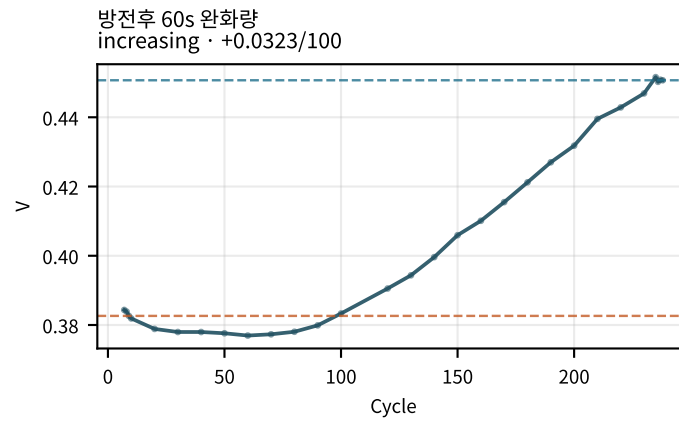
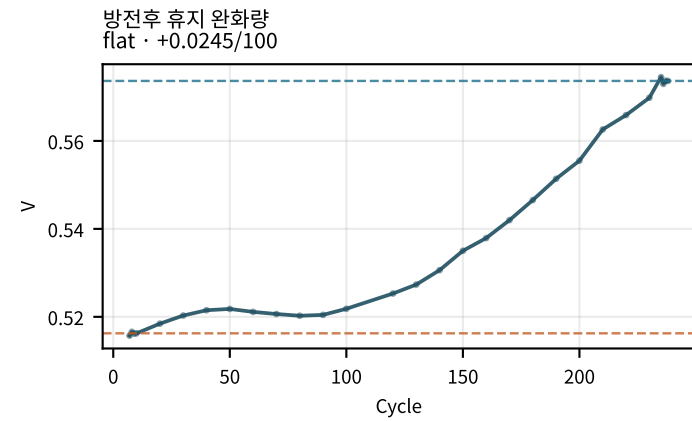
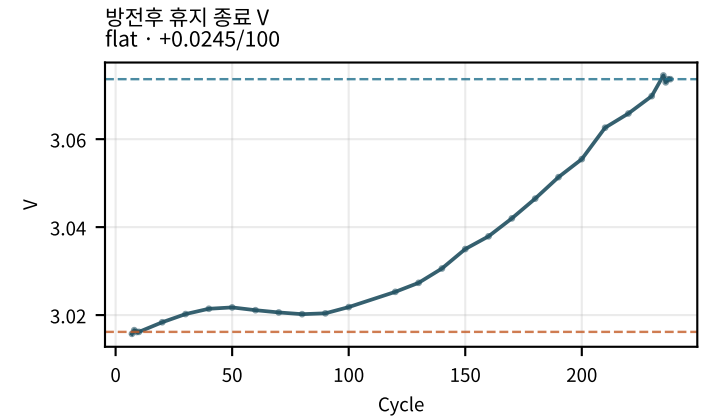
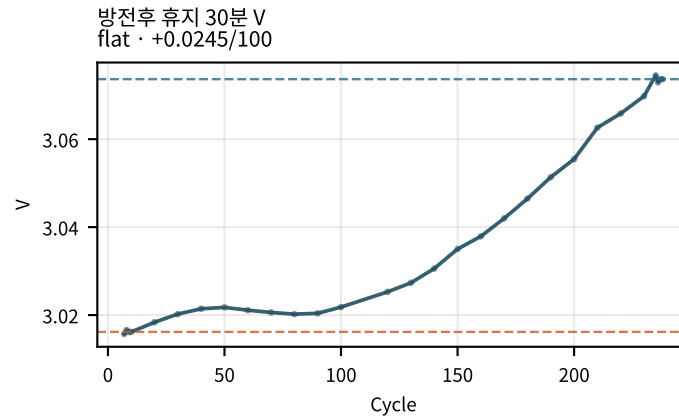
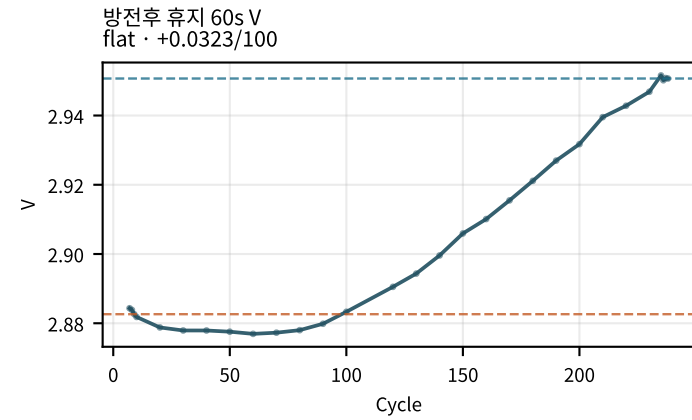
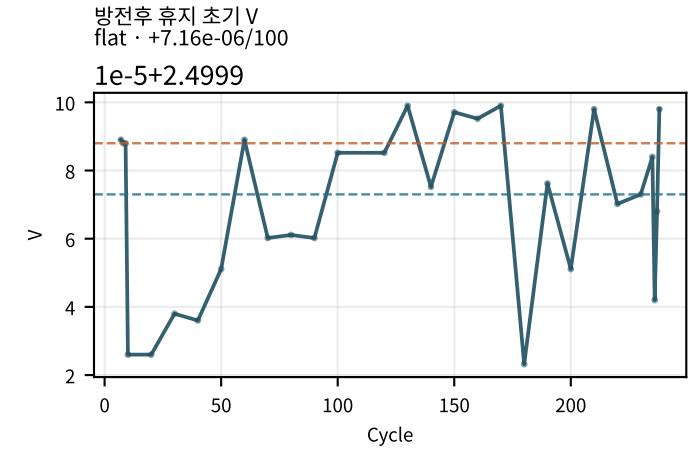
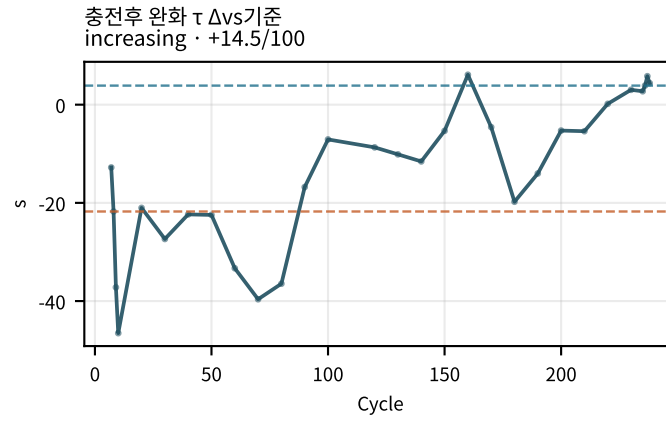
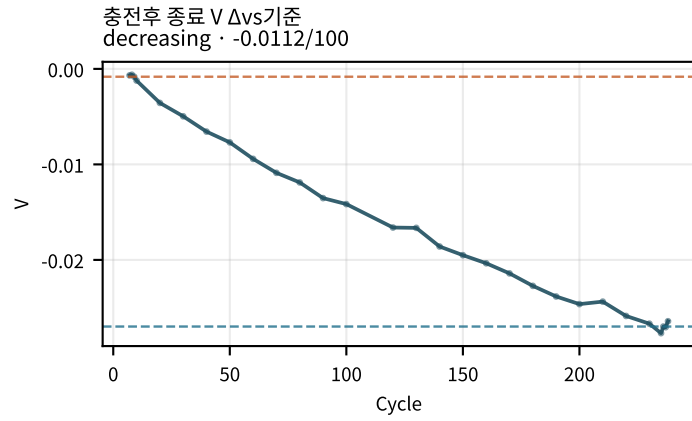
M02Ch111 · 용량 · 효율 · CV



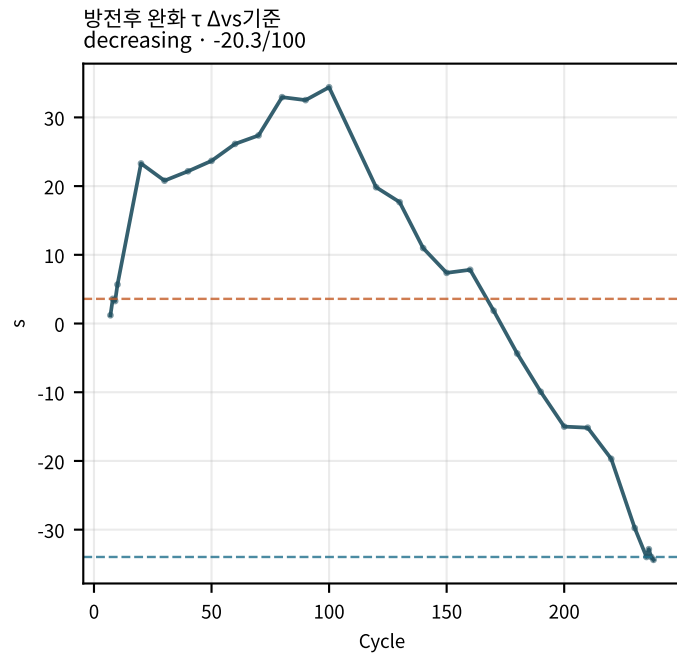
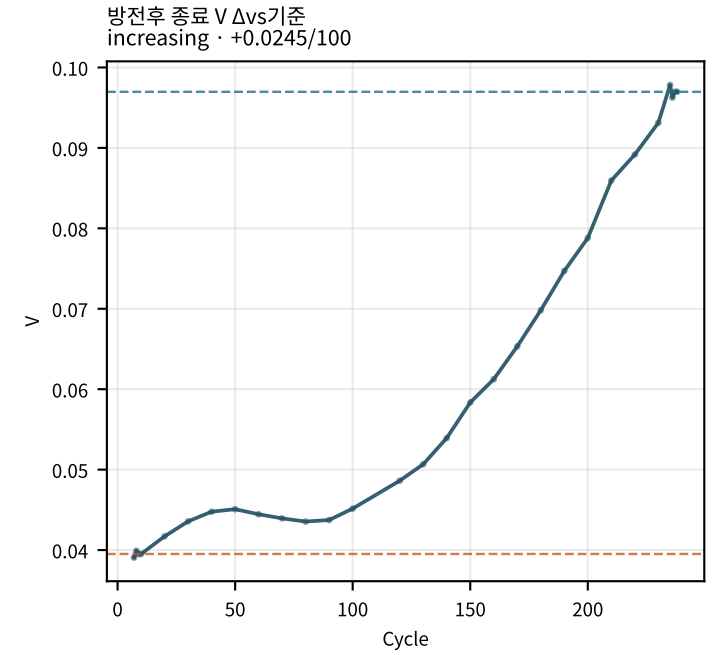
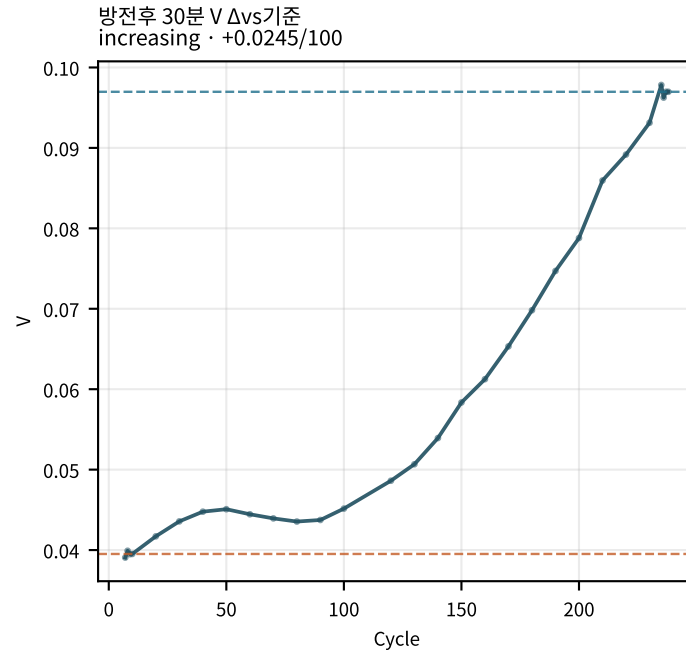
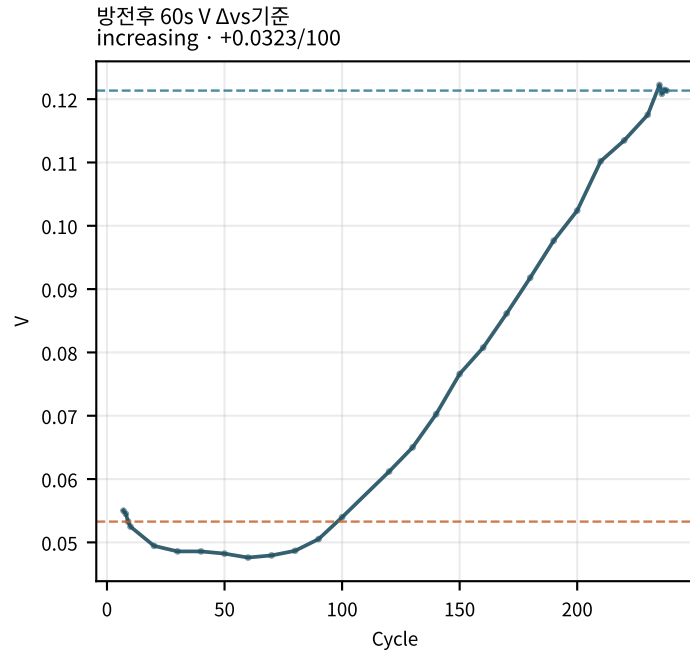
M02Ch111 · 휴지 전압 (충전/방전 후)



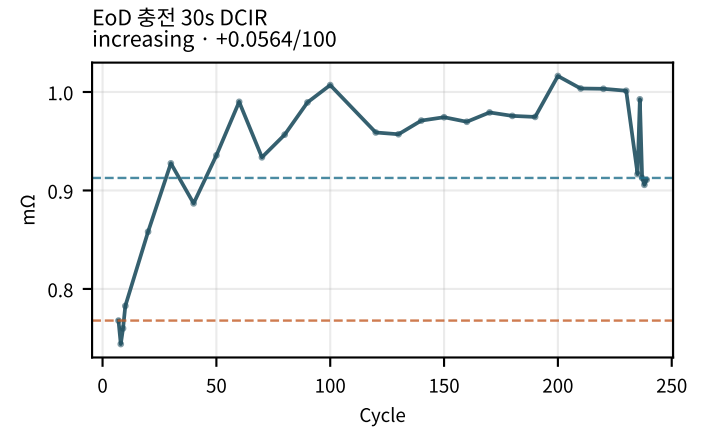
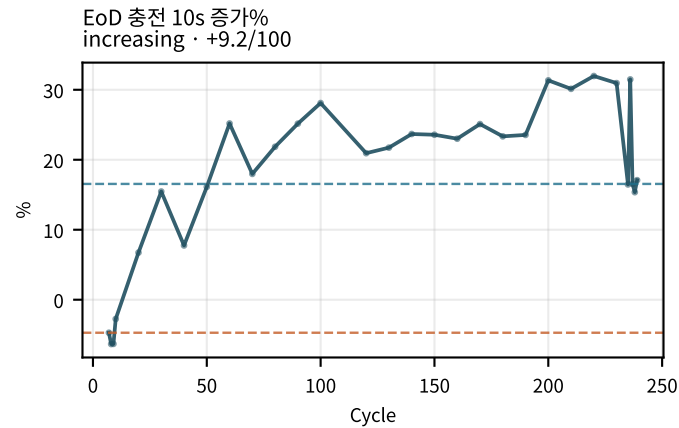
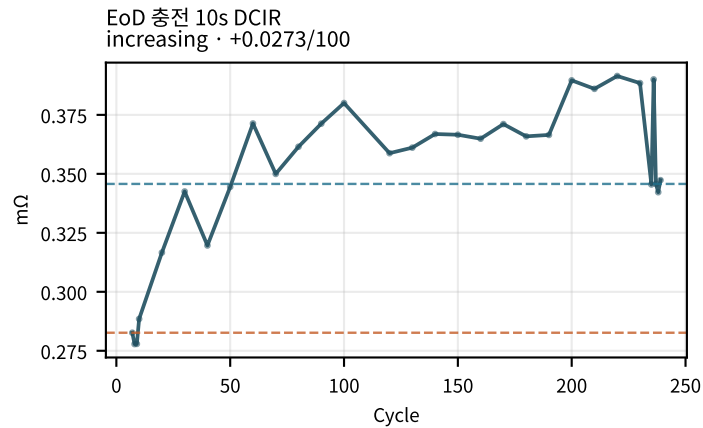
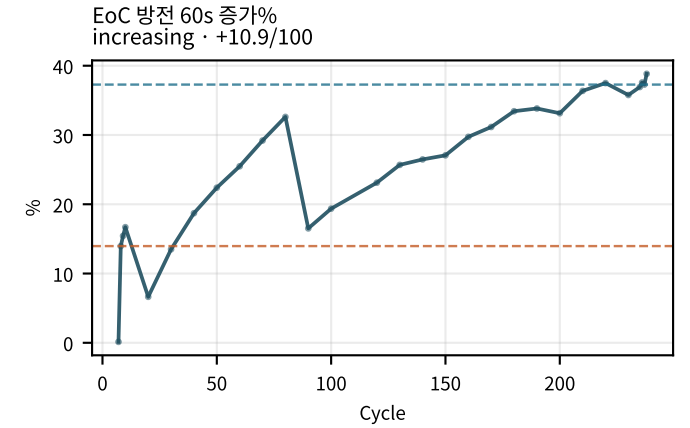
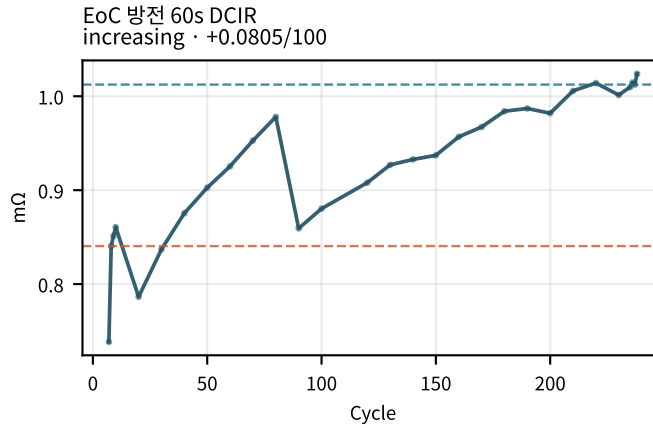
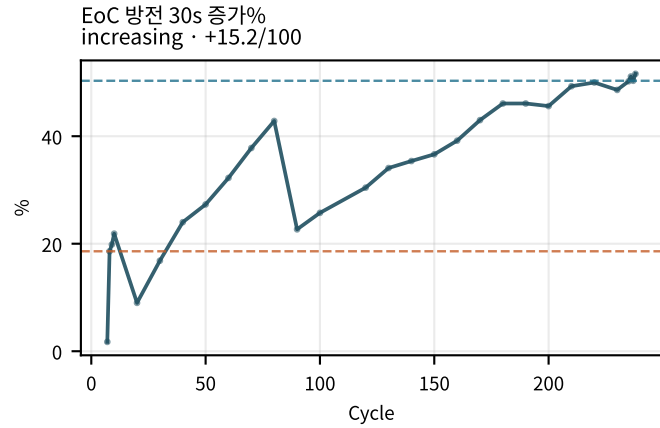
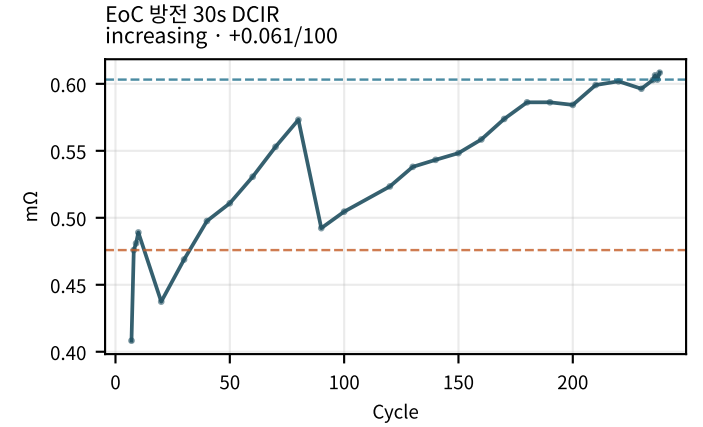
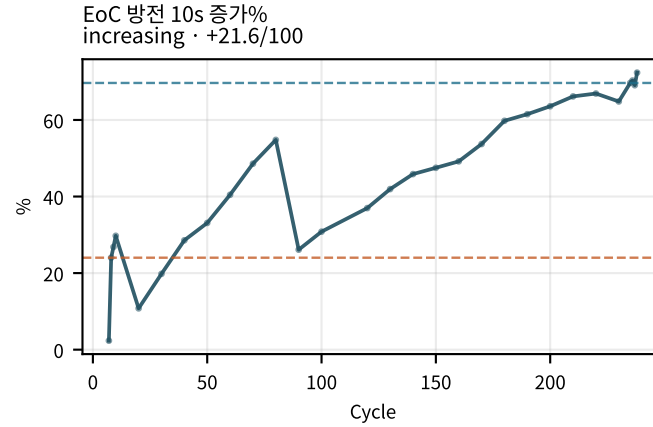
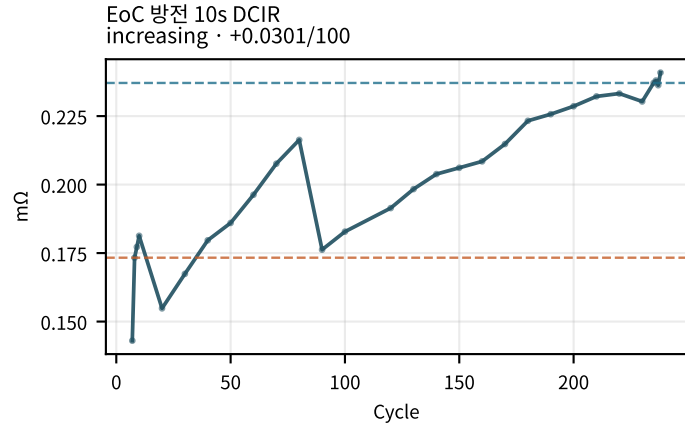
M02Ch111 · 휴지 전압 (충전/방전 후)



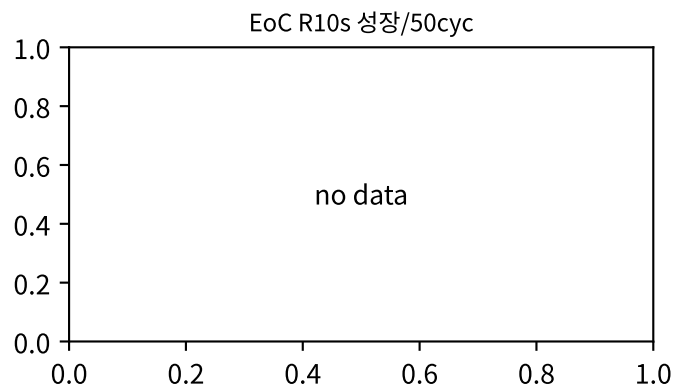
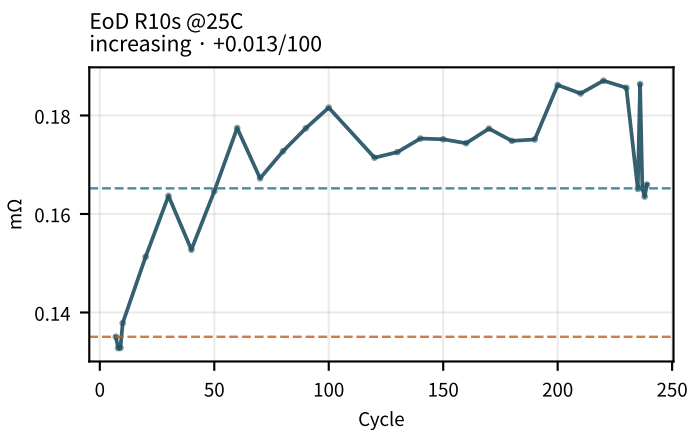
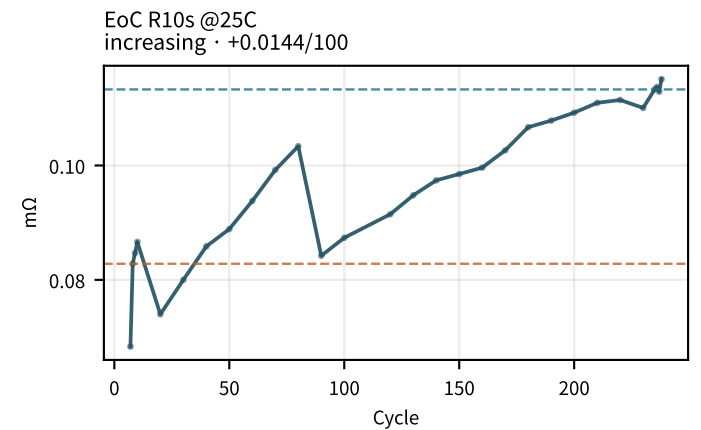
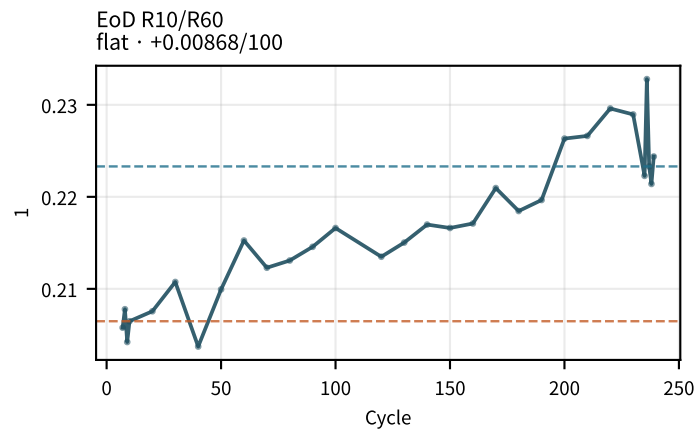
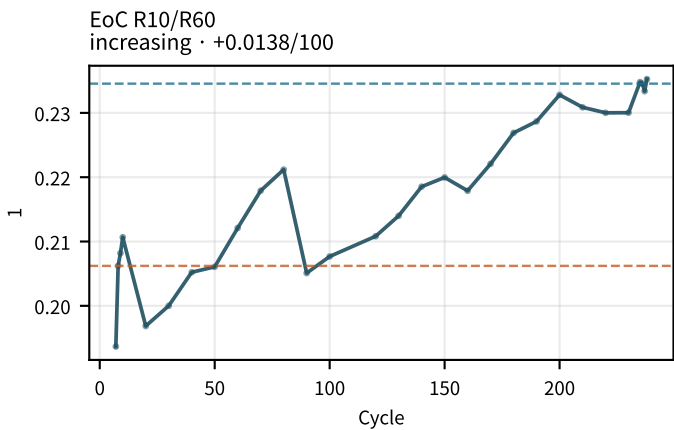
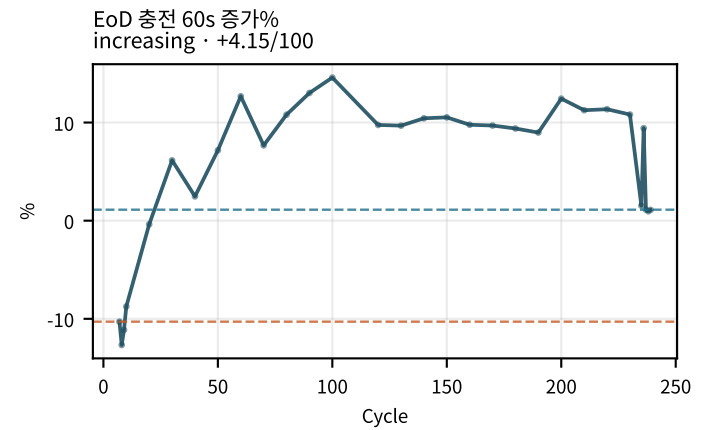
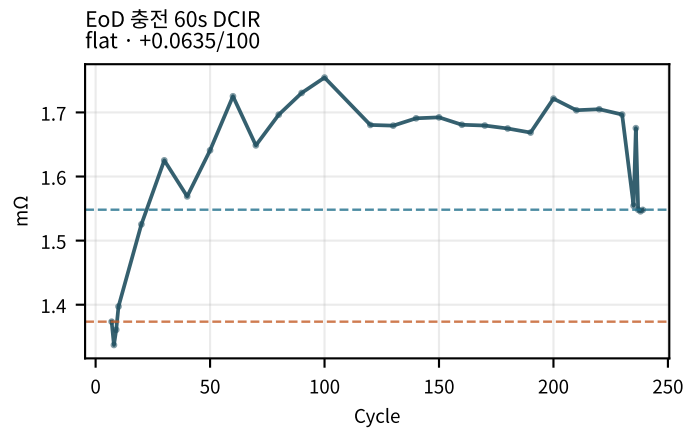
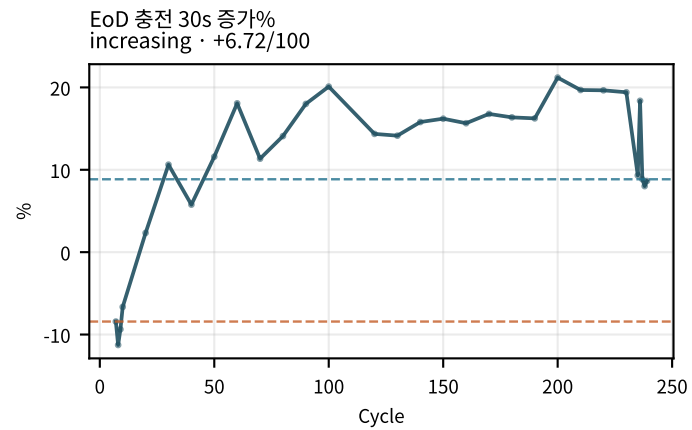
M02Ch111 · 휴지 전압 (충전/방전 후)



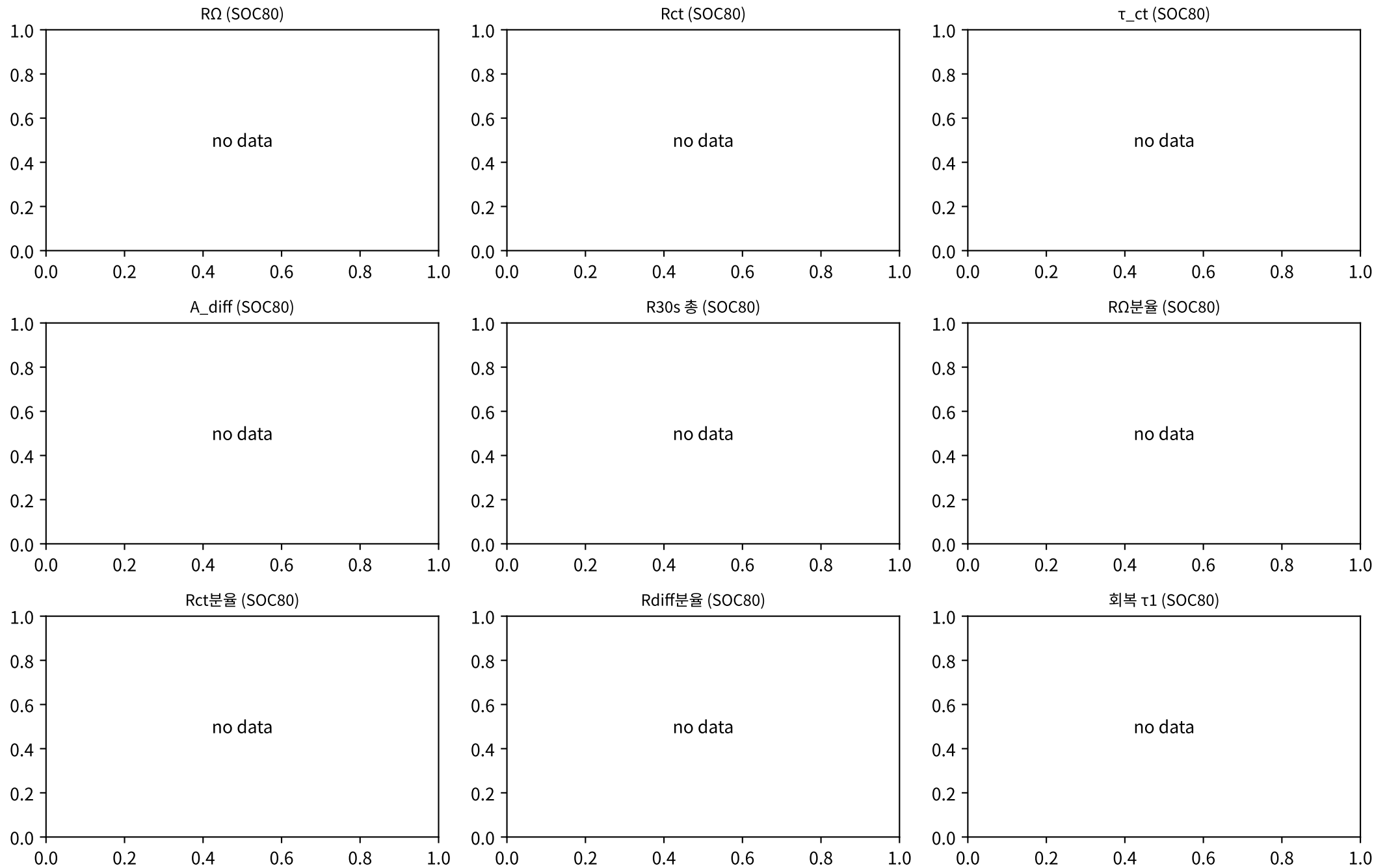
M02Ch111 · 시작 저항 (EoC/EoD)



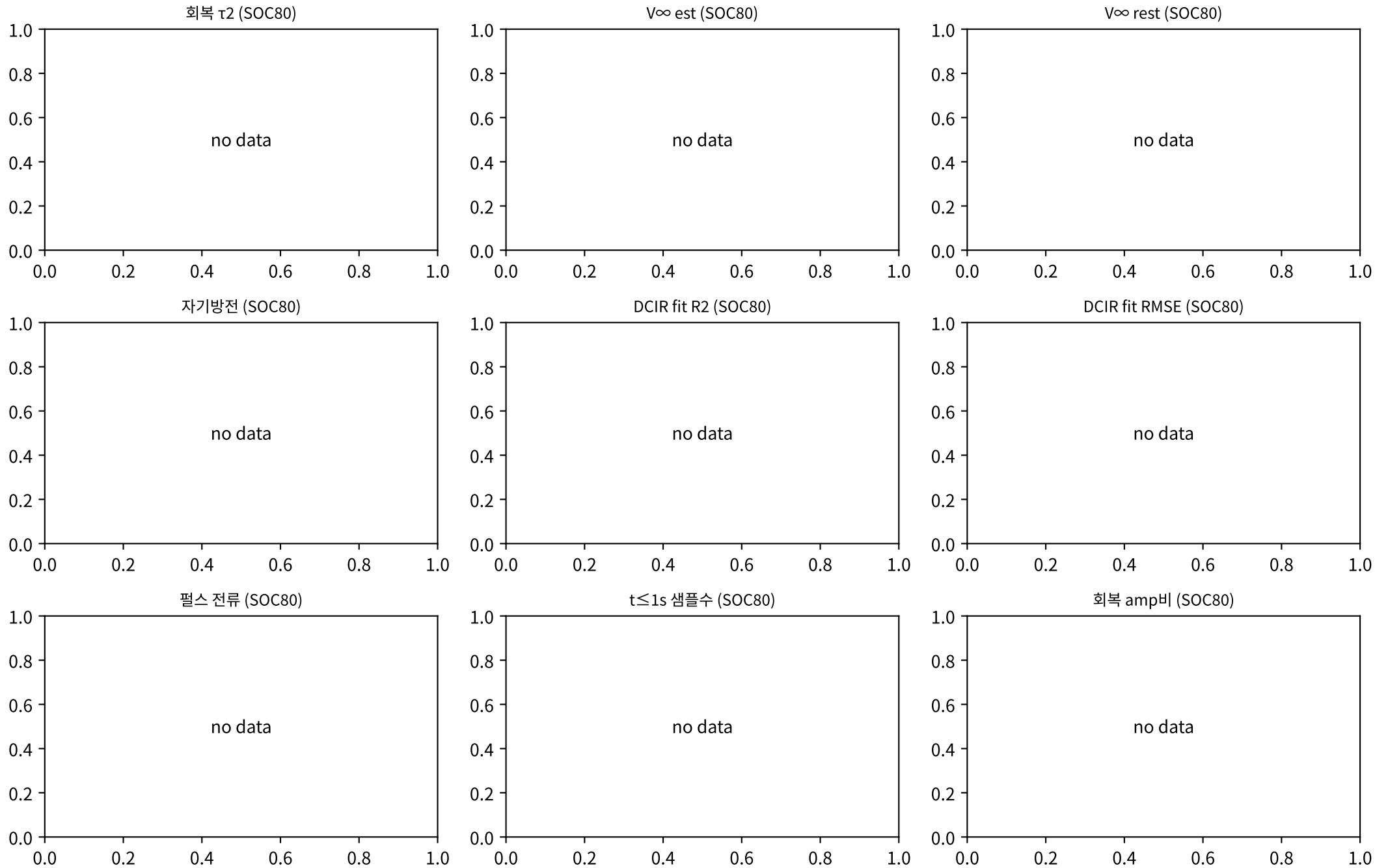
M02Ch111 · 시작 저항 (EoC/EoD)



M02Ch111 · DCIR 분해 · 성장

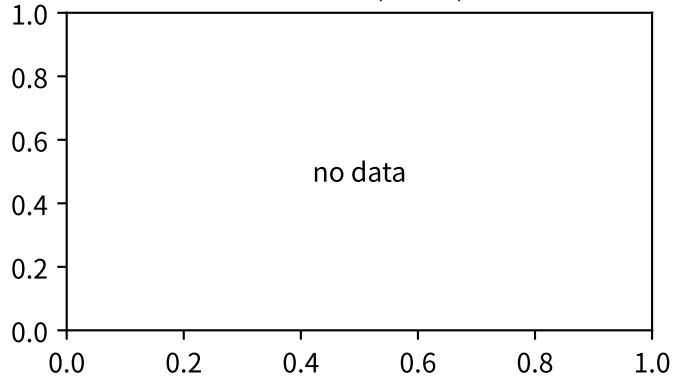


M02Ch111 · DCIR 분해 · 성장

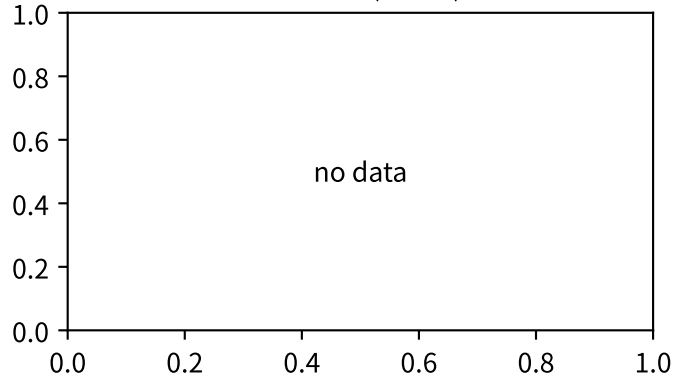


M02Ch111 · DCIR 분해 · 성장

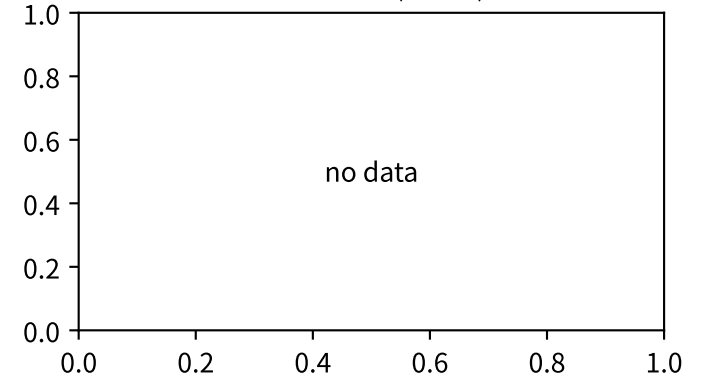
완화 완성도 (SOC80)



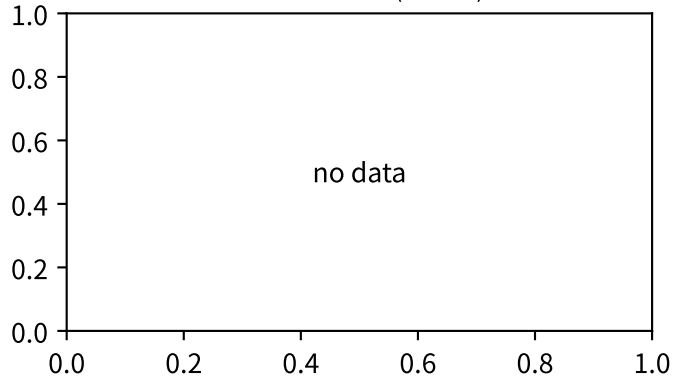
회복 fit R2 (SOC80)



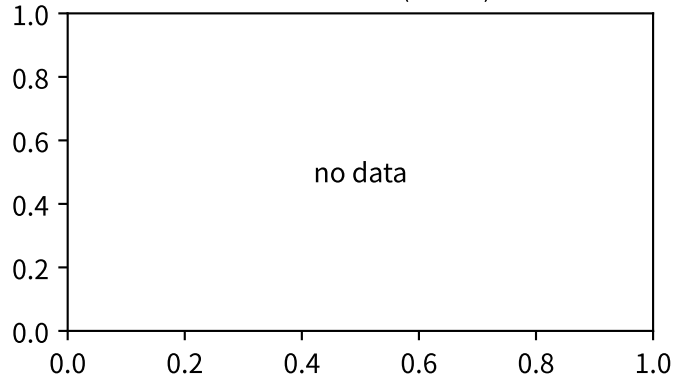
DCIR fit 유효 (SOC80)



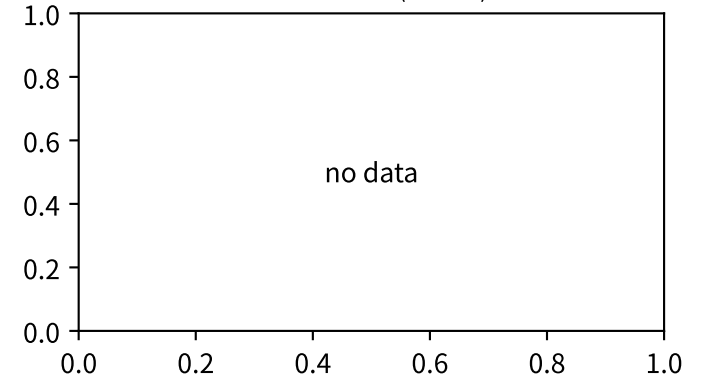
DCIR fit 조건수 (SOC80)



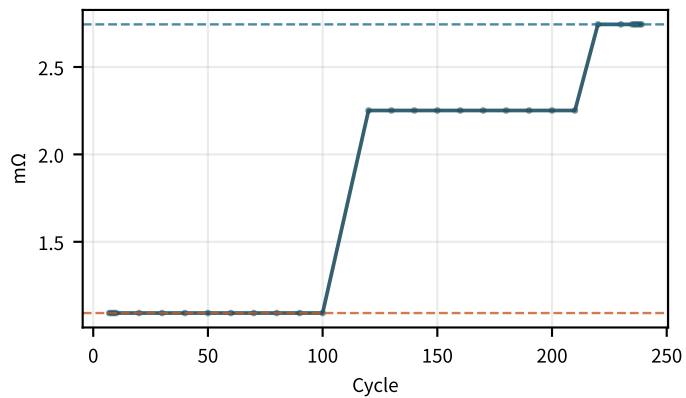
자기방전 fit 유효 (SOC80)



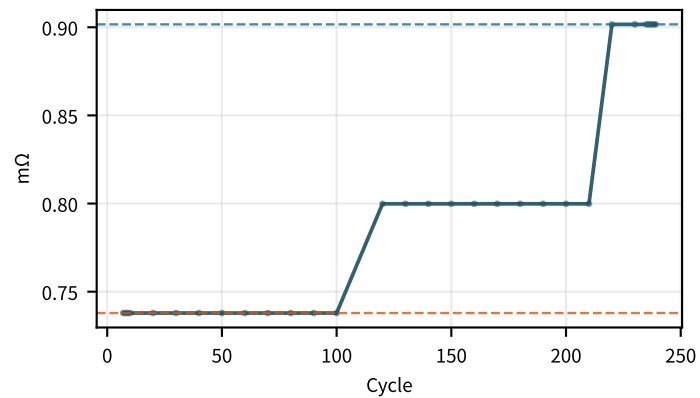
펄스 포인트수 (SOC80)



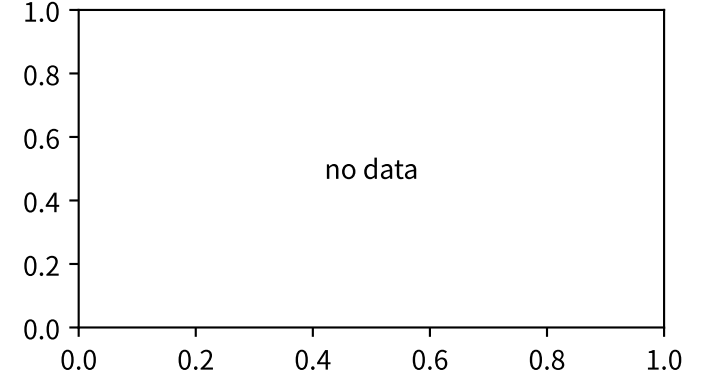
R Ω (SOC50)
increasing · +0.798/100



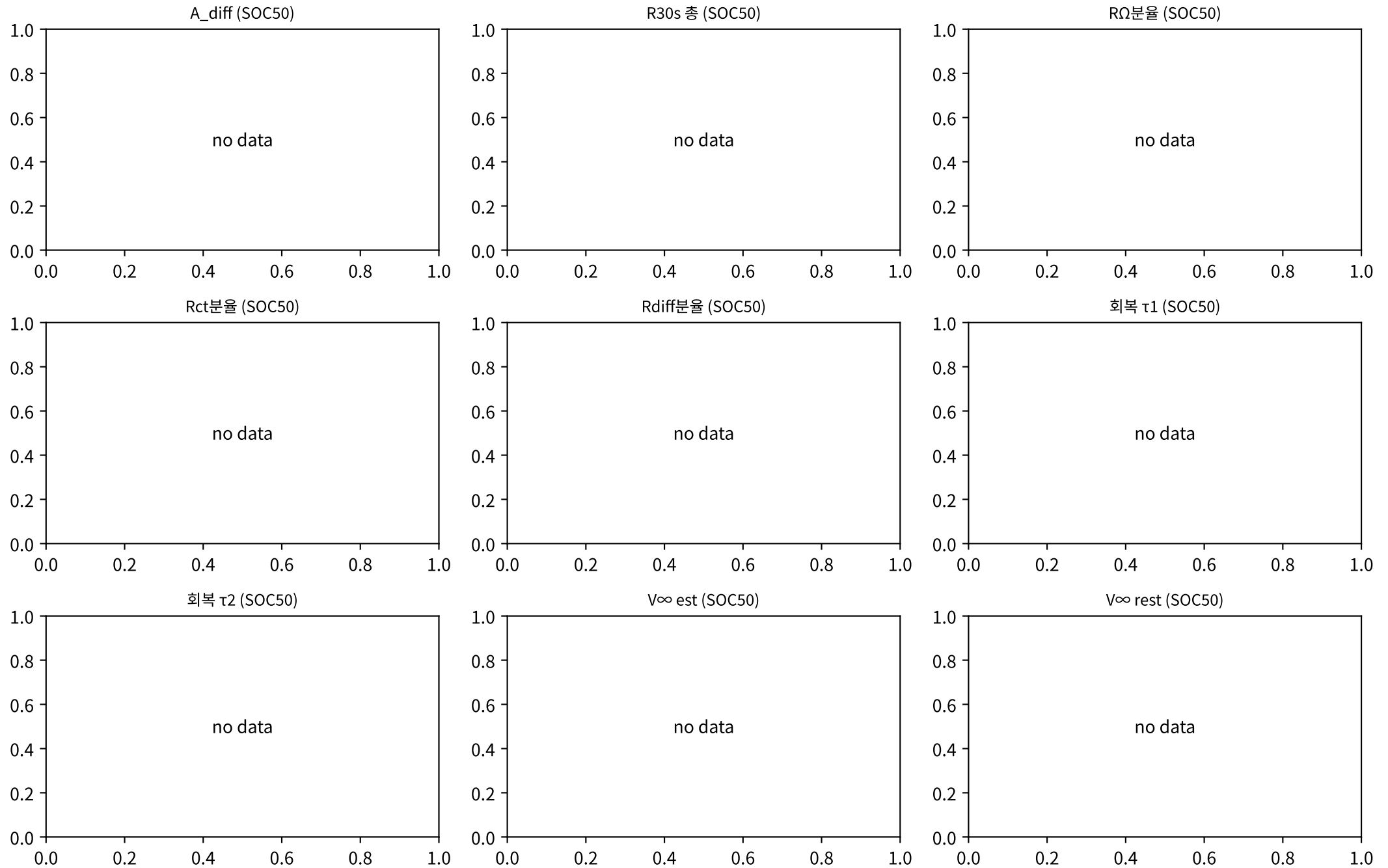
R $_{ct}$ (SOC50)
increasing · +0.0697/100



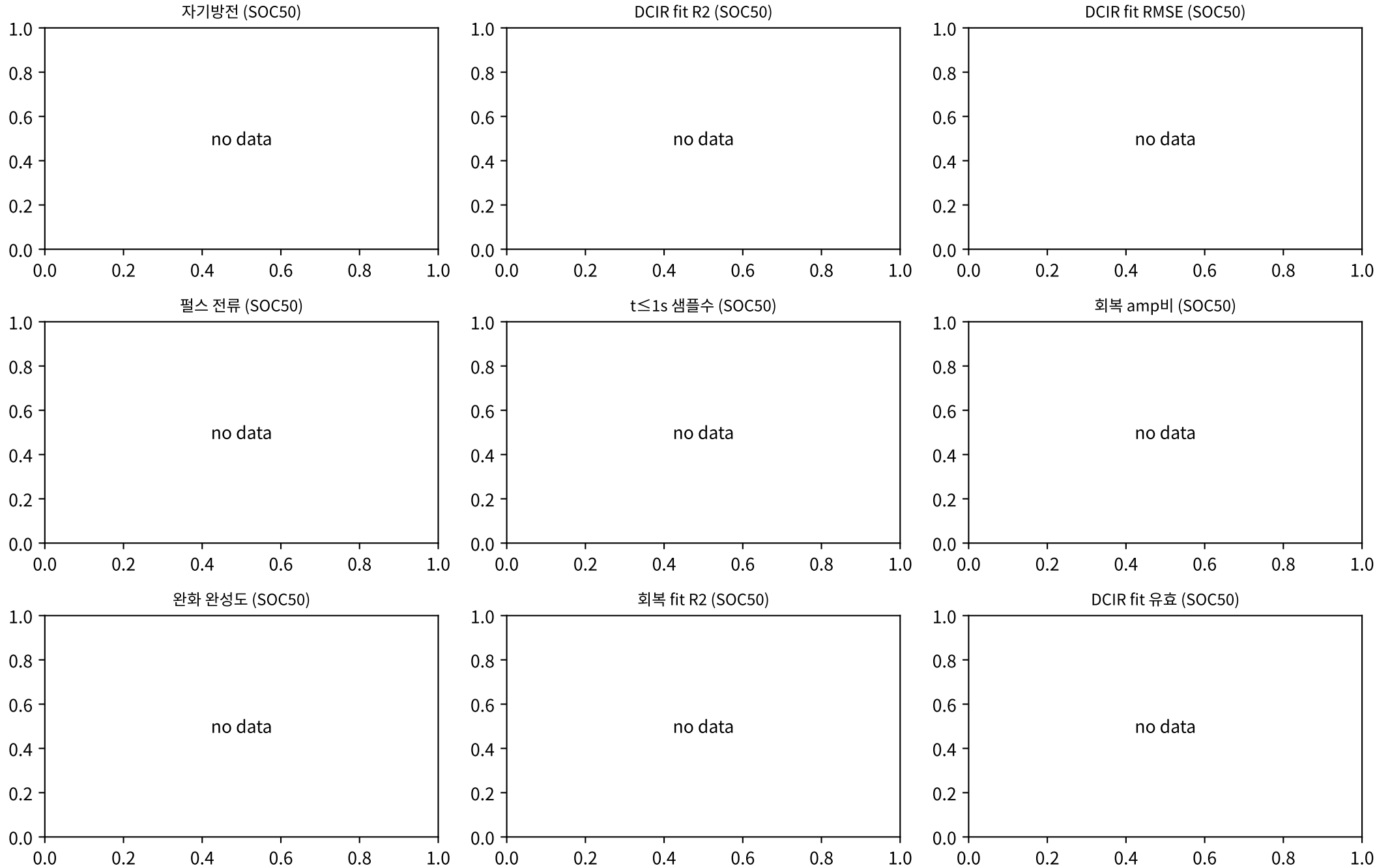
τ_{ct} (SOC50)



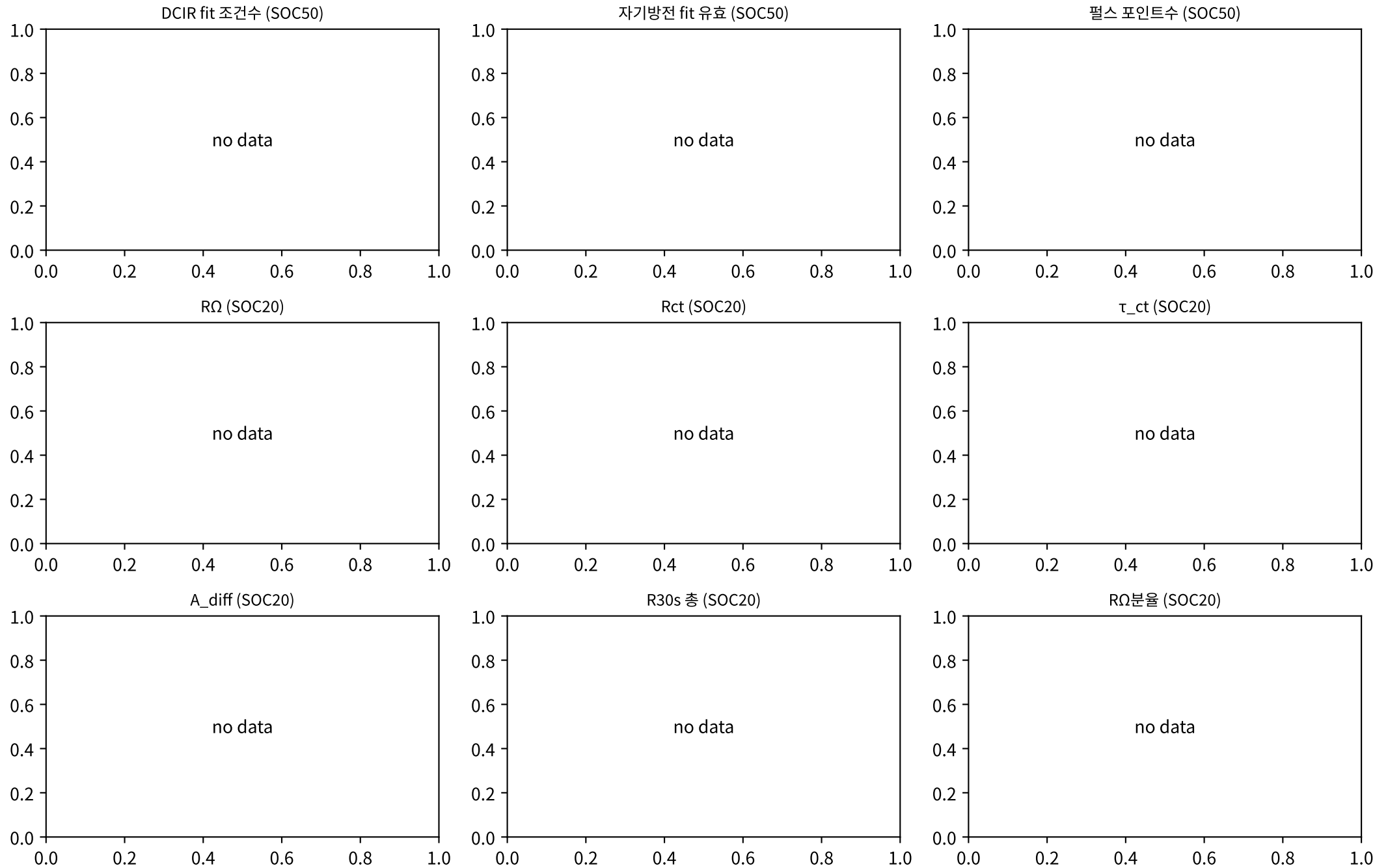
M02Ch111 · DCIR 분해 · 성장



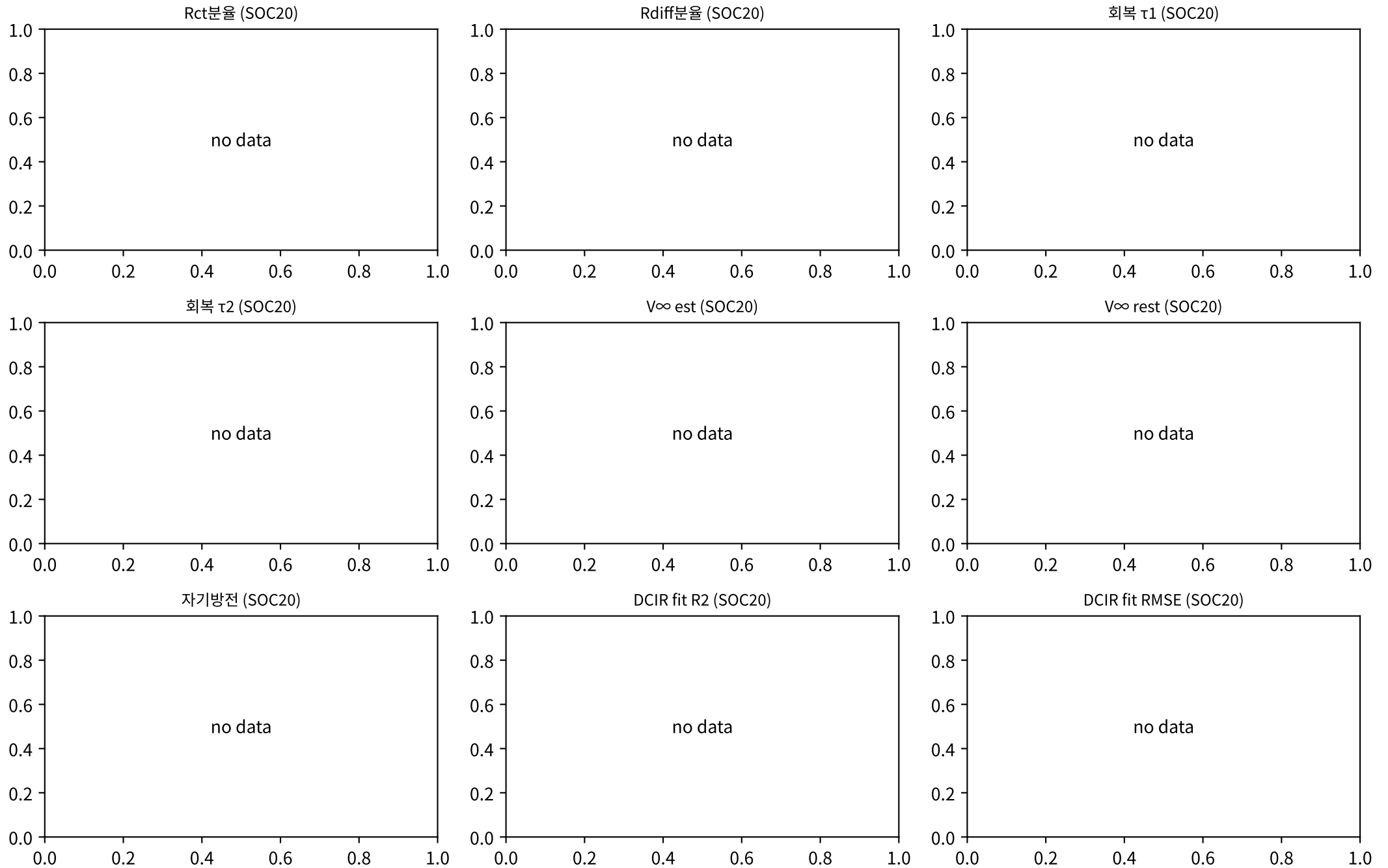
M02Ch111 · DCIR 분해 · 성장



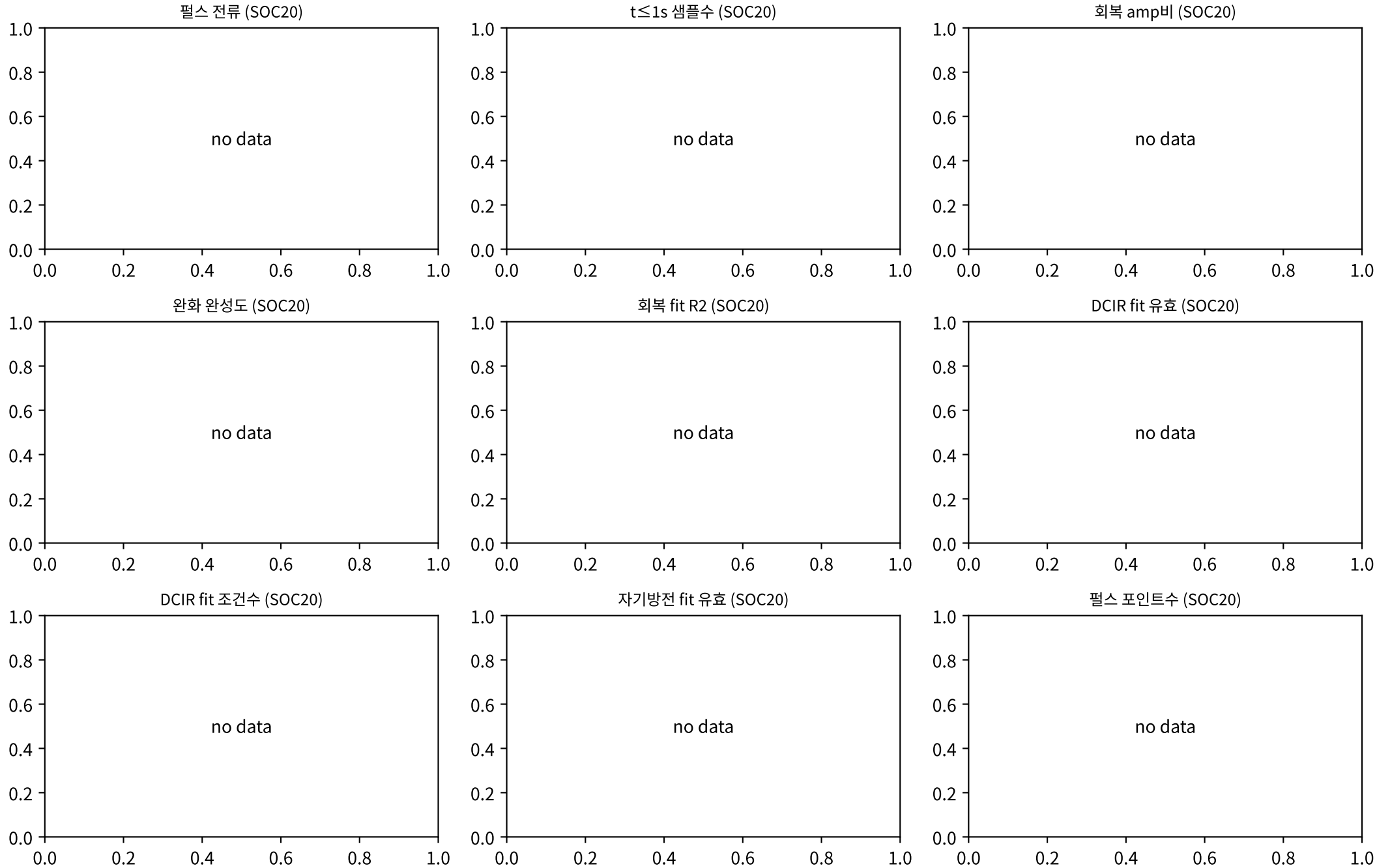
M02Ch111 · DCIR 분해 · 성장



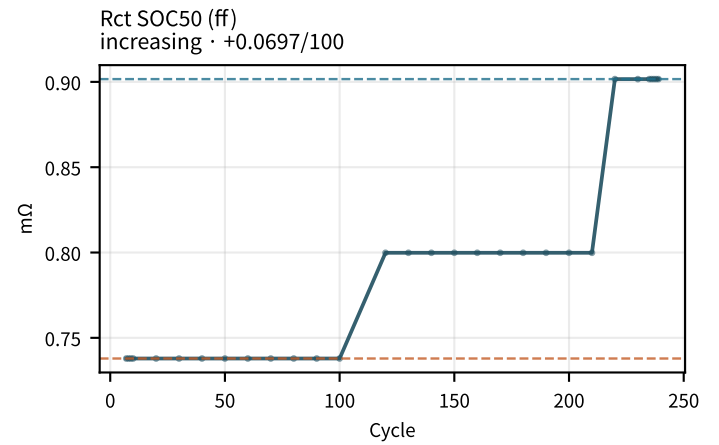
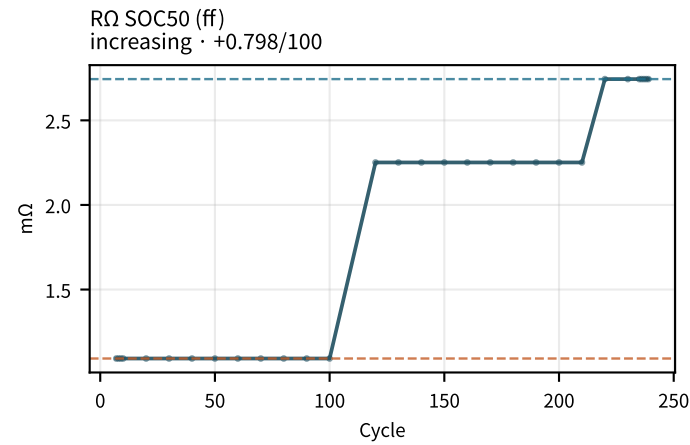
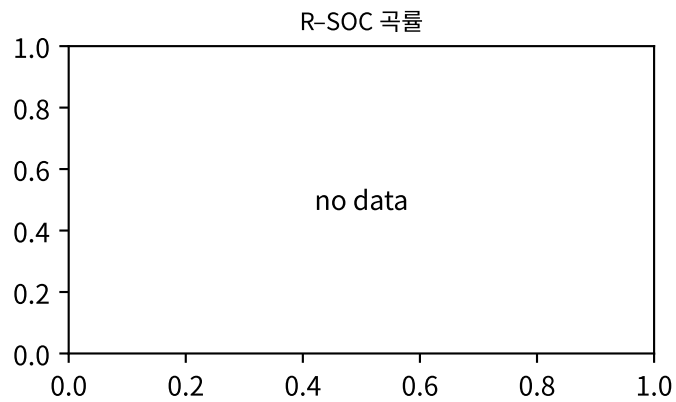
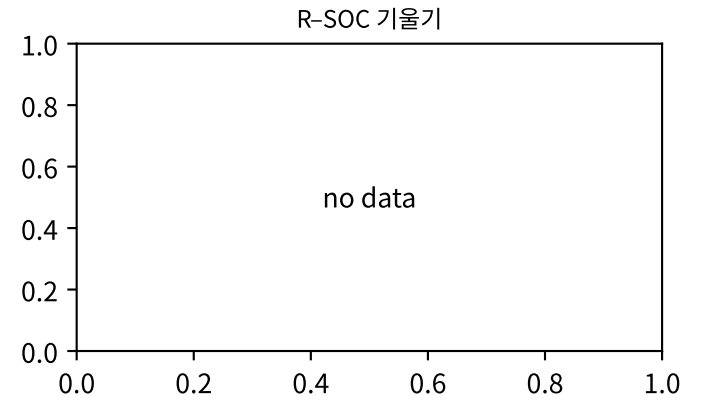
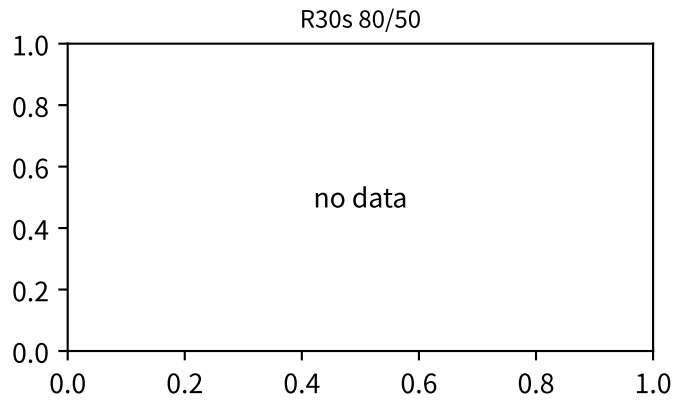
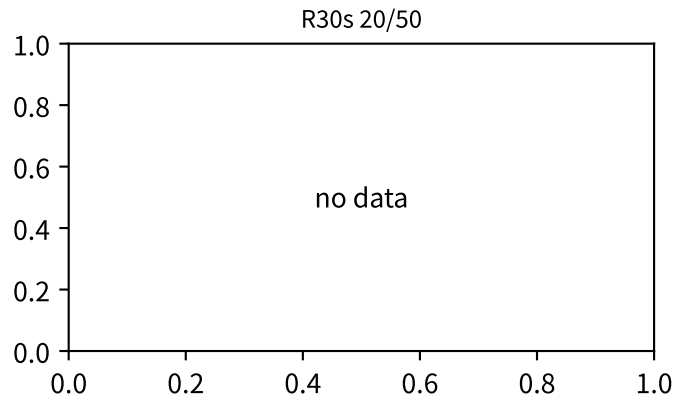
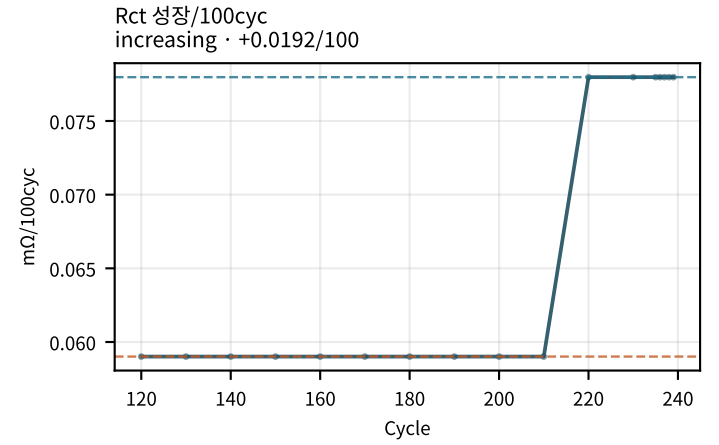
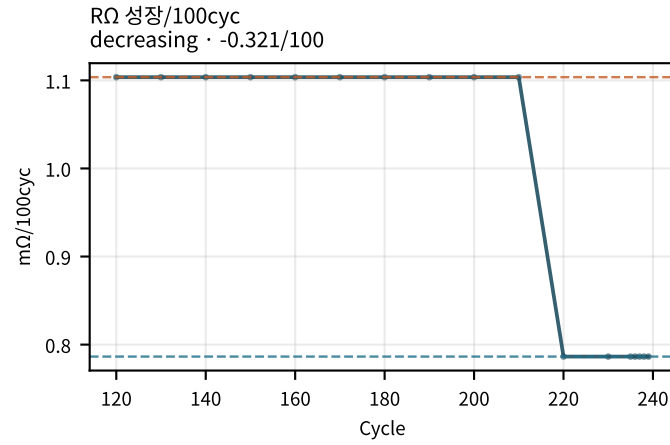
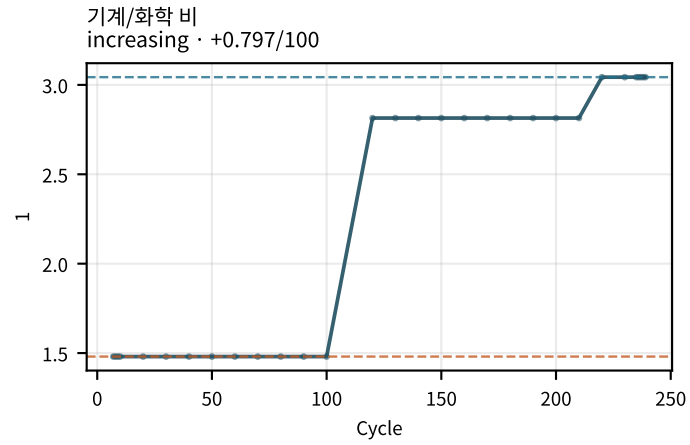
M02Ch111 · DCIR 분해 · 성장



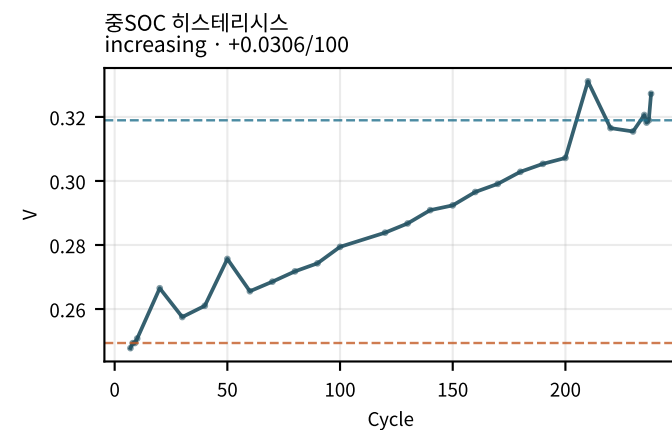
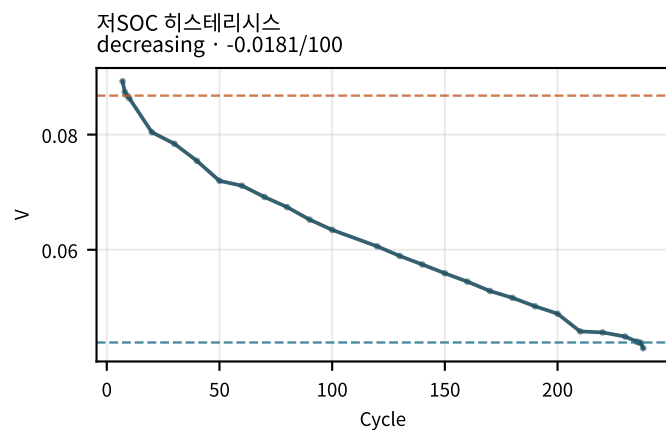
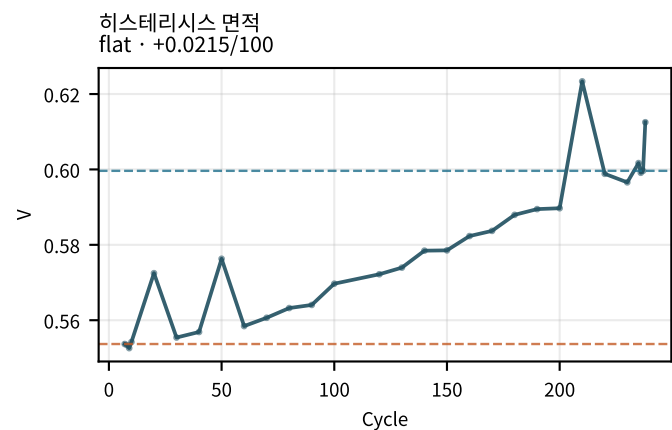
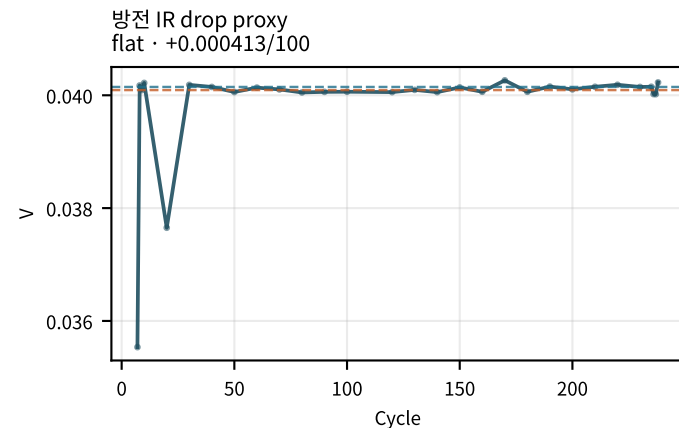
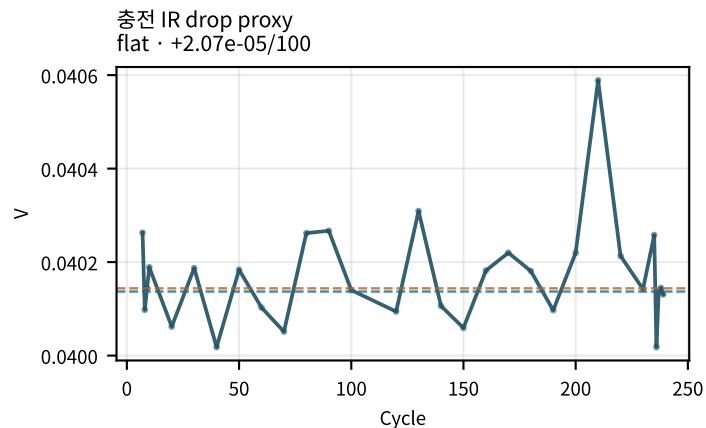
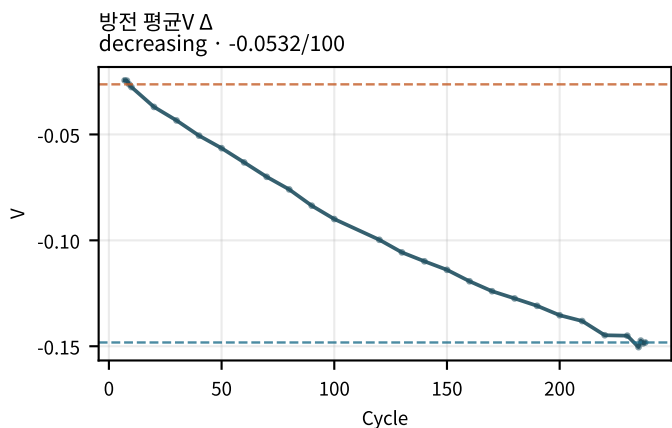
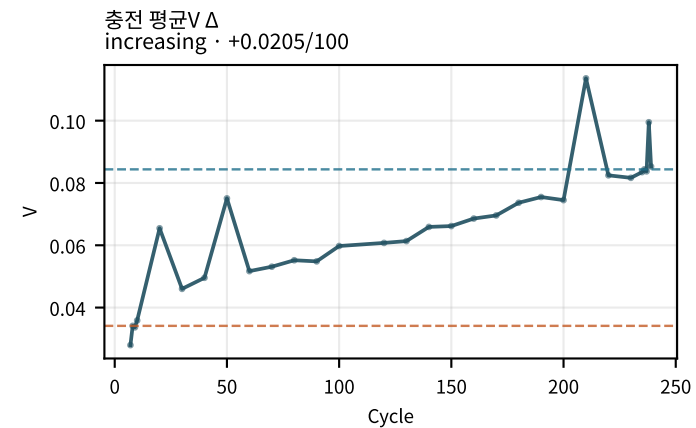
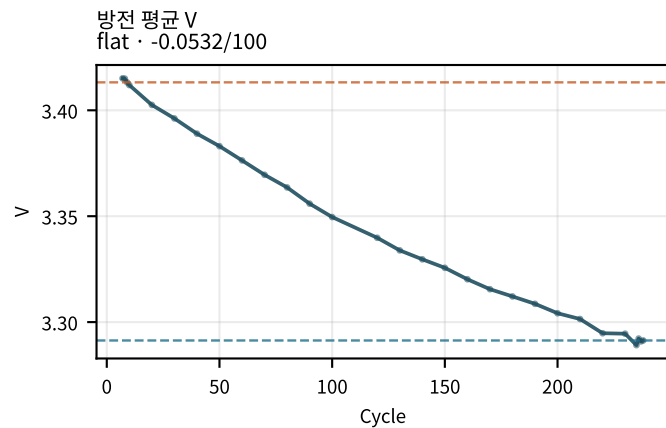
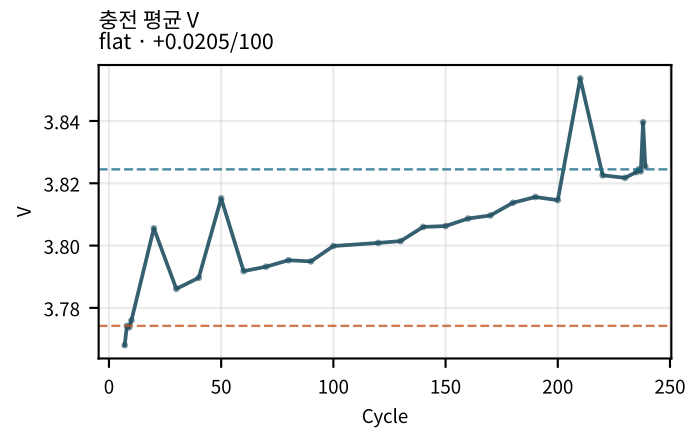
M02Ch111 · DCIR 분해 · 성장



M02Ch111 · DCIR 분해 · 성장

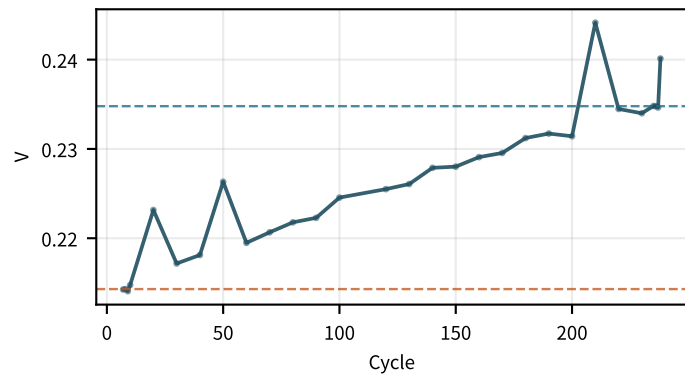


M02Ch111 · 곡선 형상 · 히스테리시스

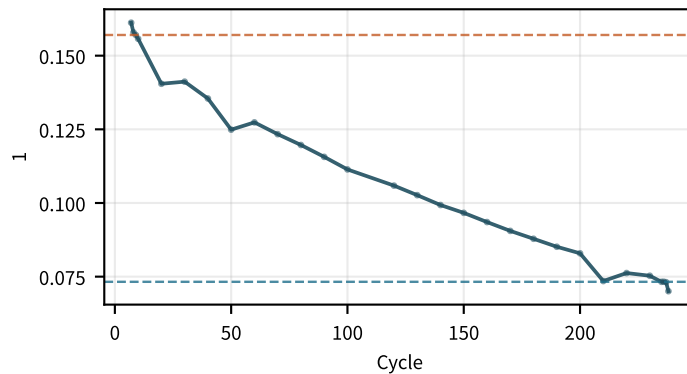


M02Ch111 · 곡선 형상 · 히스테리시스

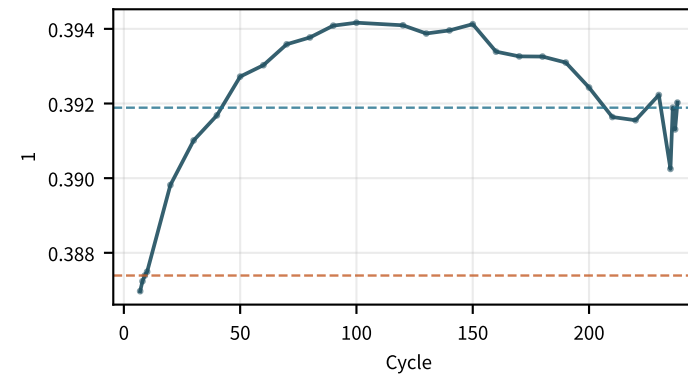
고SOC 히스테리시스
flat · +0.00904/100



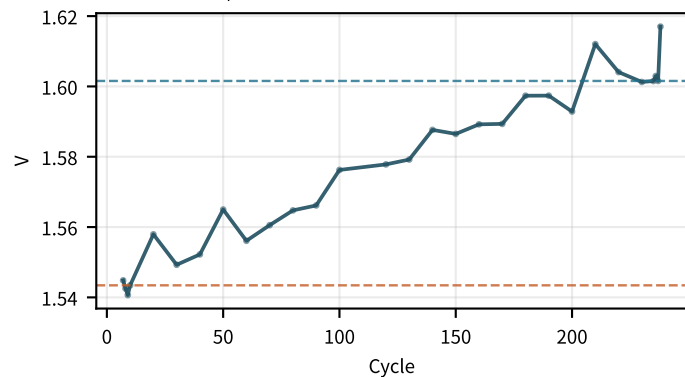
히스테리시스 저SOC분율
decreasing · -0.0353/100



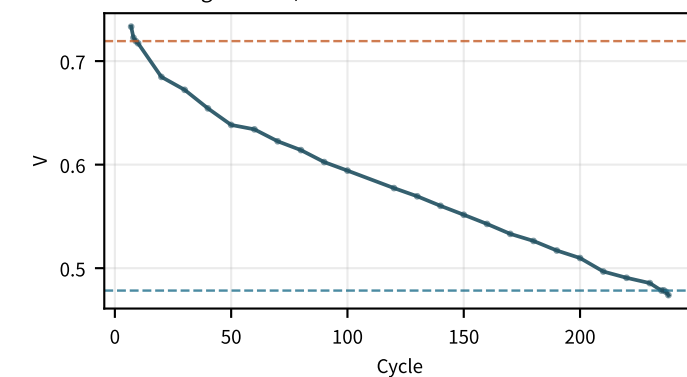
히스테리시스 고SOC분율
flat · +0.00113/100



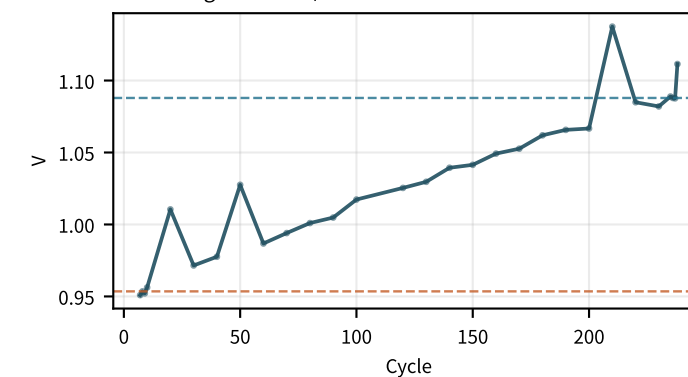
최대 히스테리시스 dV
flat · +0.0274/100



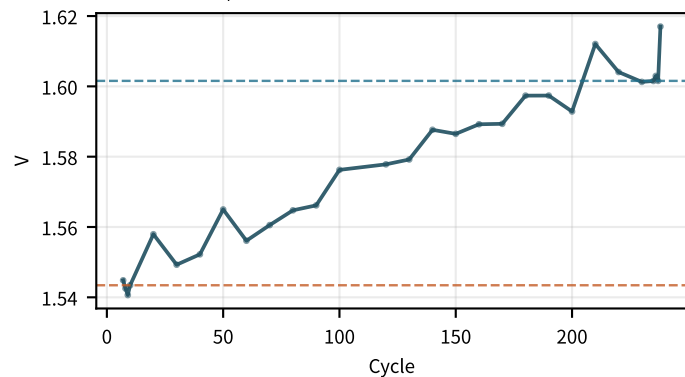
max dV 저SOC
decreasing · -0.101/100



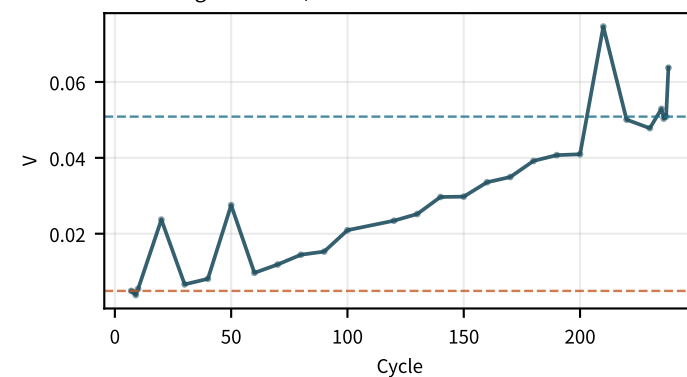
max dV 중SOC
increasing · +0.0586/100



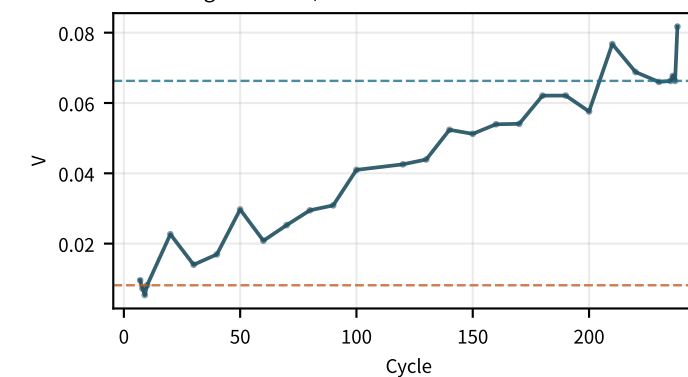
max dV 고SOC
flat · +0.0274/100



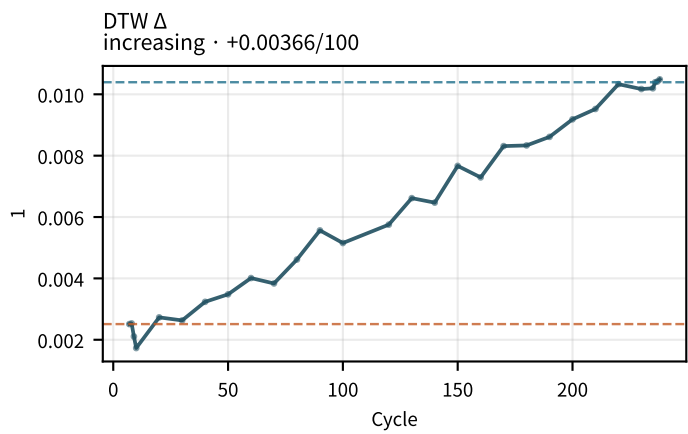
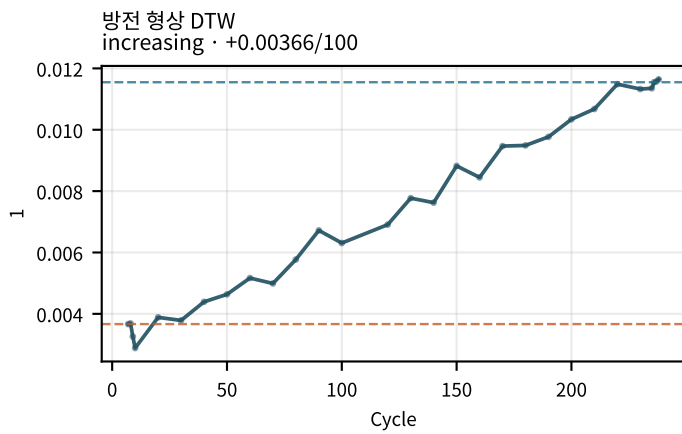
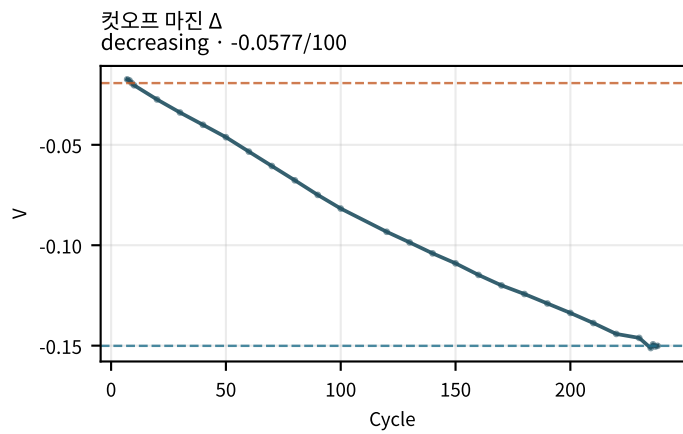
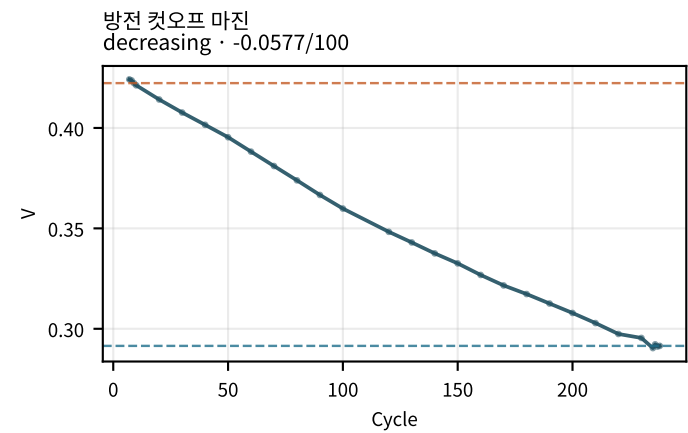
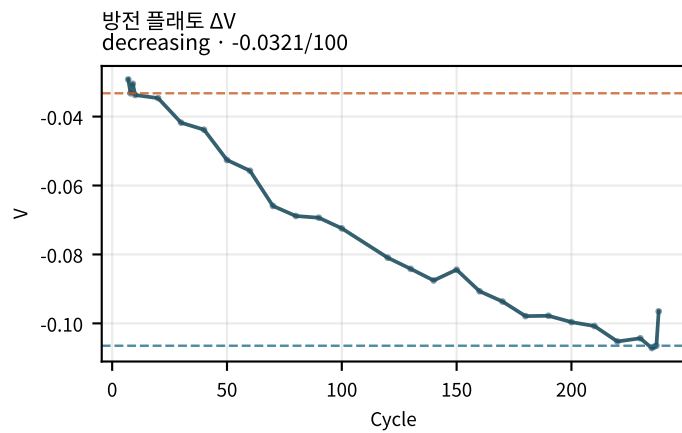
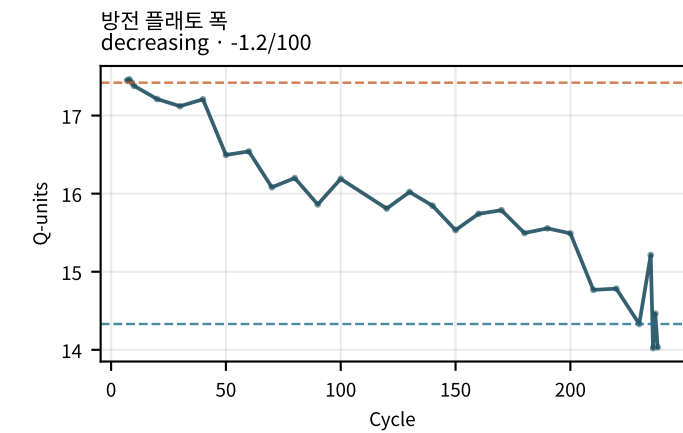
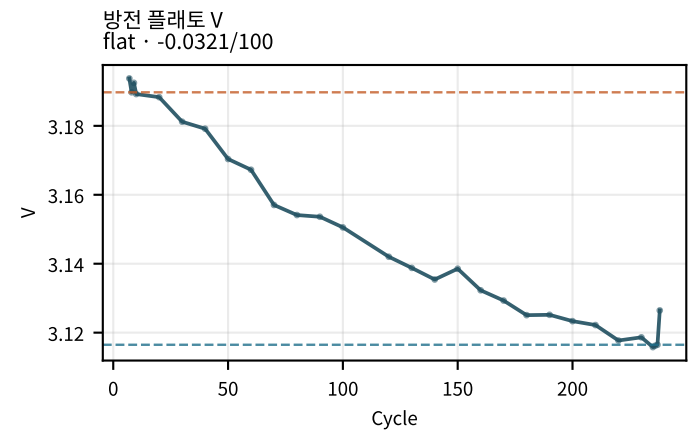
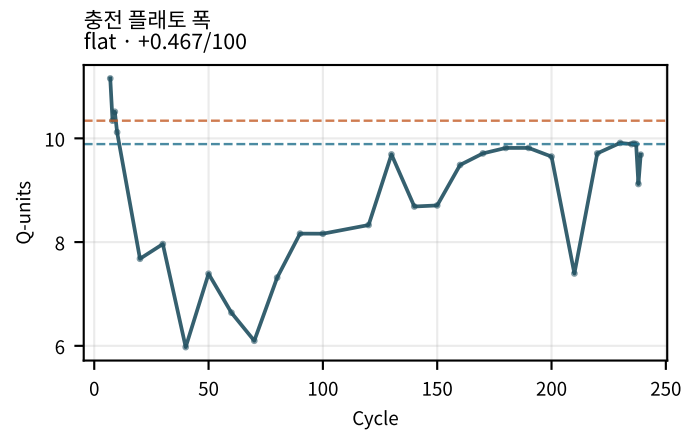
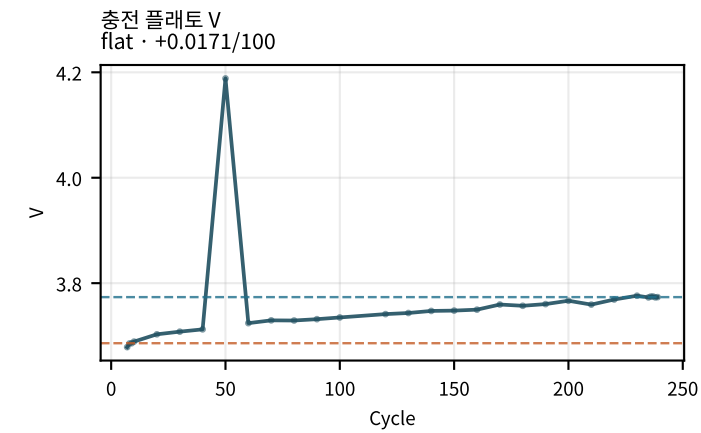
히스테리시스 Δ
increasing · +0.0215/100



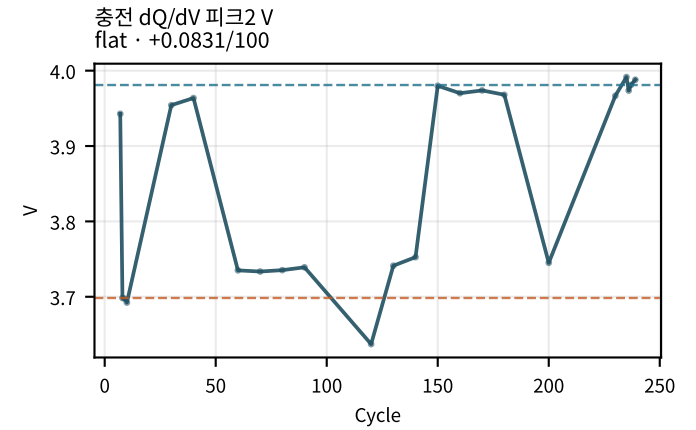
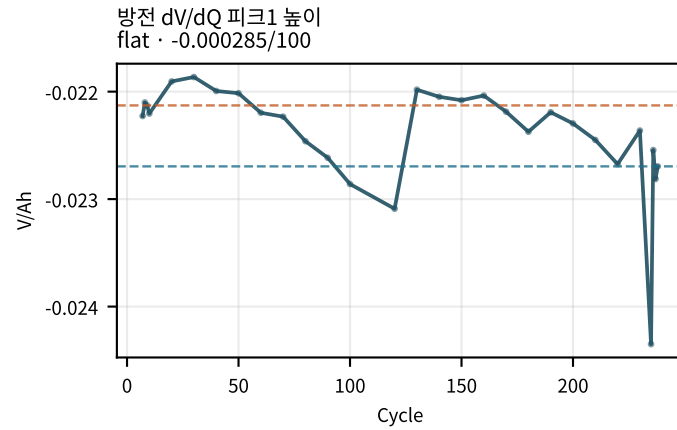
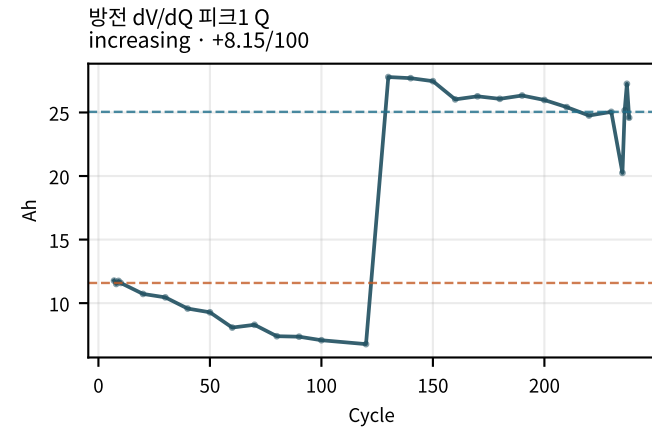
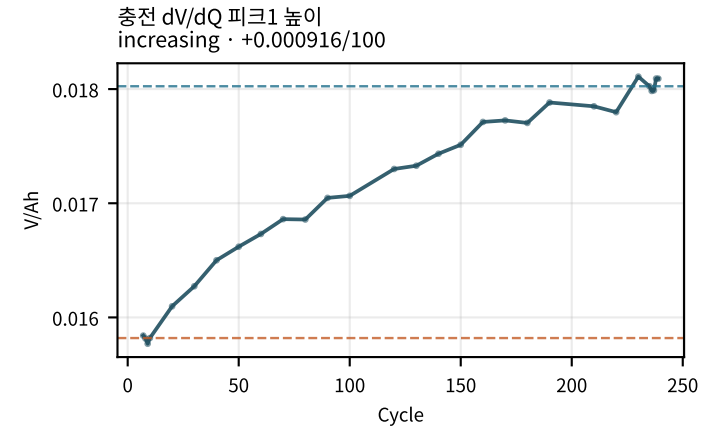
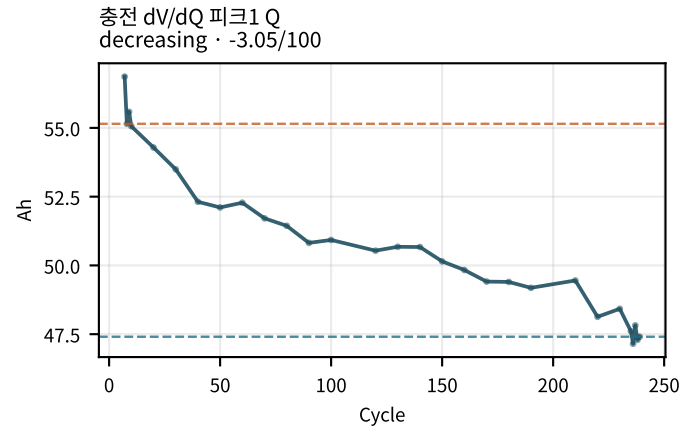
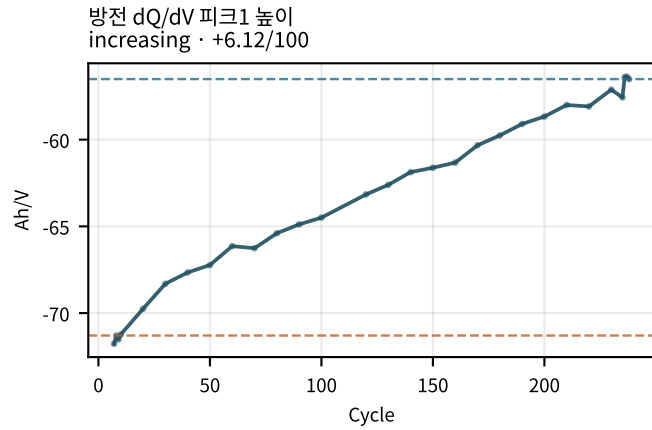
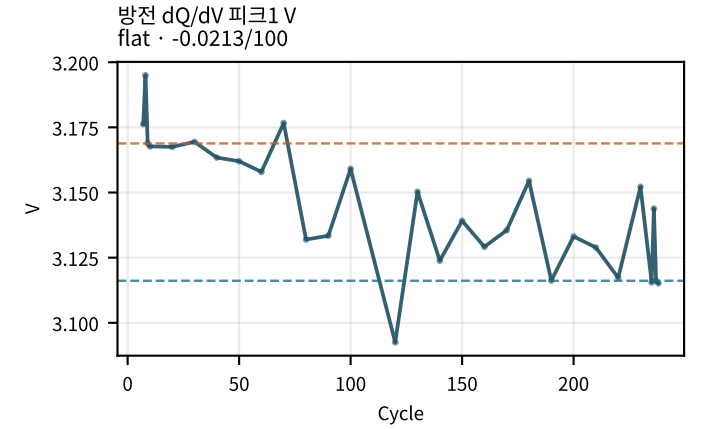
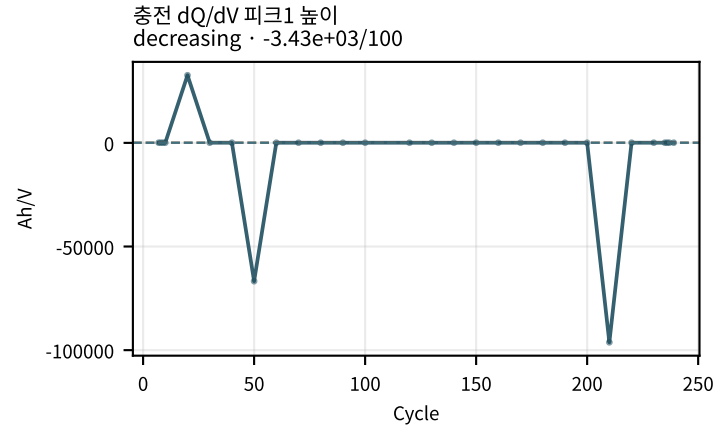
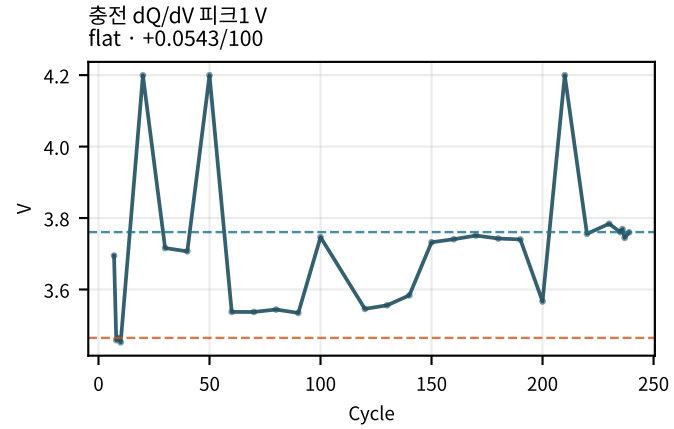
max dV Δ
increasing · +0.0274/100



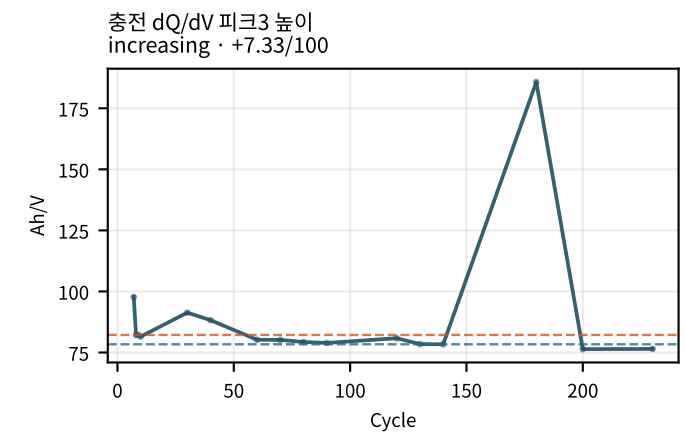
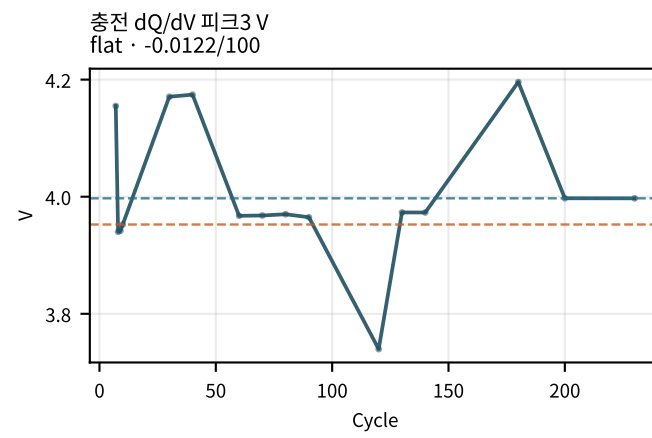
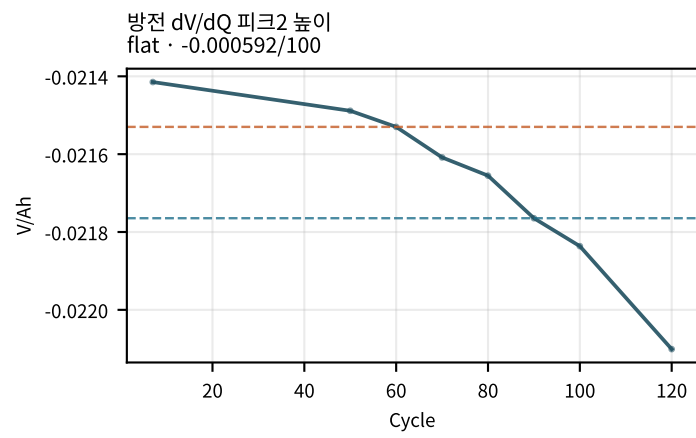
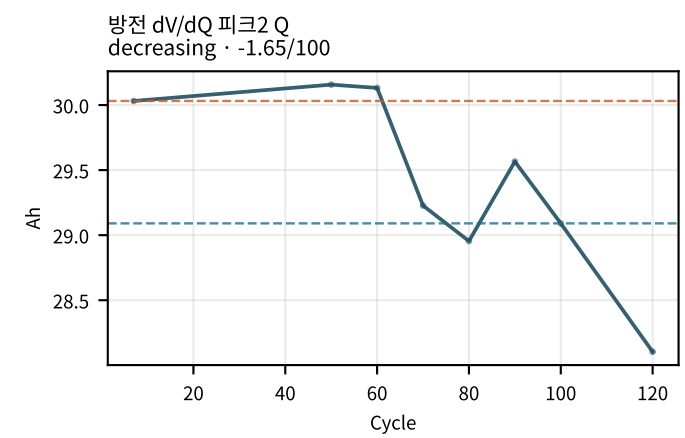
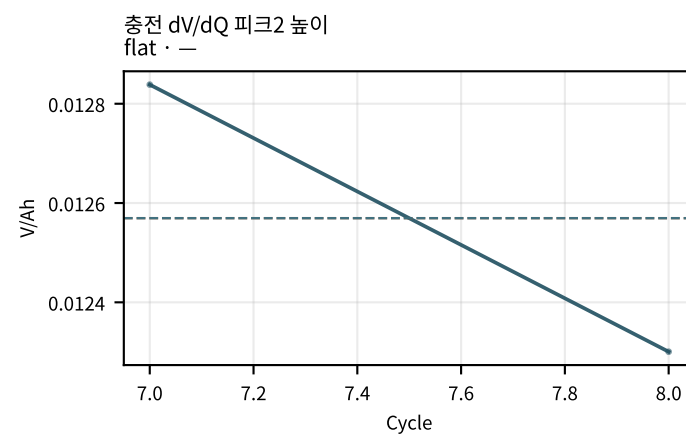
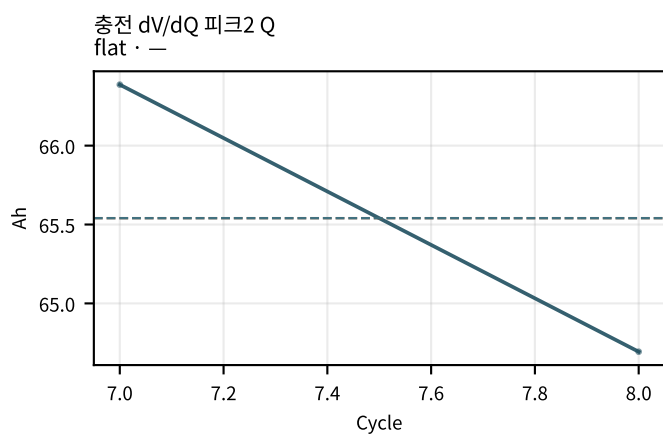
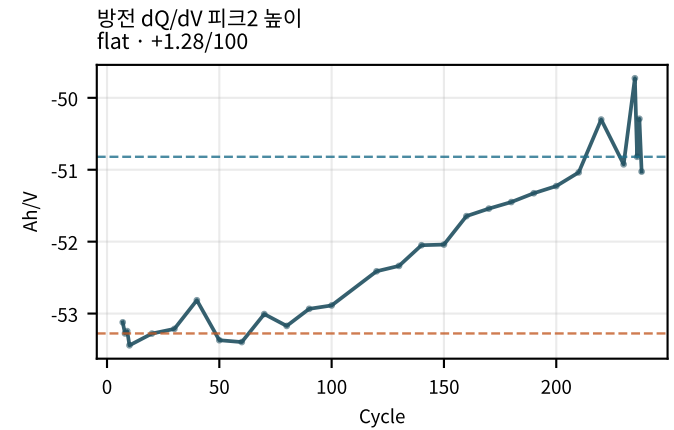
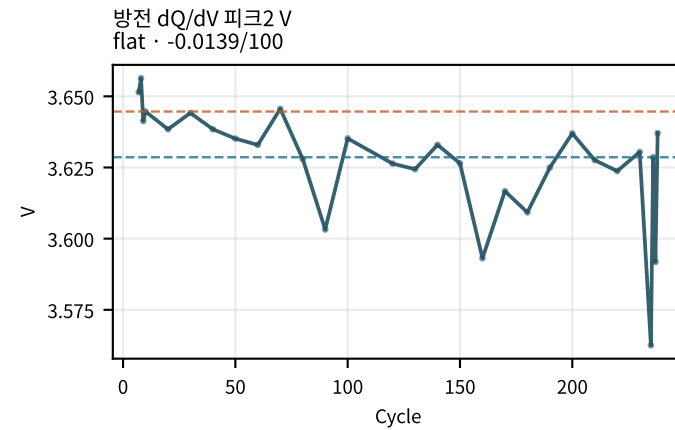
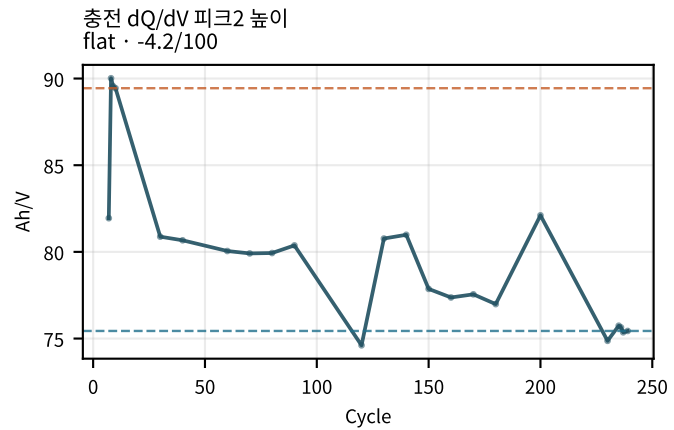
M02Ch111 · 곡선 형상 · 히스테리시스



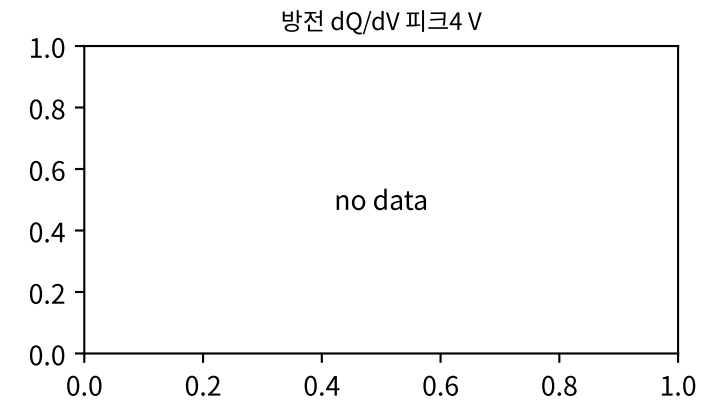
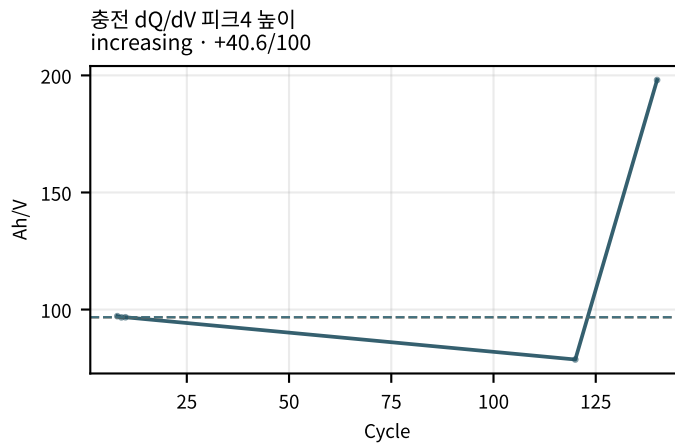
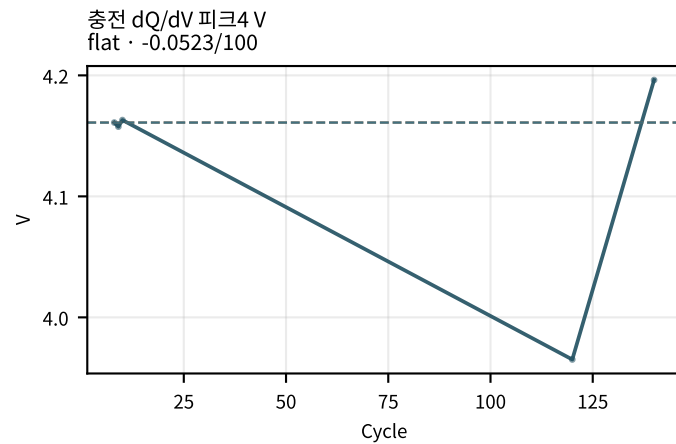
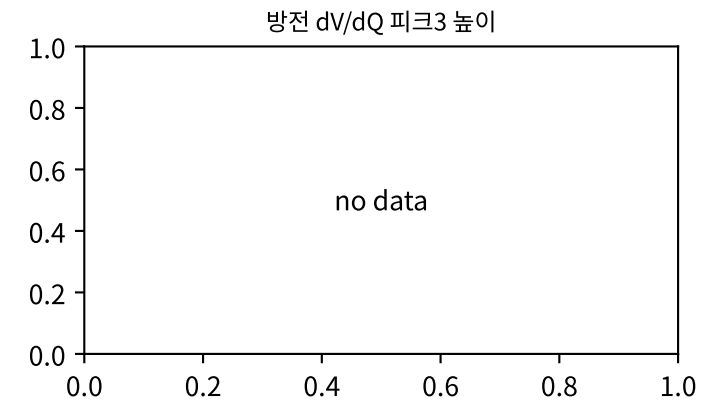
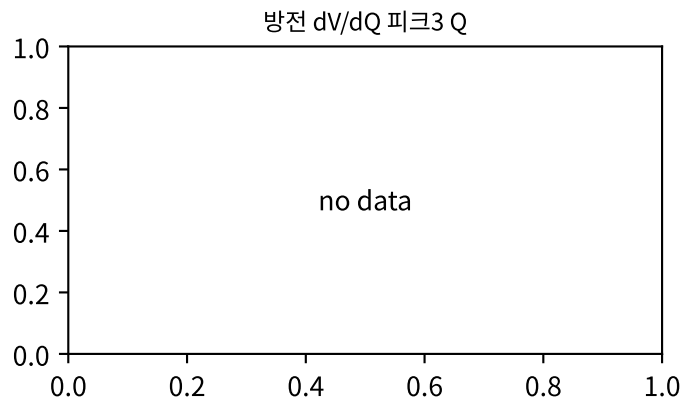
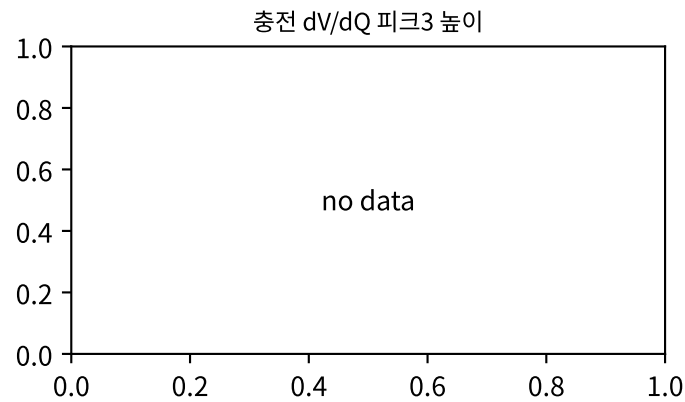
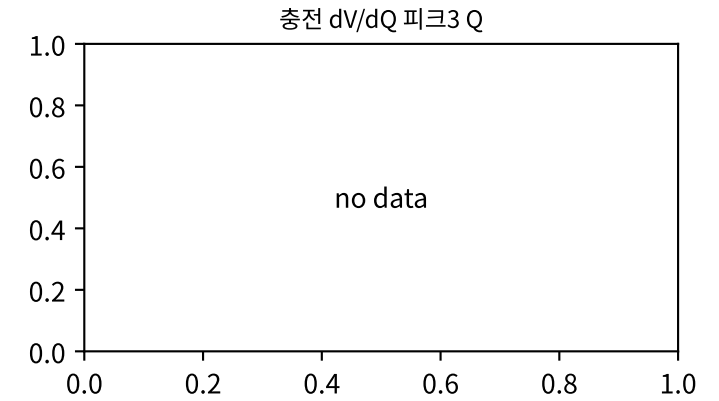
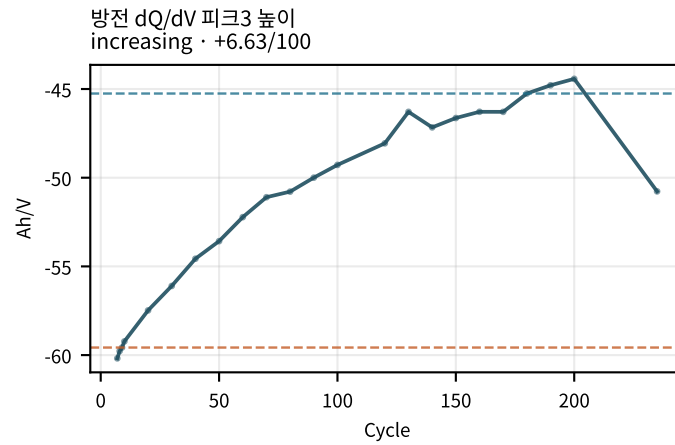
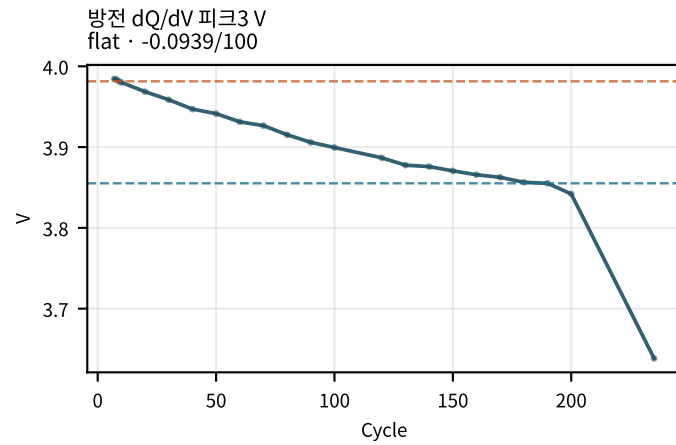
M02Ch111 · dQ/dV · dV/dQ 피크



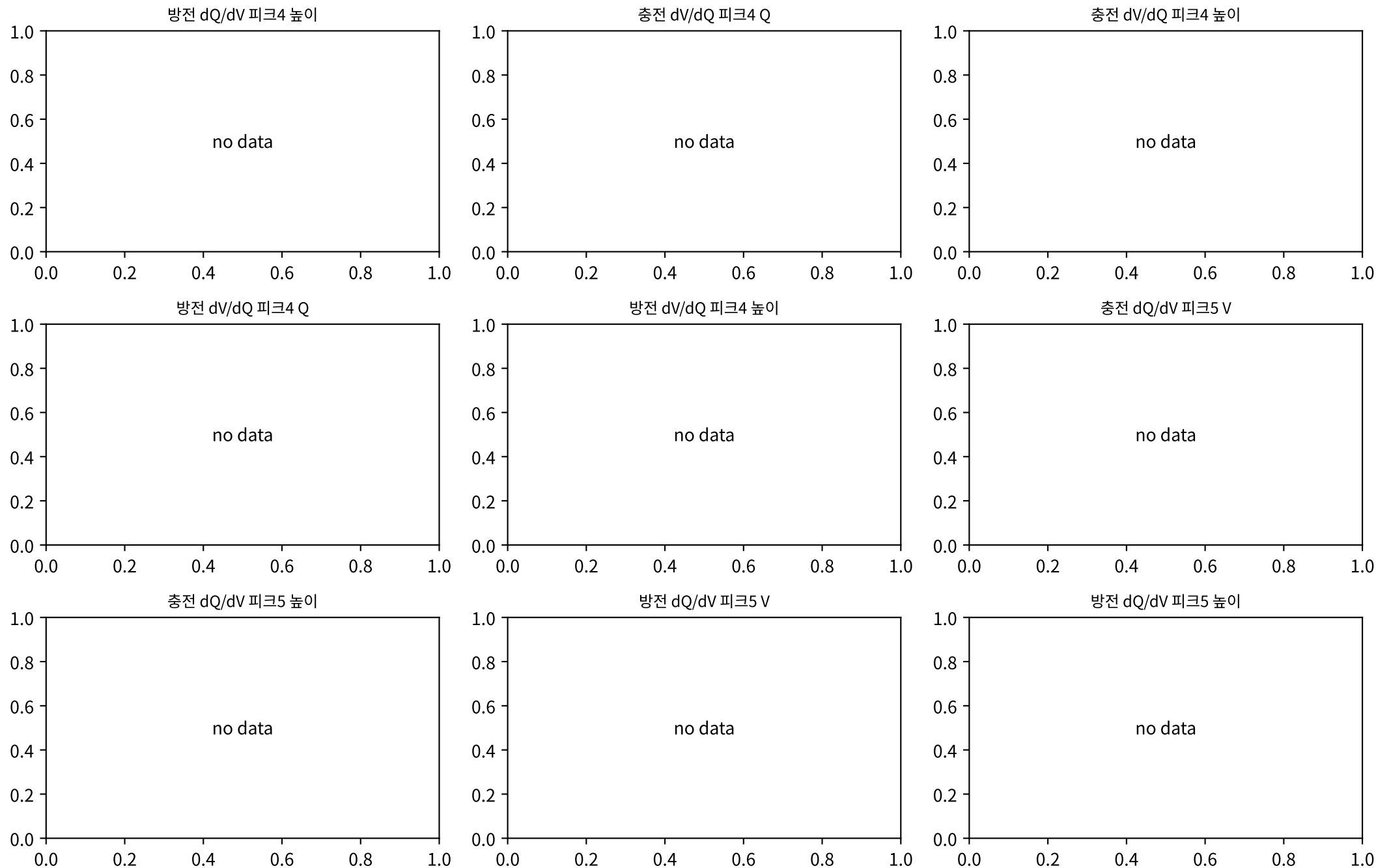
M02Ch111 · dQ/dV · dV/dQ 피크



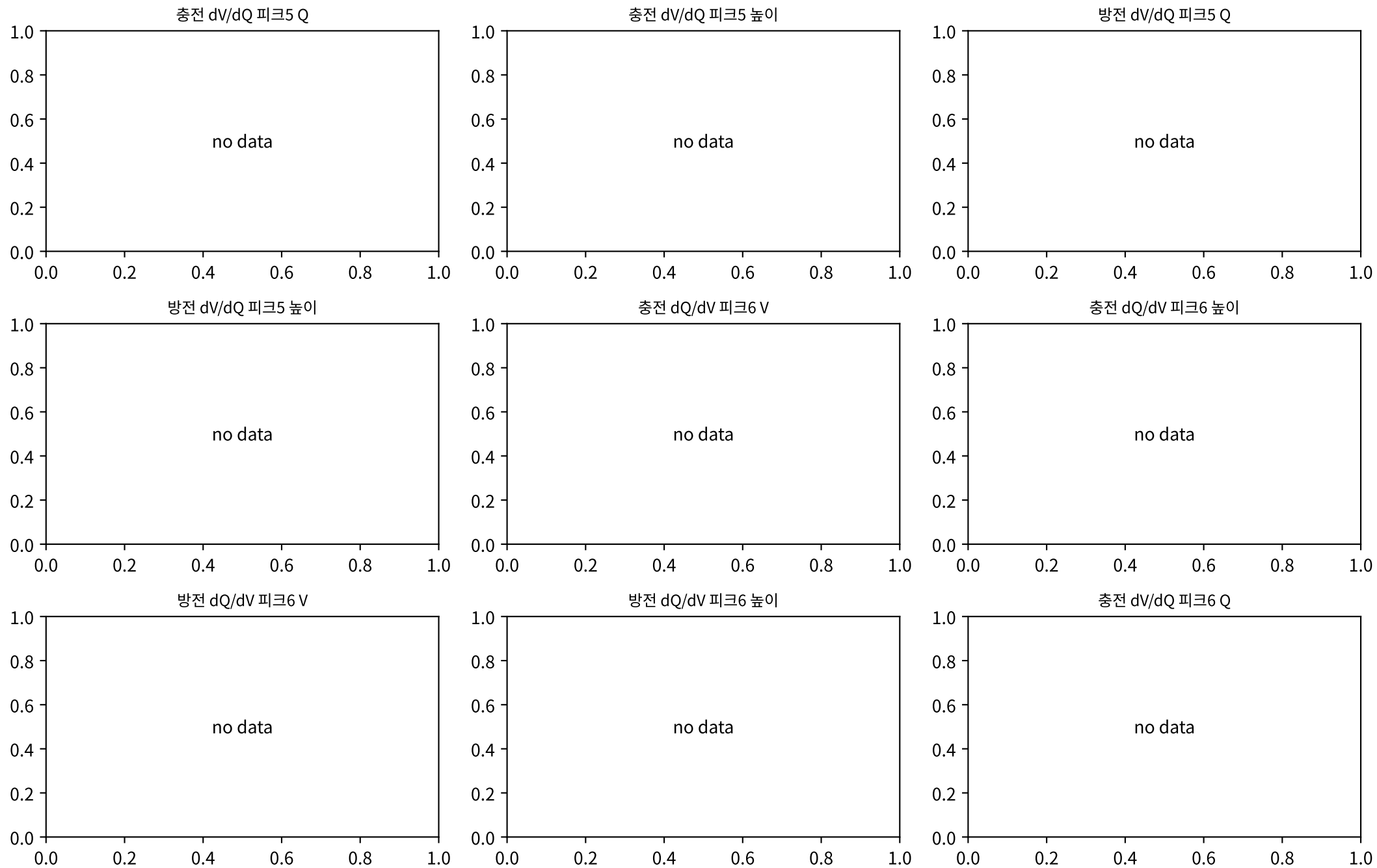
M02Ch111 · dQ/dV · dV/dQ 피크



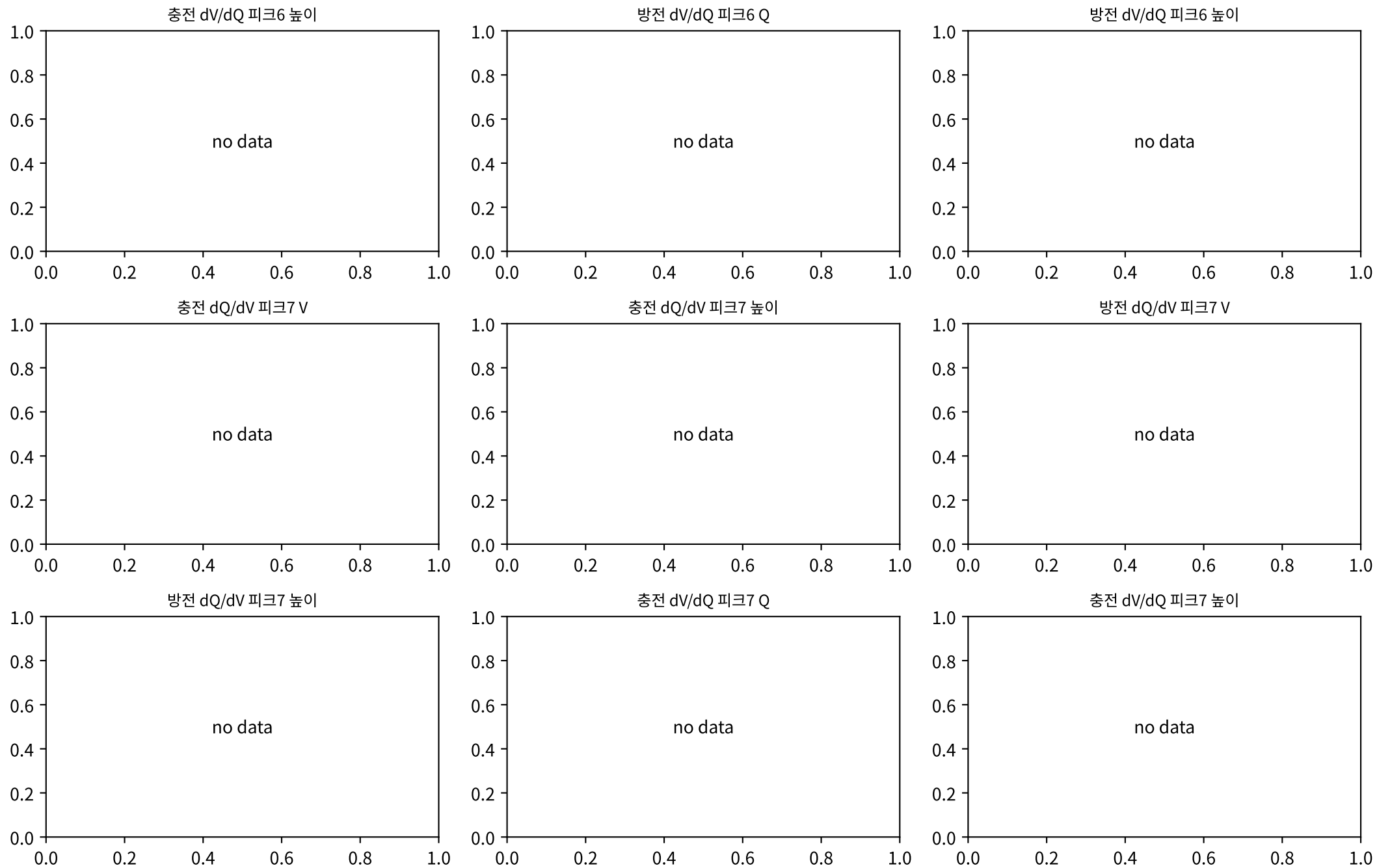
M02Ch111 · dQ/dV · dV/dQ 피크



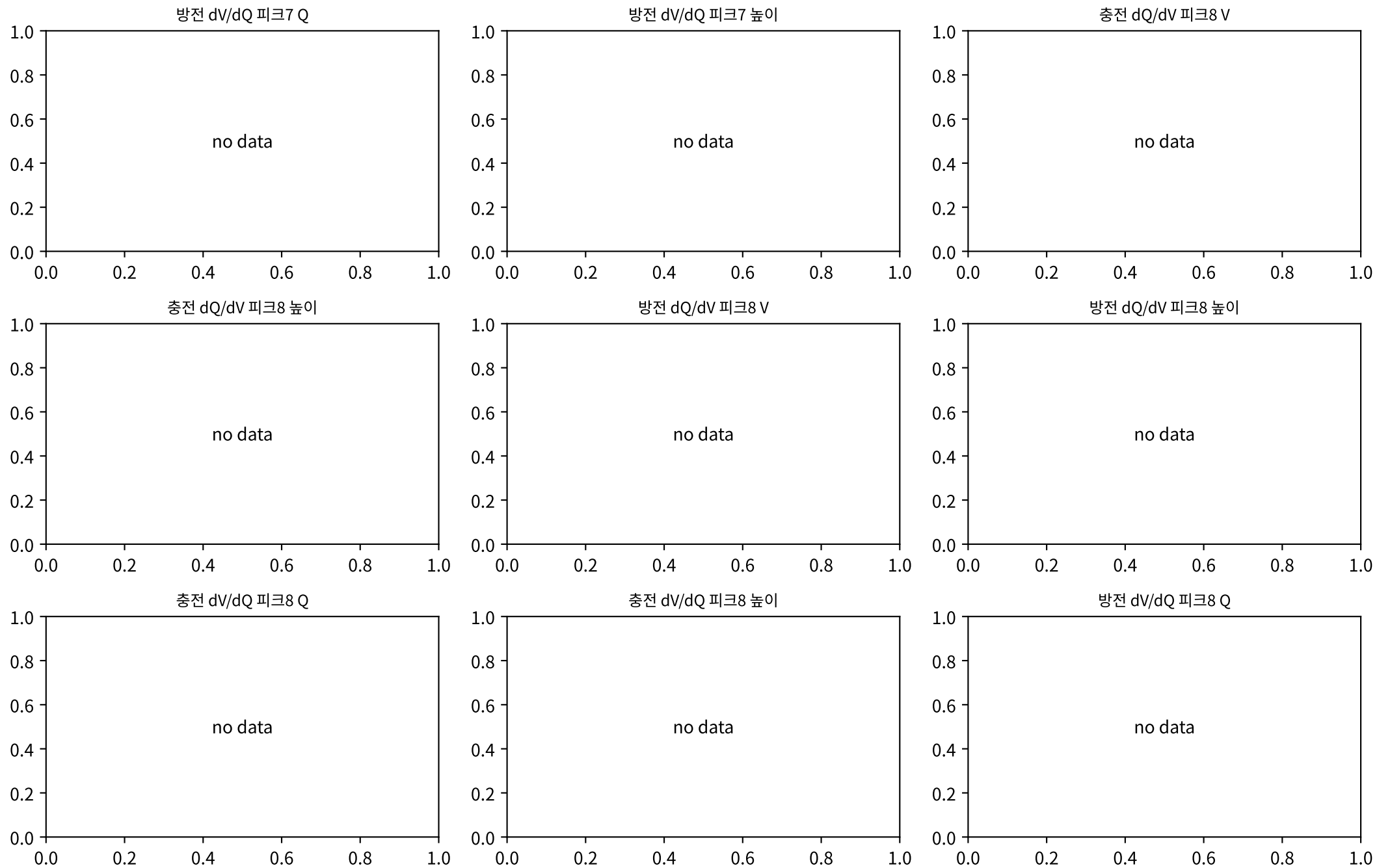
M02Ch111 · dQ/dV · dV/dQ 피크



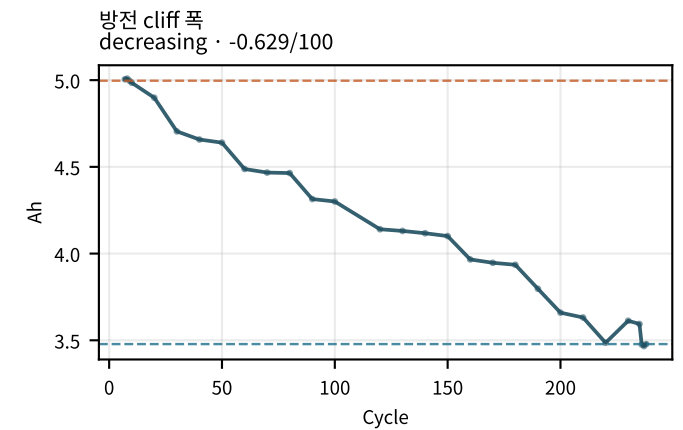
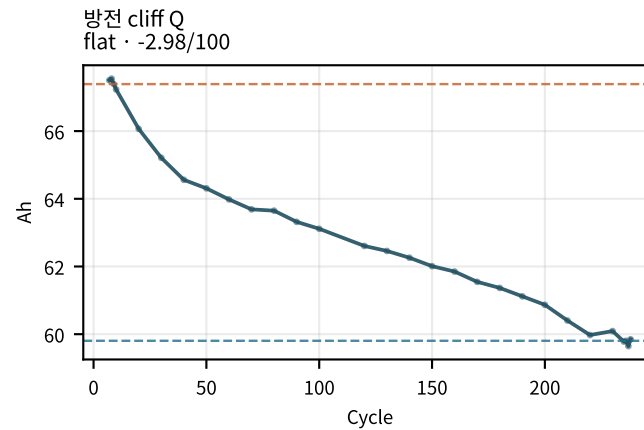
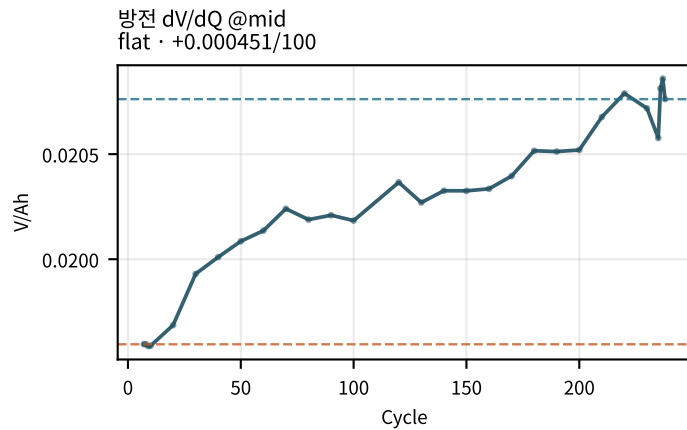
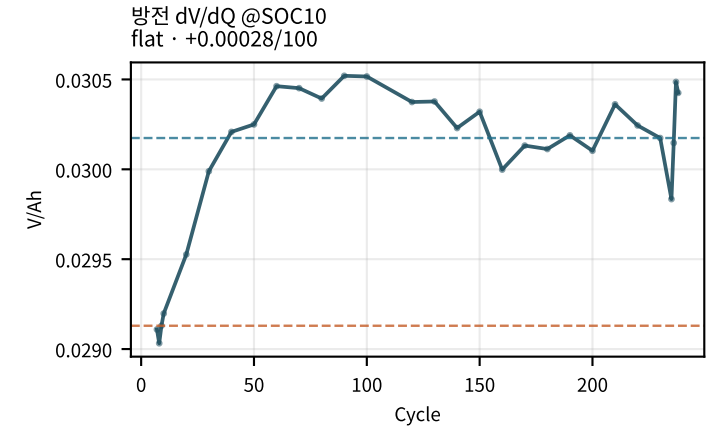
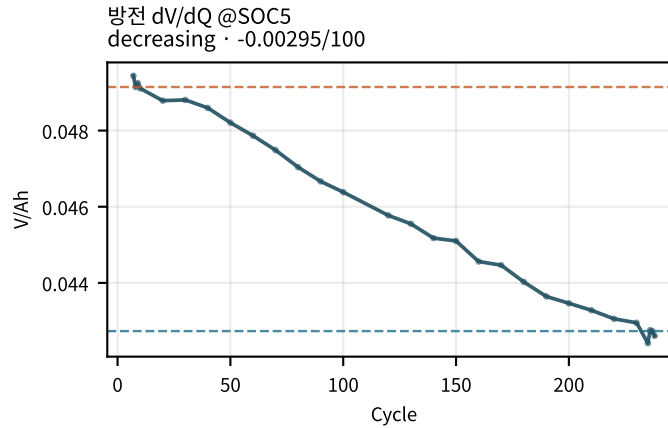
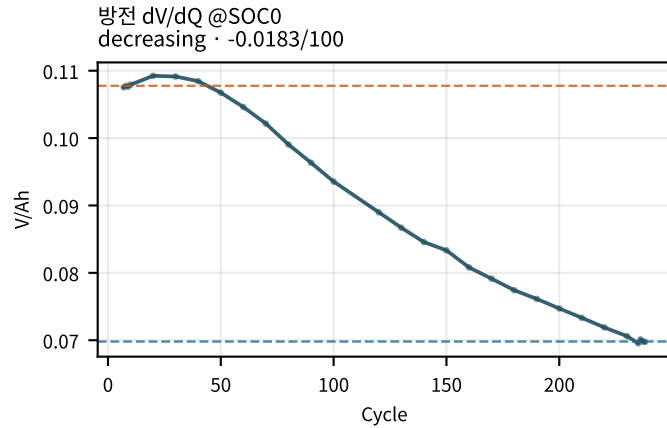
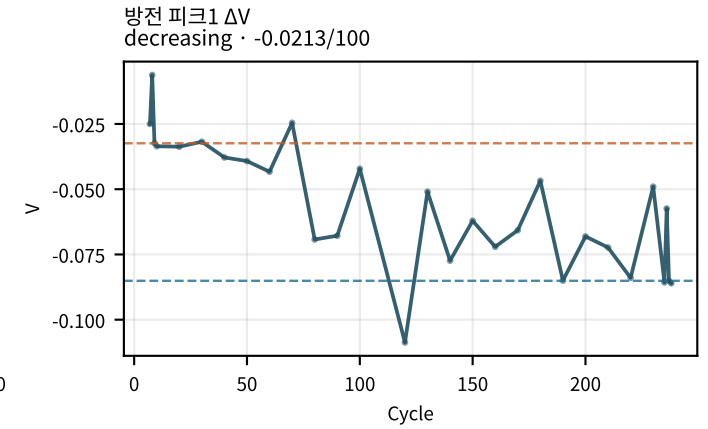
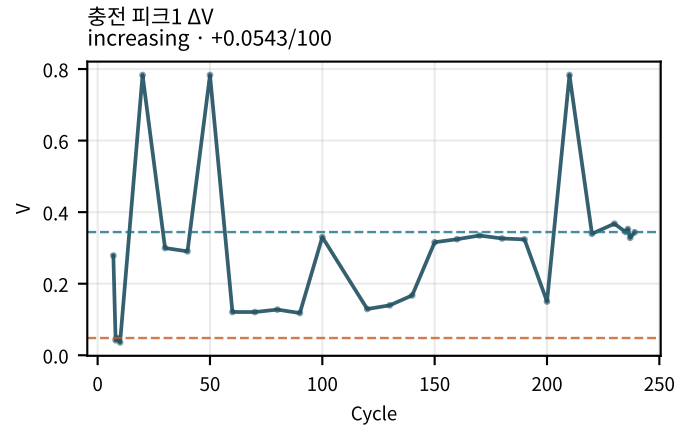
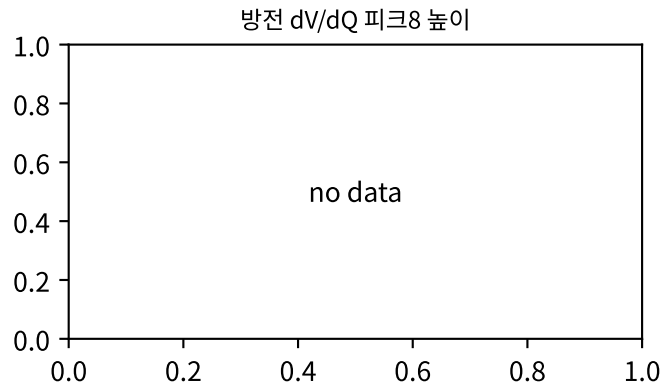
M02Ch111 · dQ/dV · dV/dQ 피크



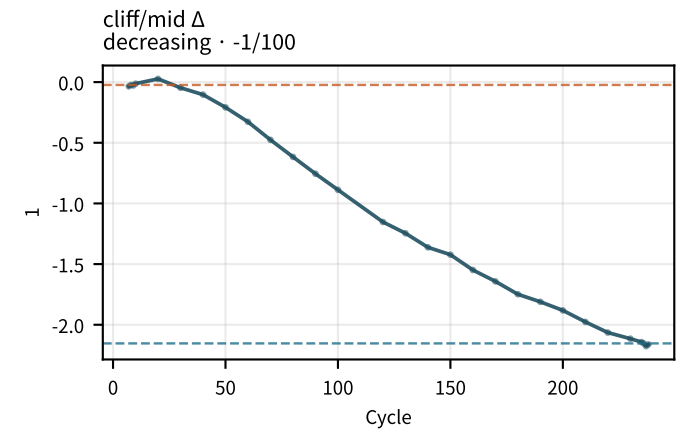
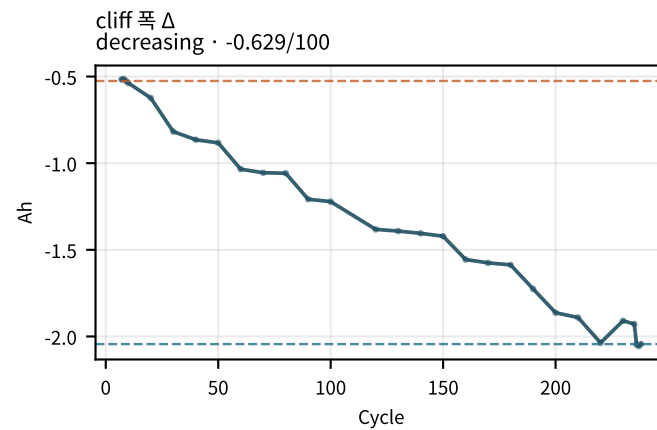
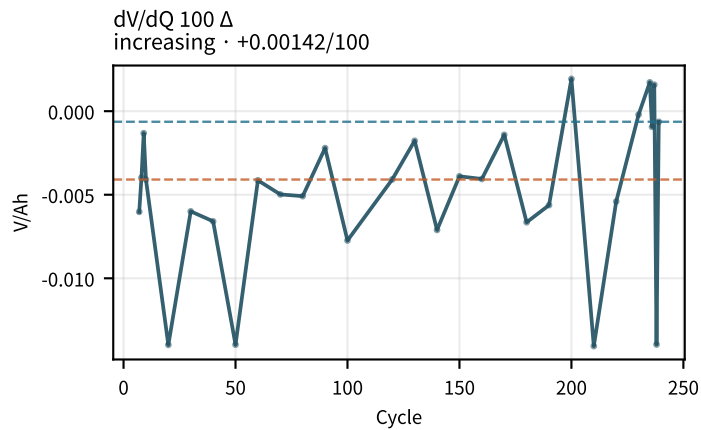
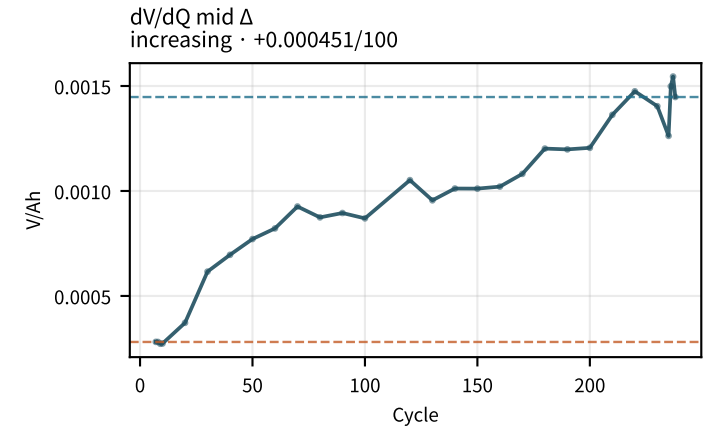
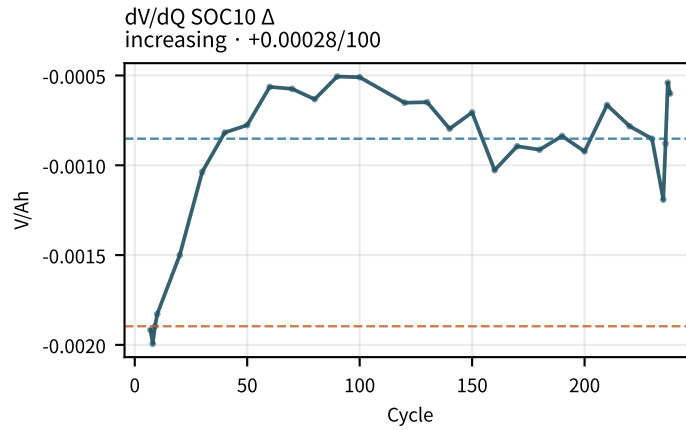
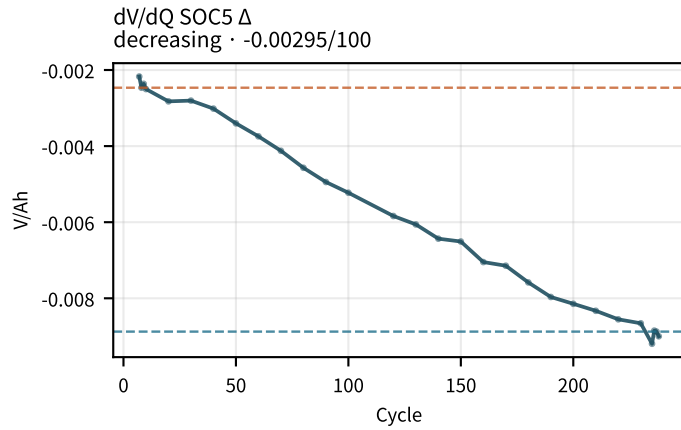
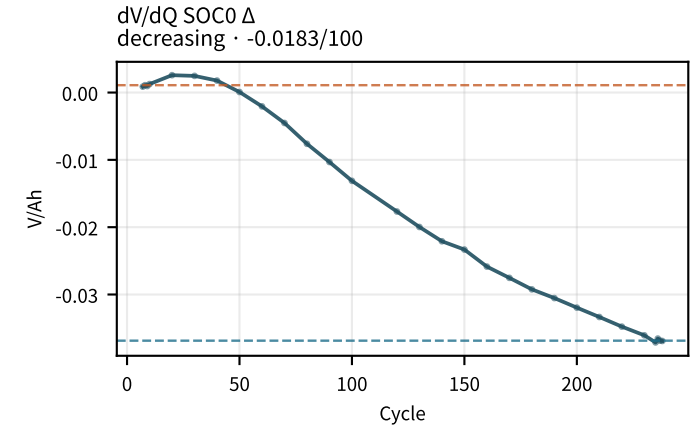
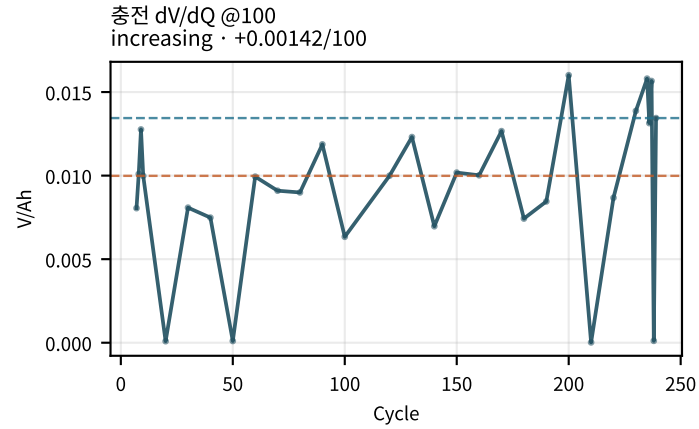
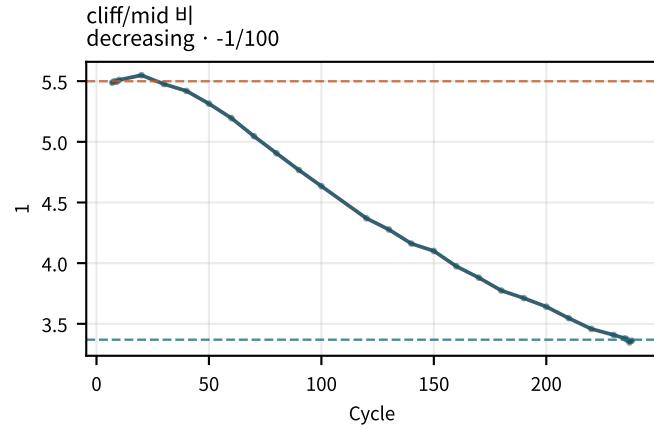
M02Ch111 · dQ/dV · dV/dQ 피크



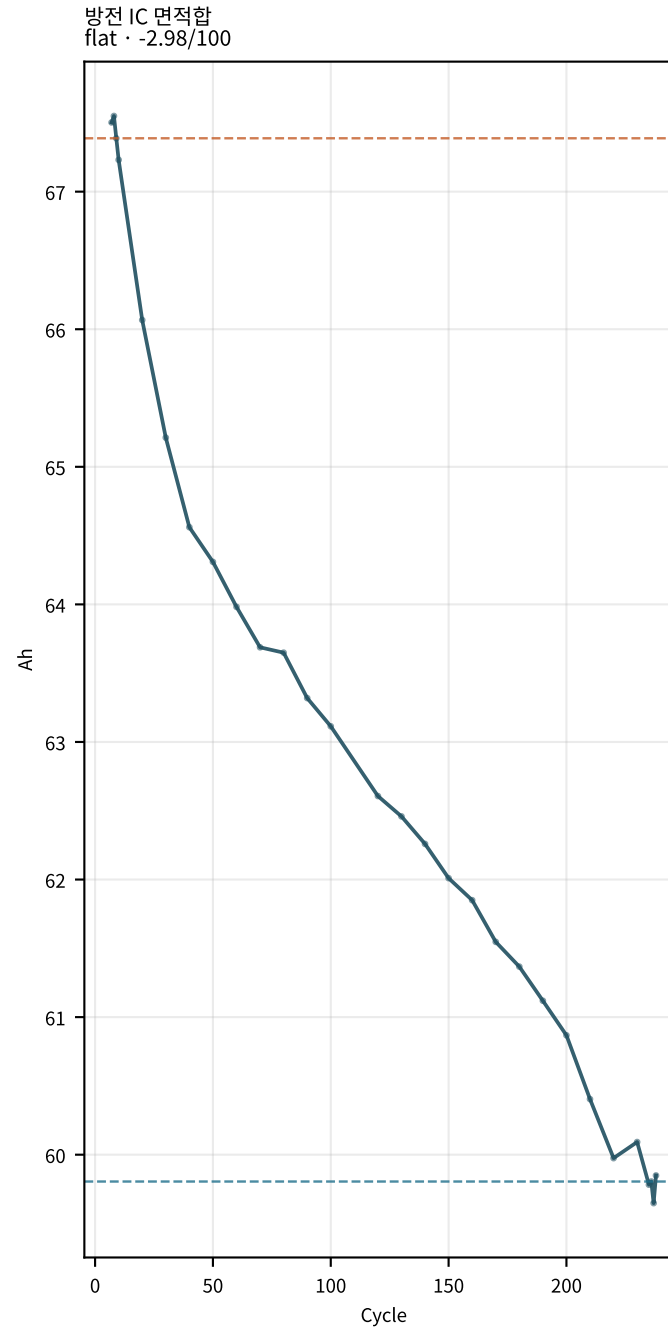
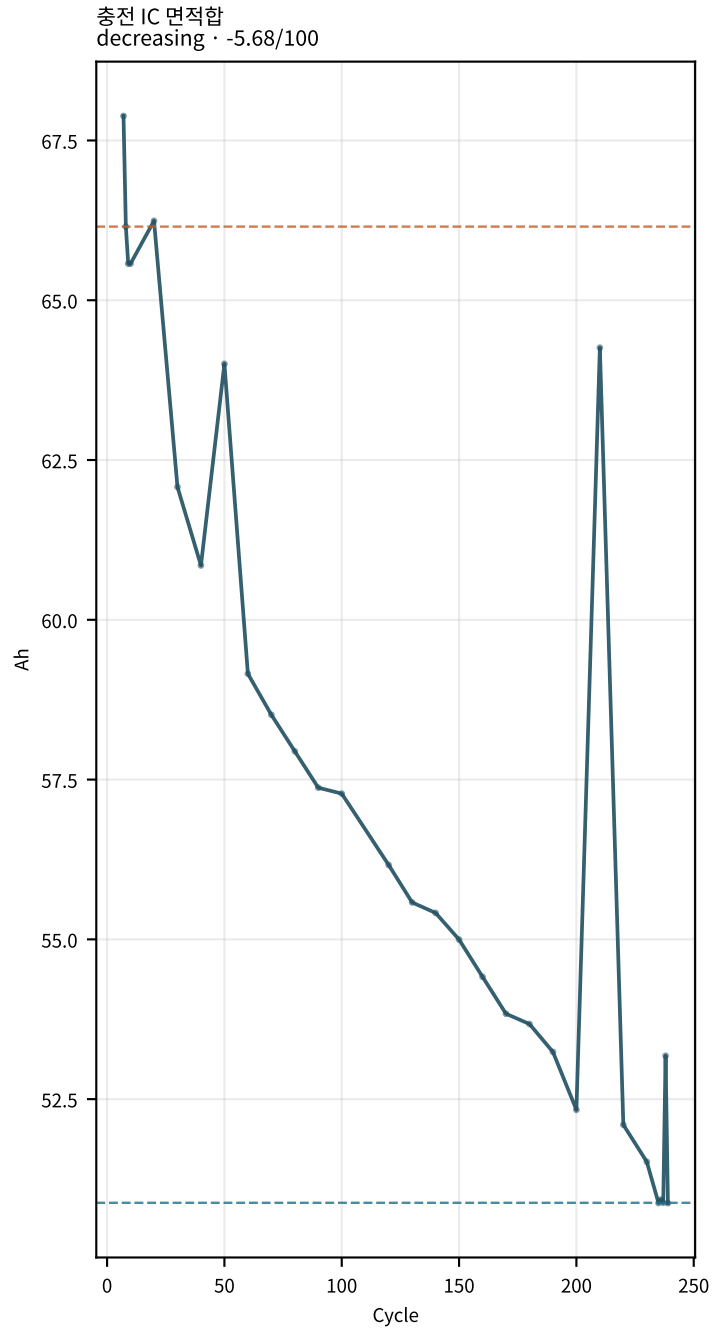
M02Ch111 · dQ/dV · dV/dQ 피크



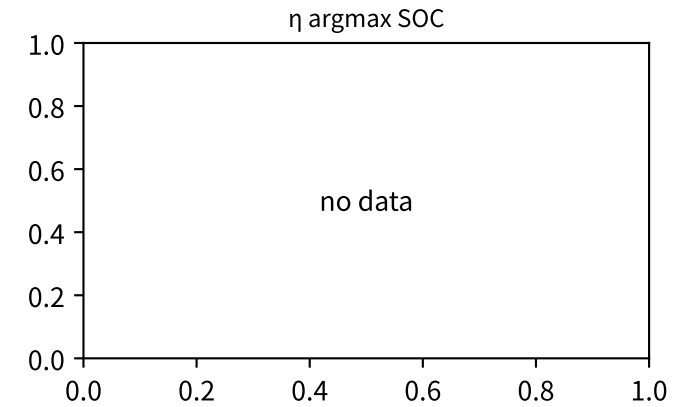
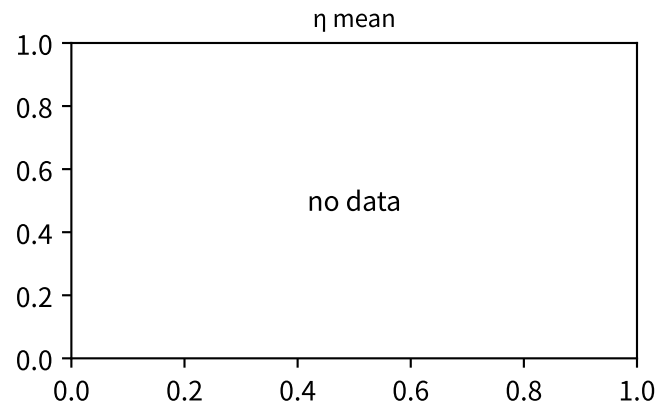
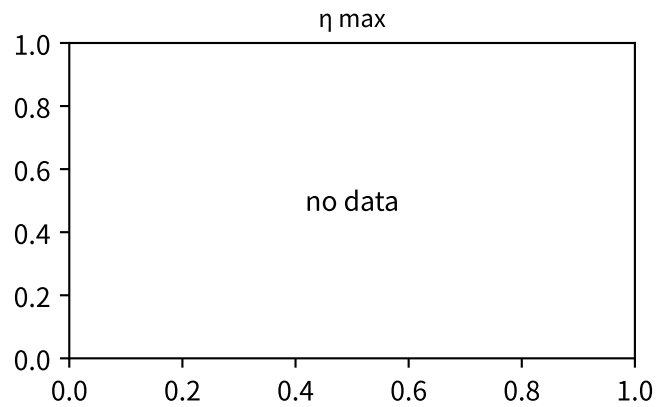
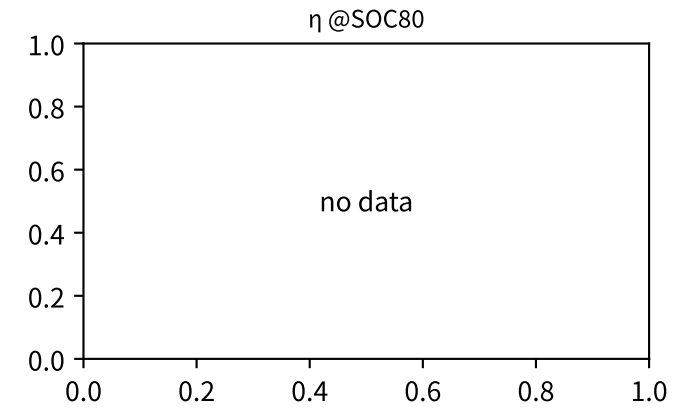
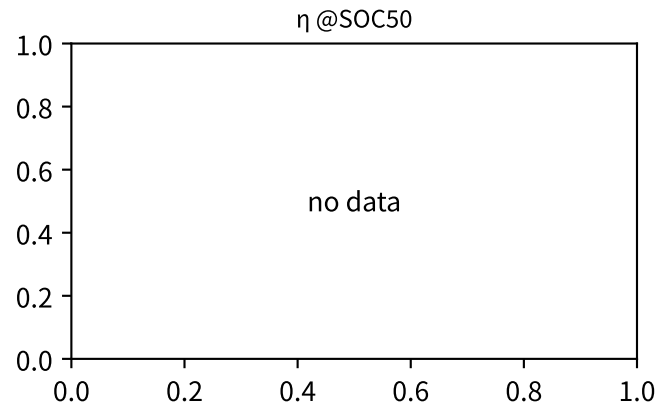
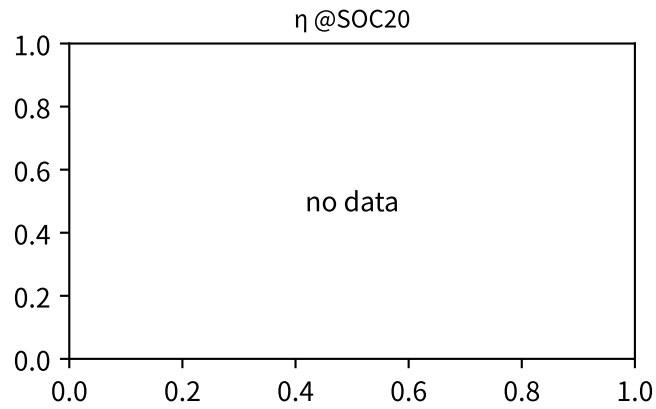
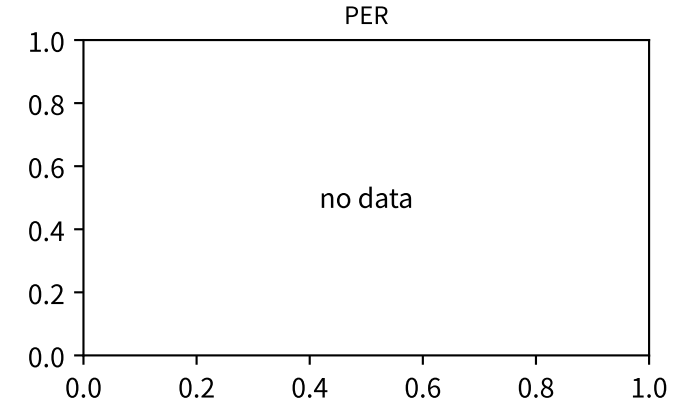
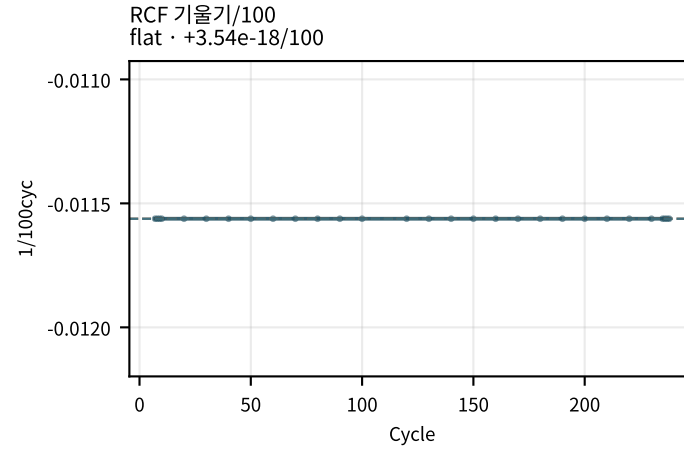
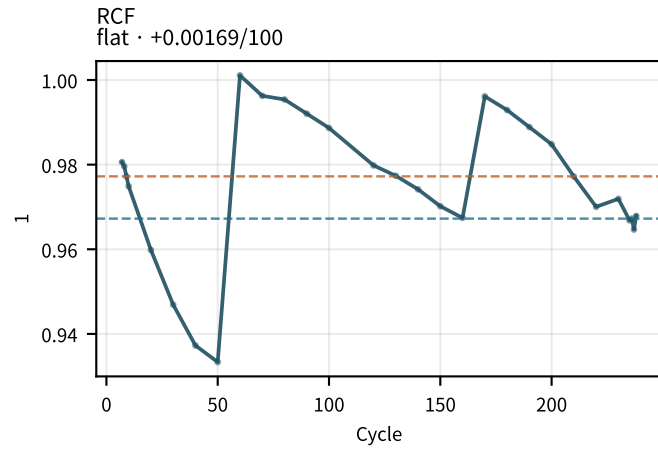
M02Ch111 · dQ/dV · dV/dQ 피크



M02Ch111 · dQ/dV · dV/dQ 피크

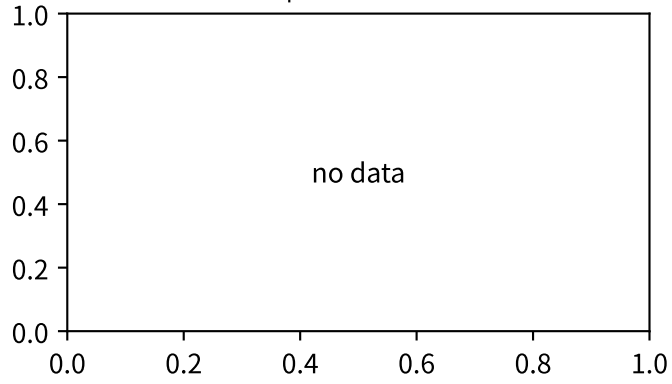


M02Ch111 · 수송 · rate · η

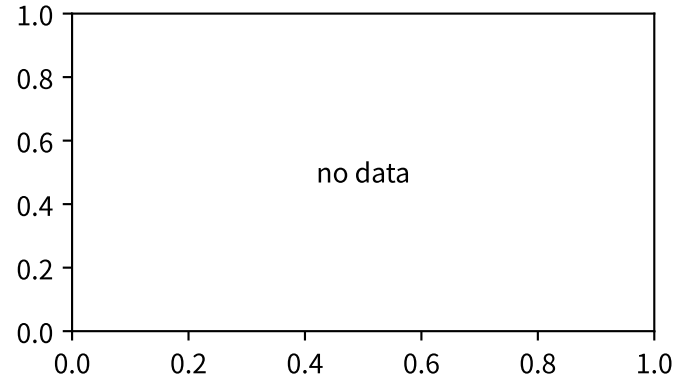


M02Ch111 · 수송 · rate · η

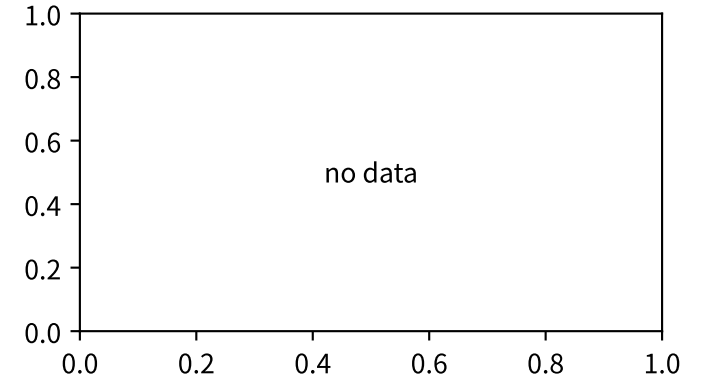
η 저SOC 기울기



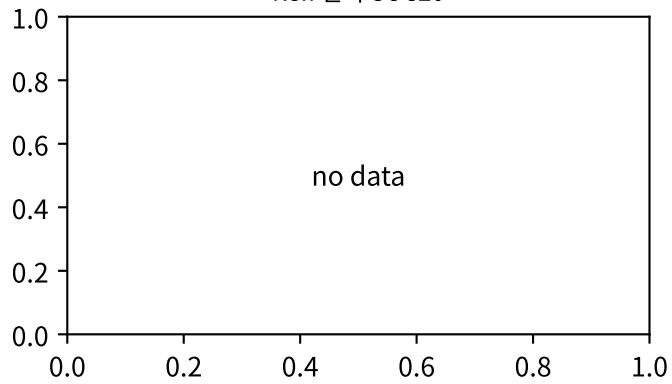
Reff scale



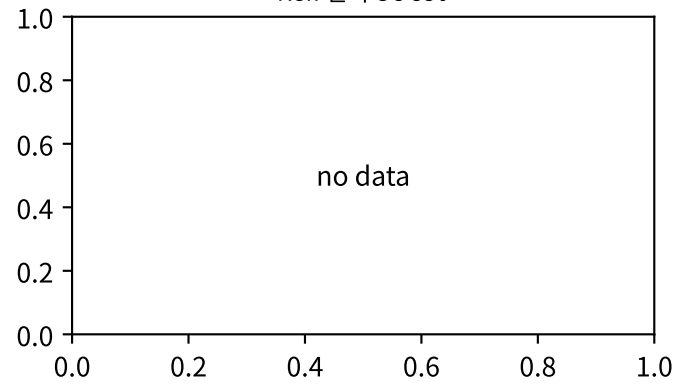
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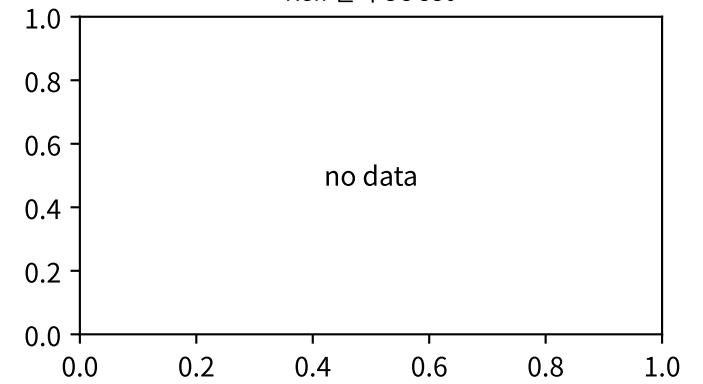
Reff 잔차 SOC20



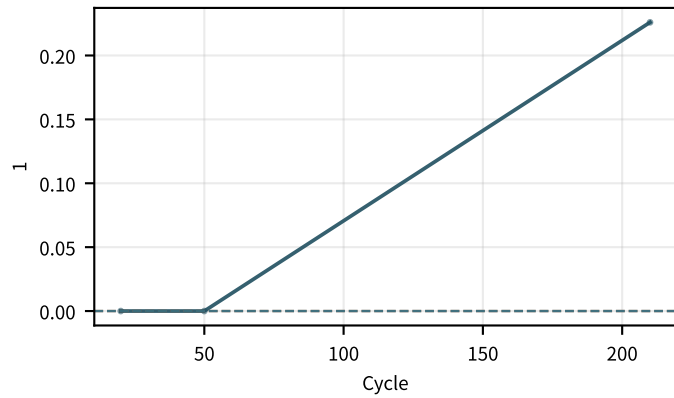
Reff 잔차 SOC50



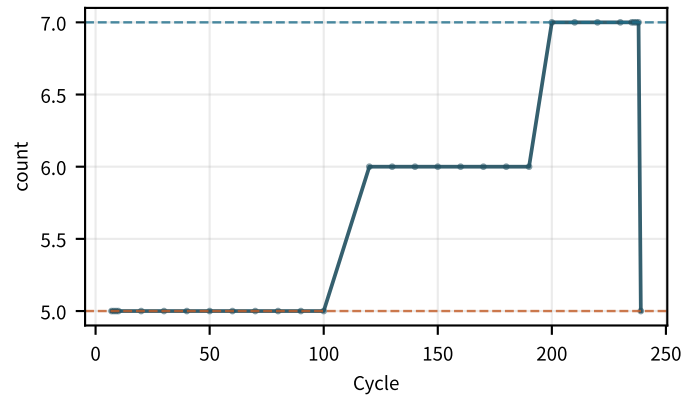
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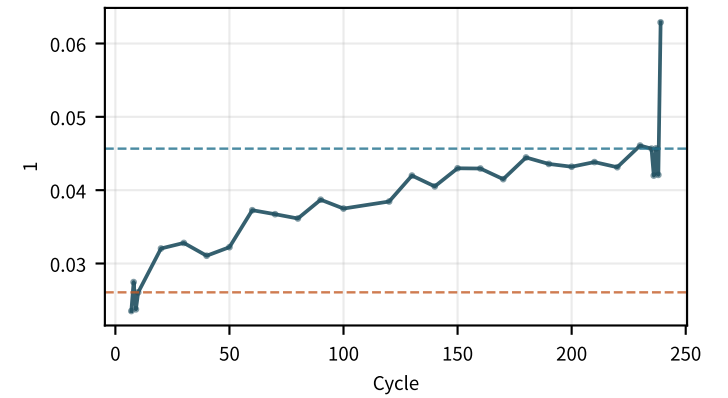
I_{∞} 정규화
increasing · +0.126/100



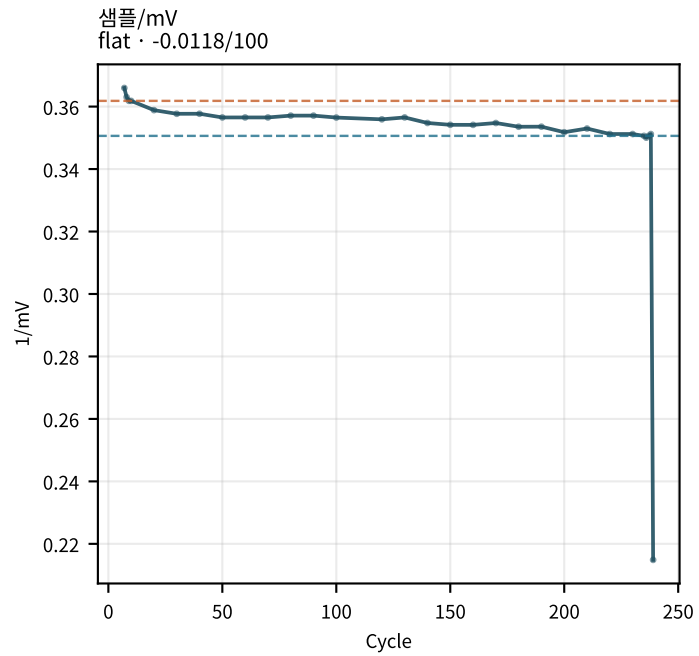
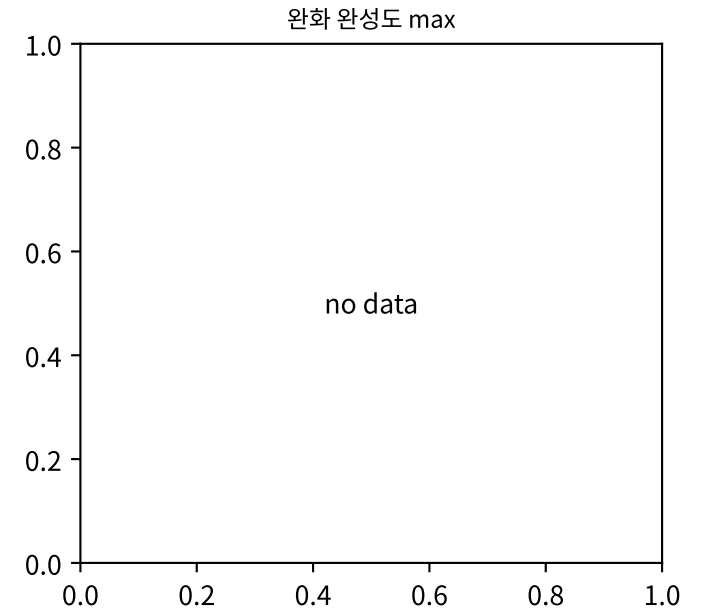
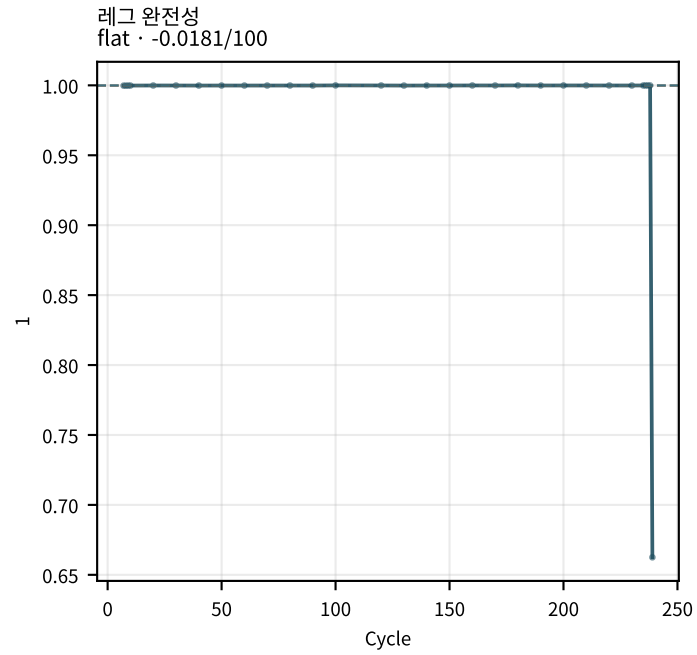
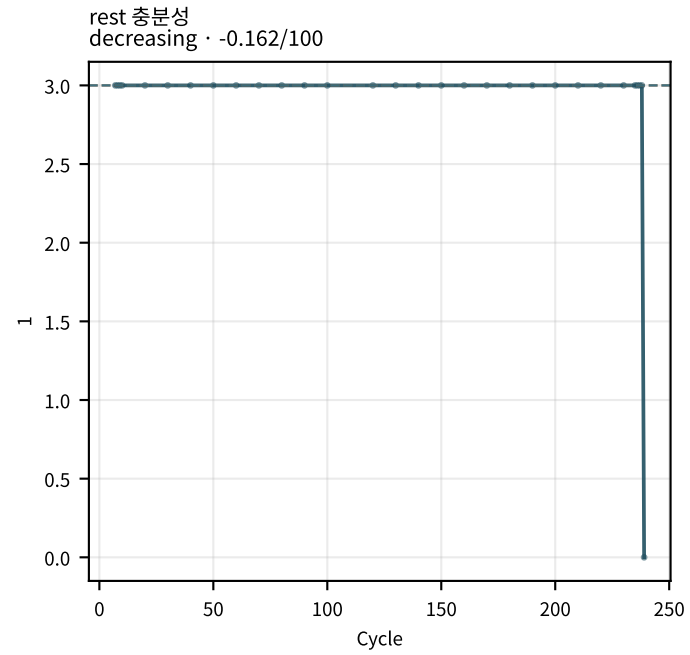
펄스 1s 샘플수
increasing · +0.861/100



펄스 전류 안정도
increasing · +0.00834/100

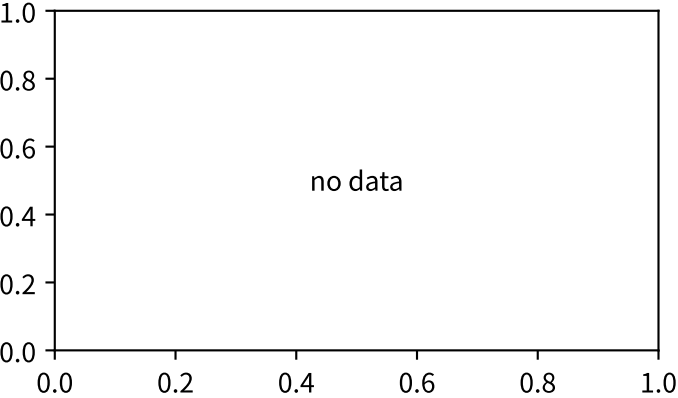


M02Ch111 · 수송 · rate · η

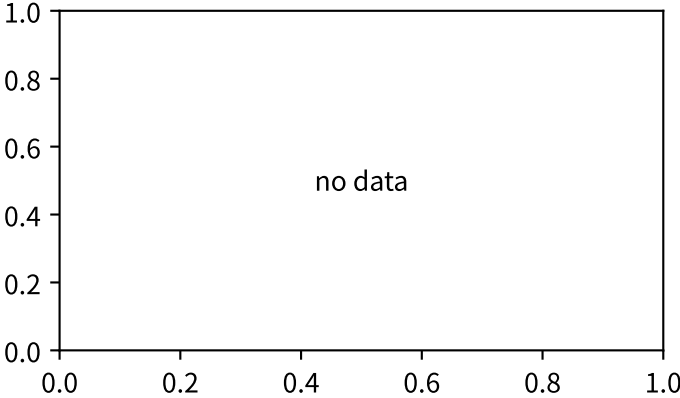


M02Ch111 · OCV · 자기방전

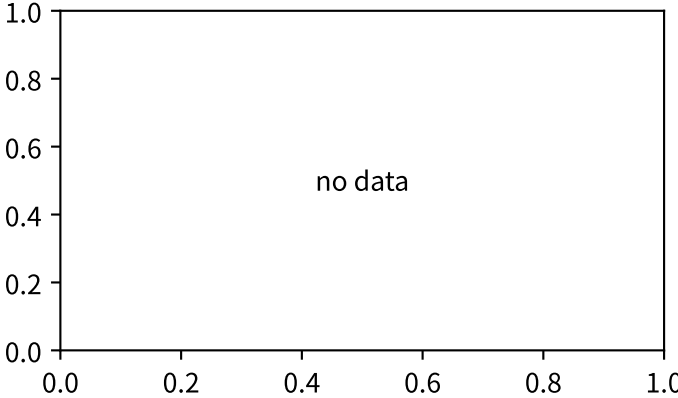
OCV $V \infty$ SOC80



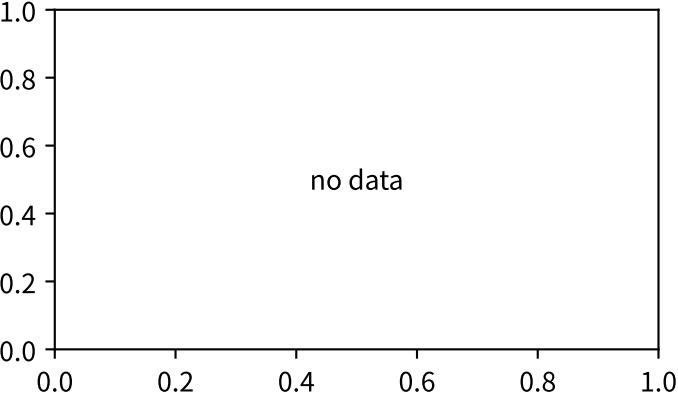
OCV $V \infty$ SOC50



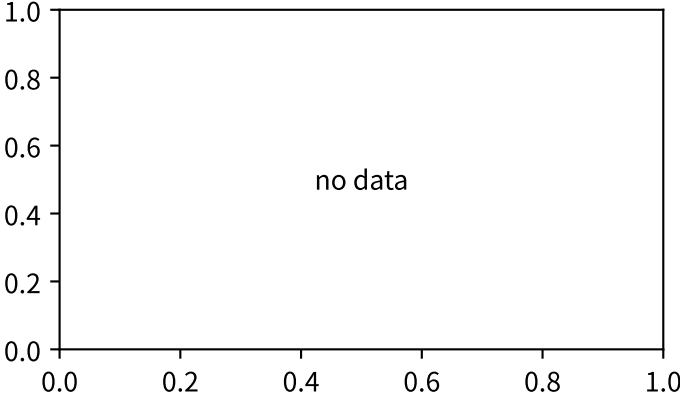
OCV $V \infty$ SOC20



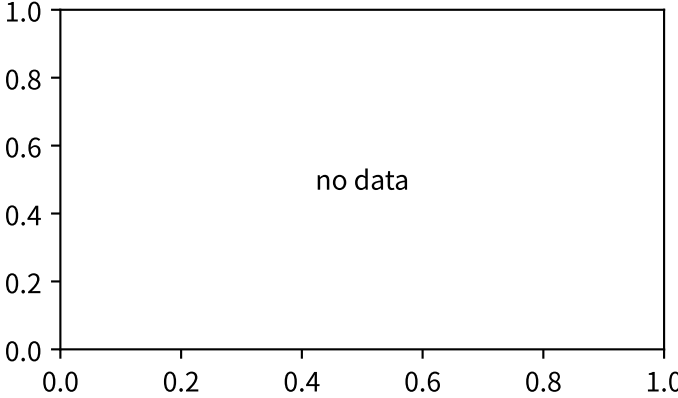
OCV spread 20–80



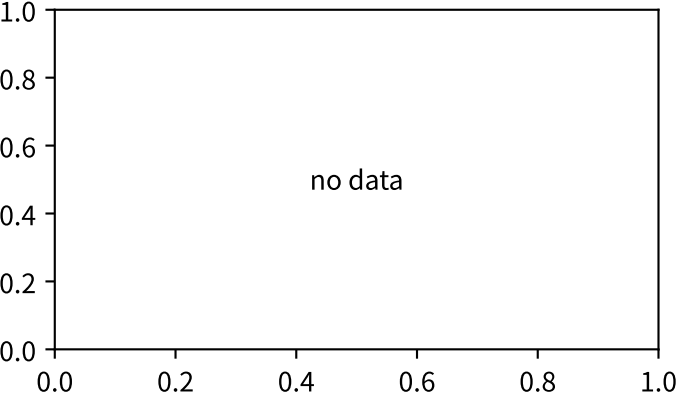
OCV spread 50–80



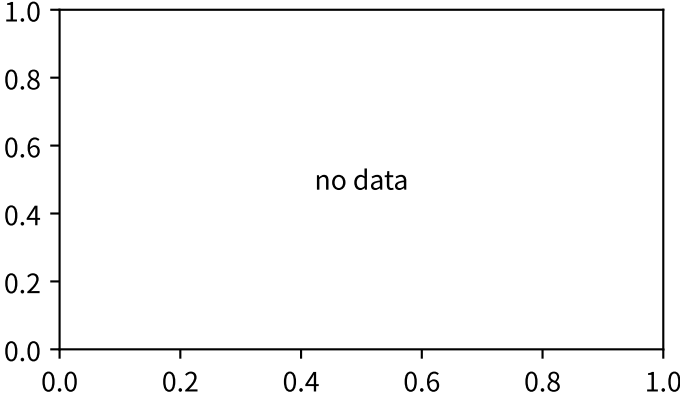
OCV spread 20–50



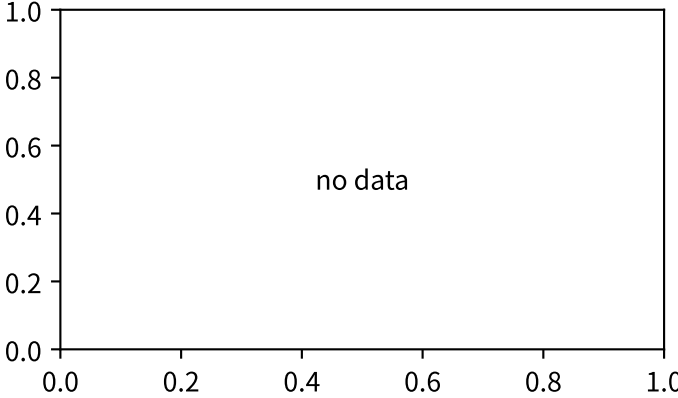
OCV 평행이동



OCV 폭 압축

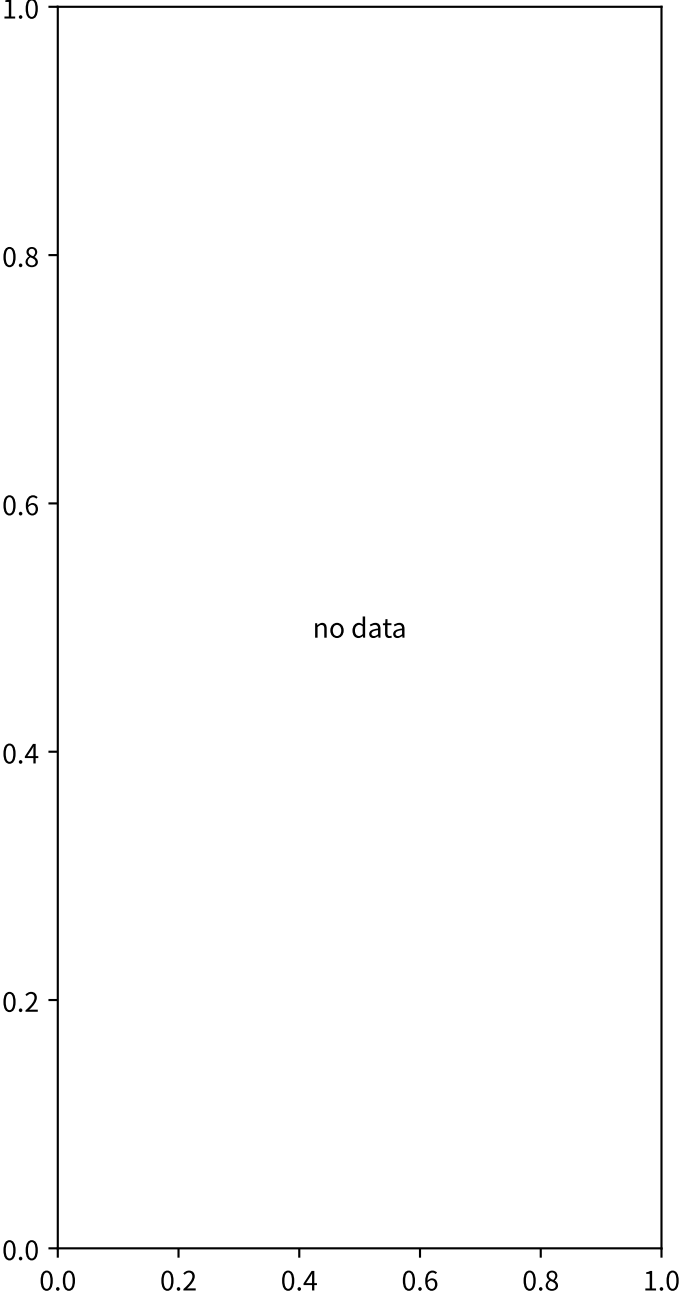


OCV80 Δ

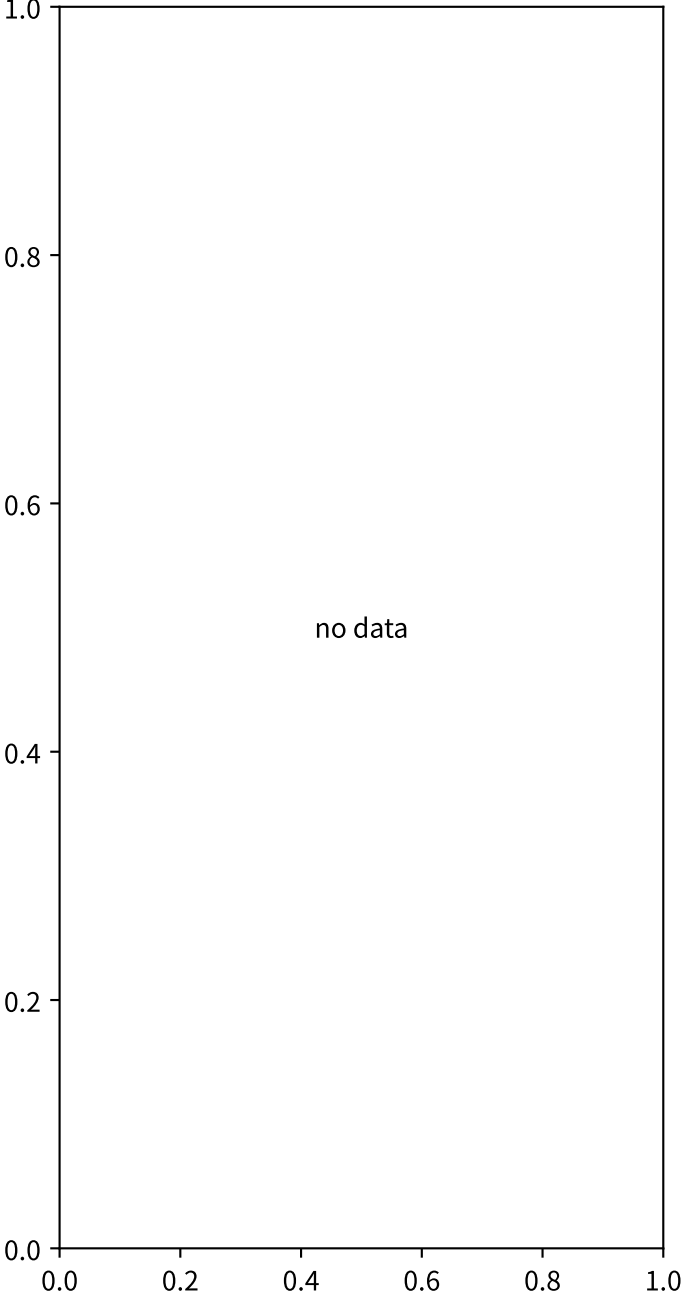


M02Ch111 · OCV · 자기방전

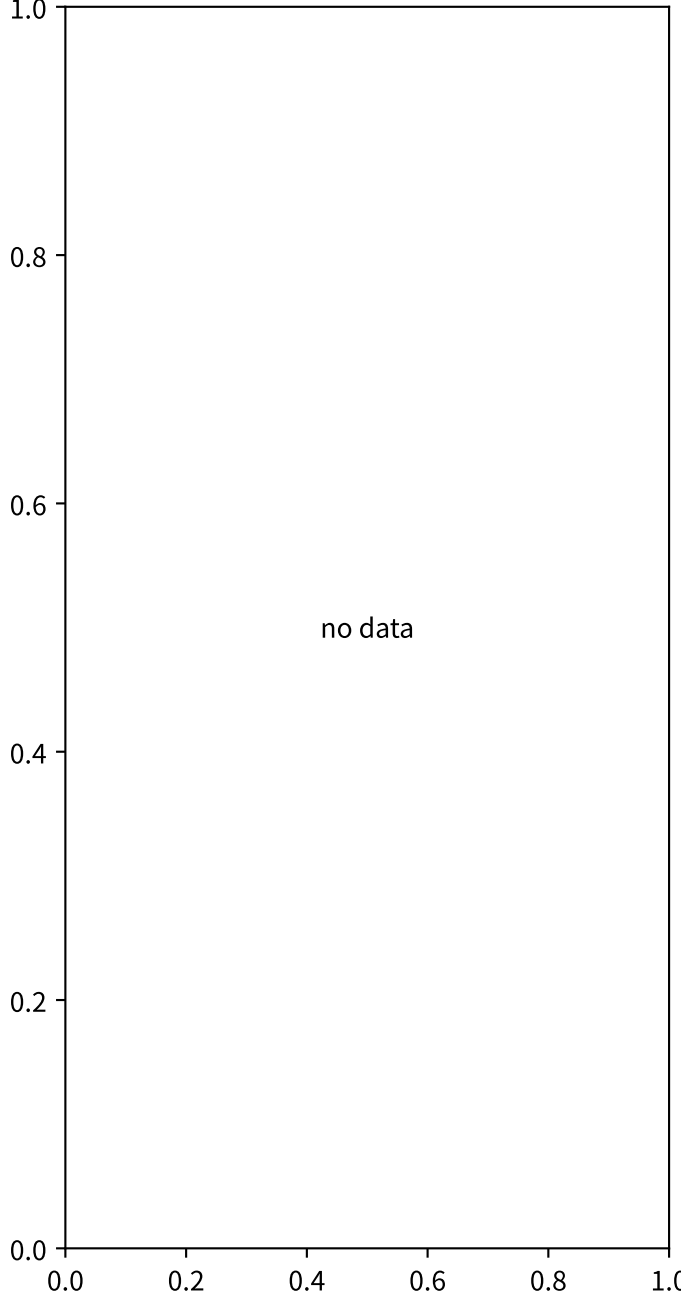
OCV50 Δ



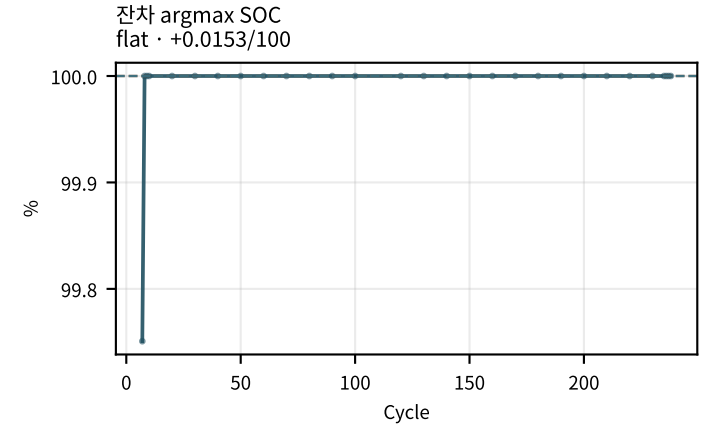
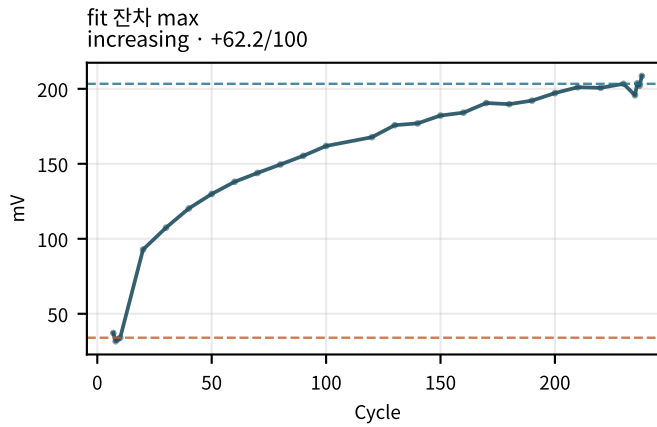
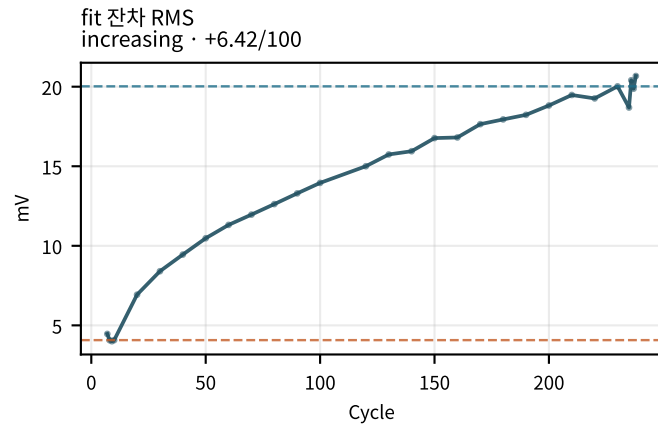
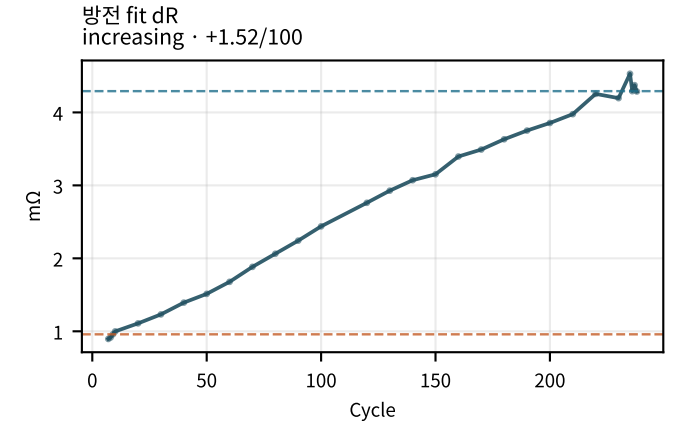
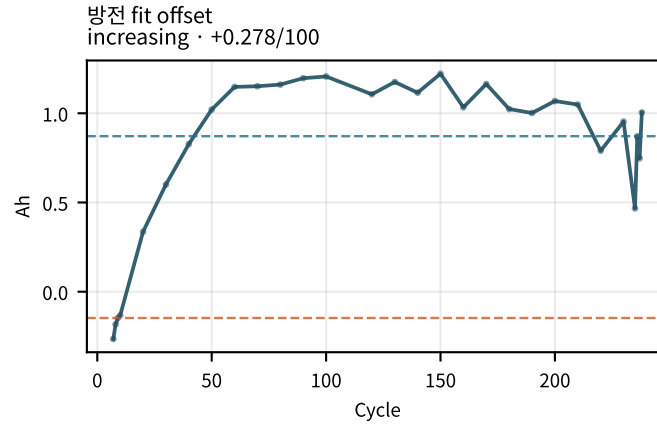
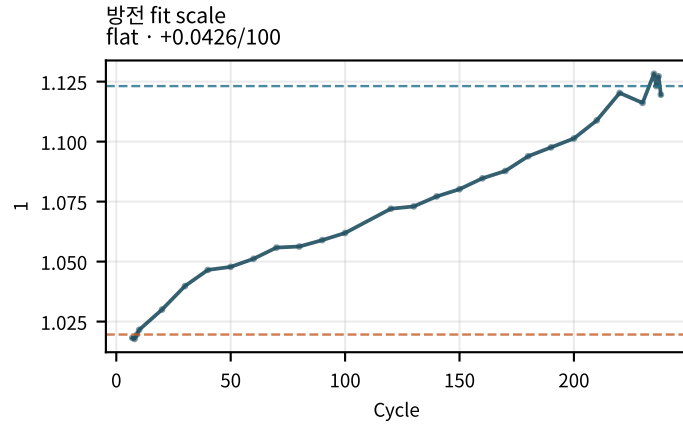
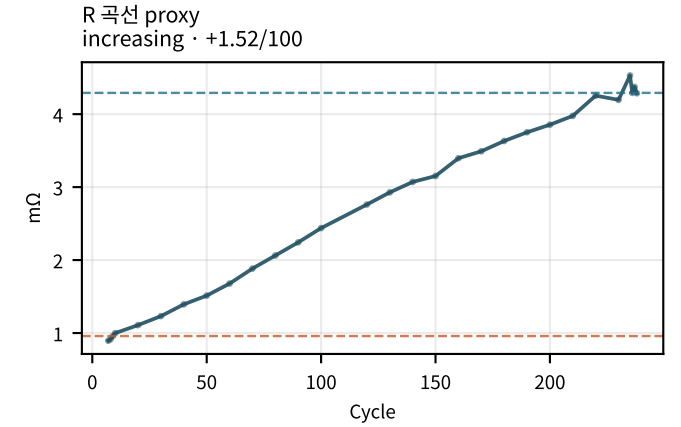
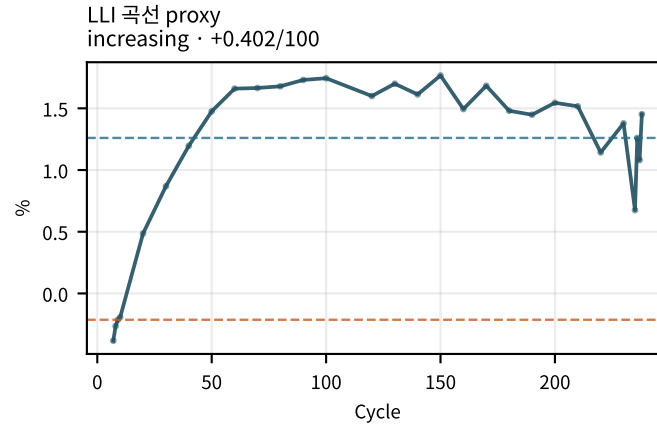
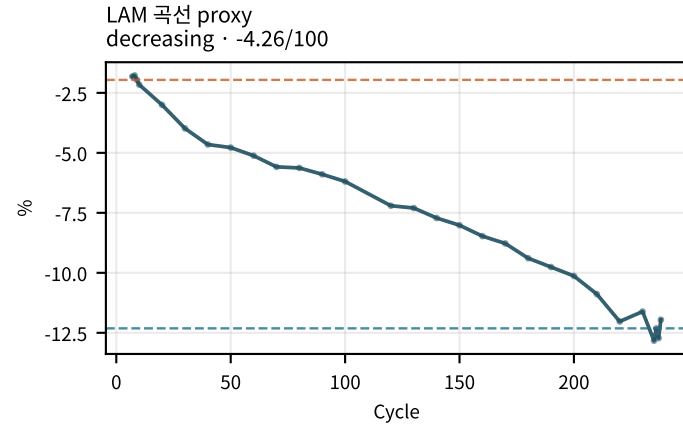
OCV20 Δ



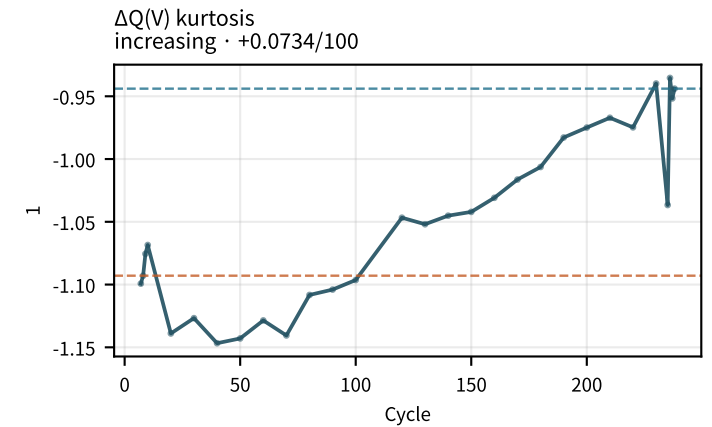
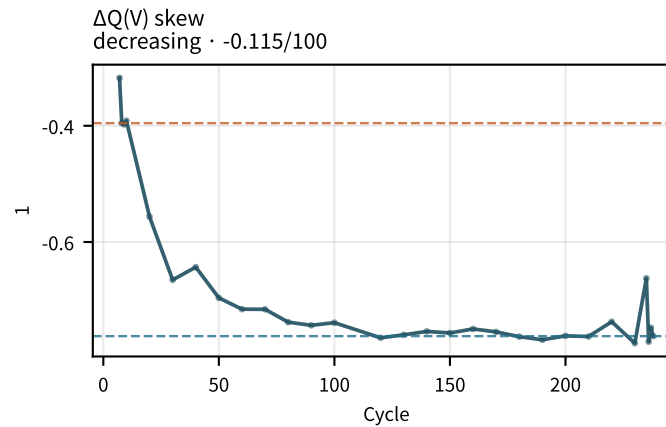
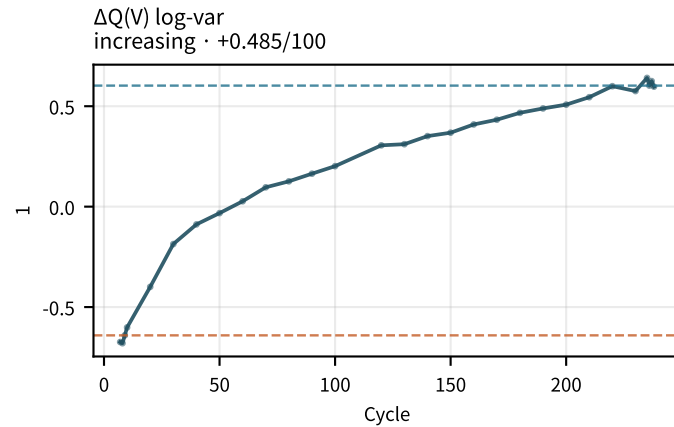
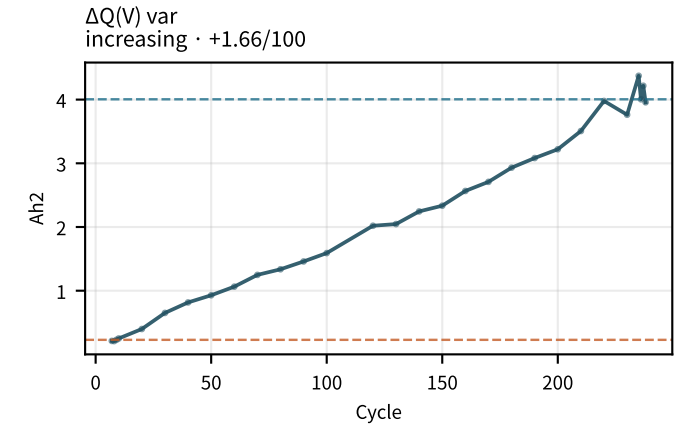
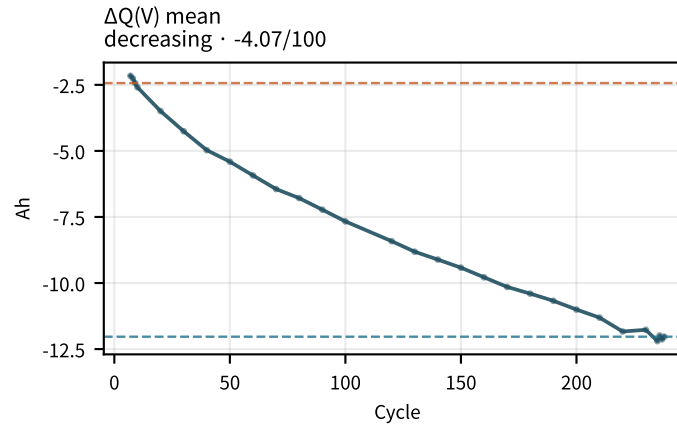
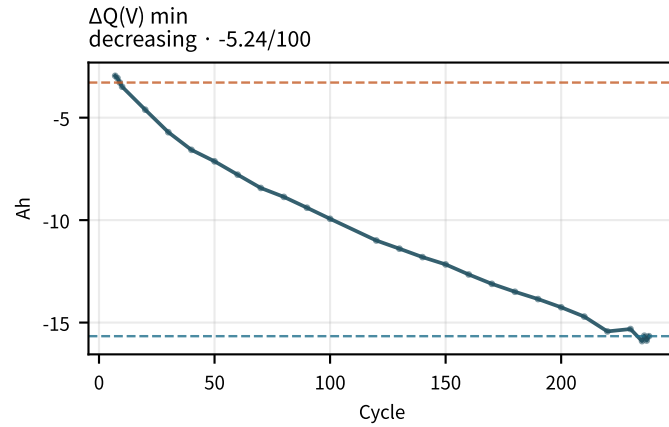
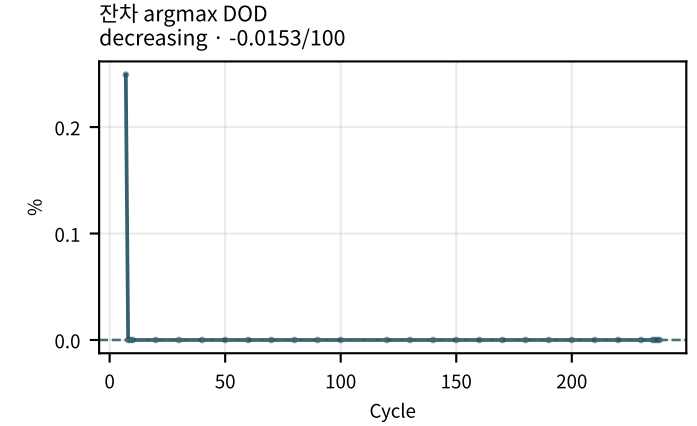
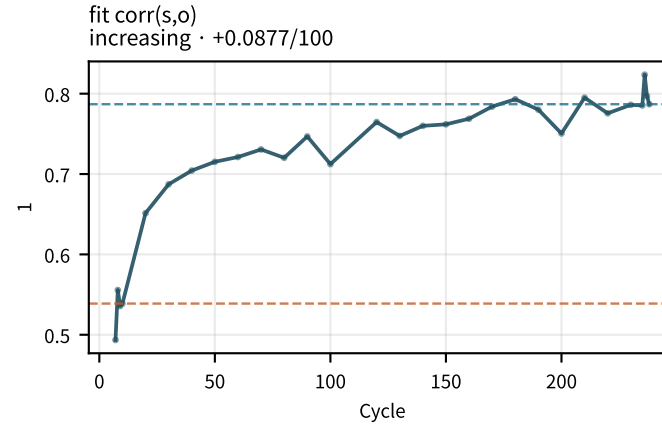
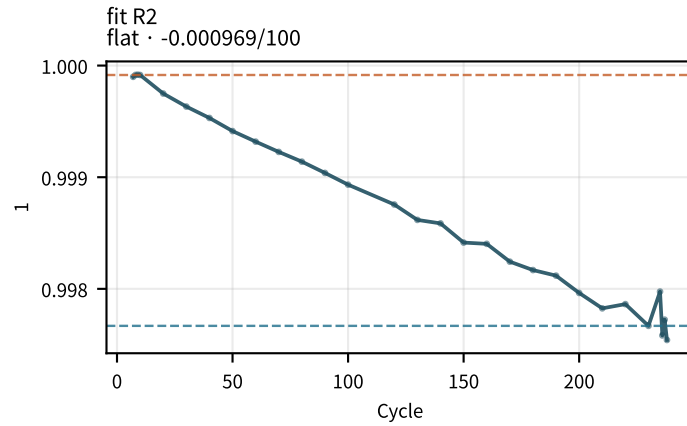
OCV spread Δ



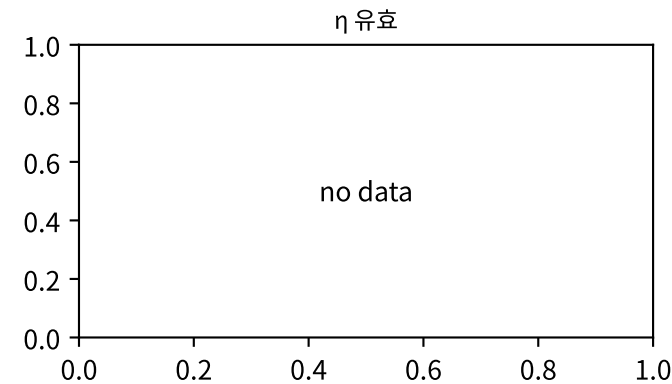
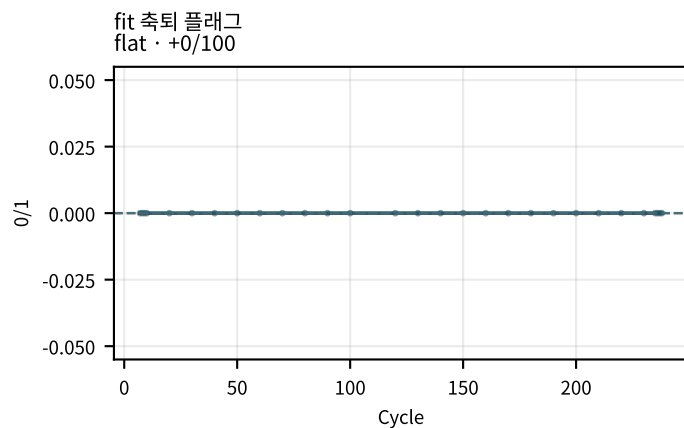
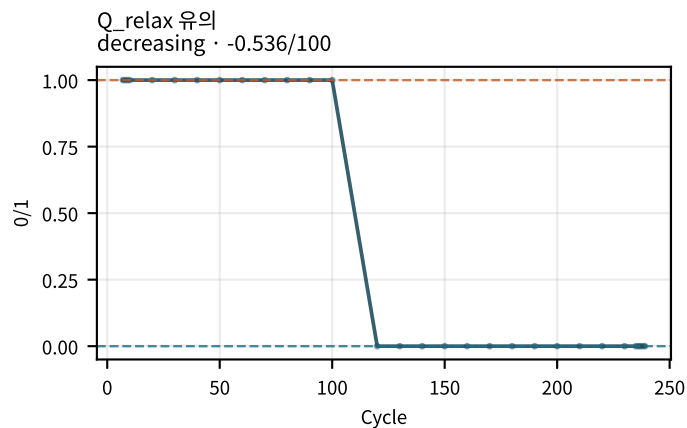
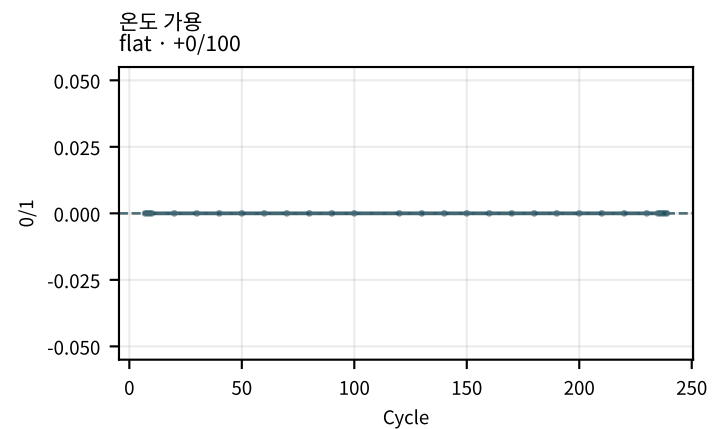
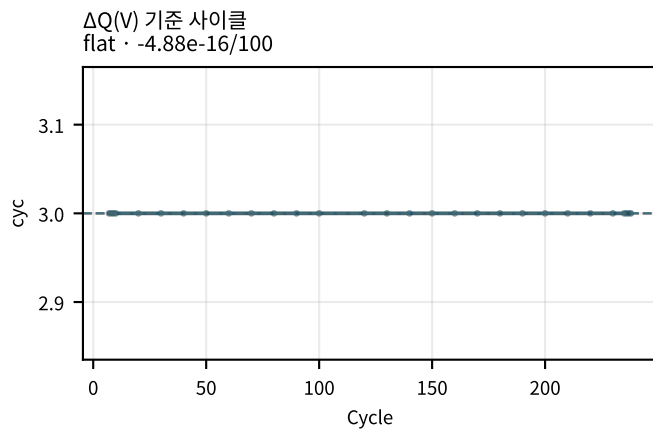
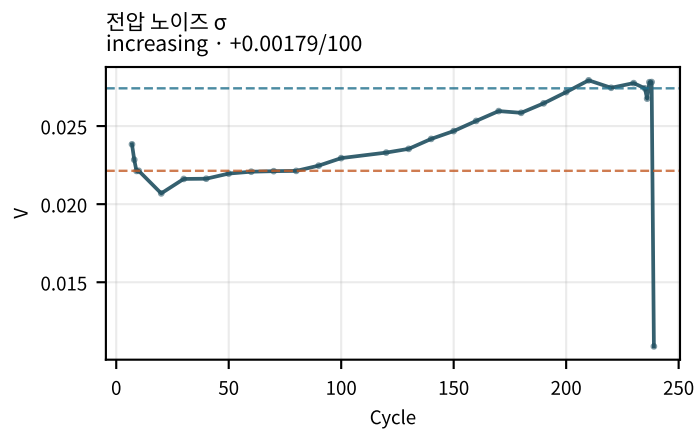
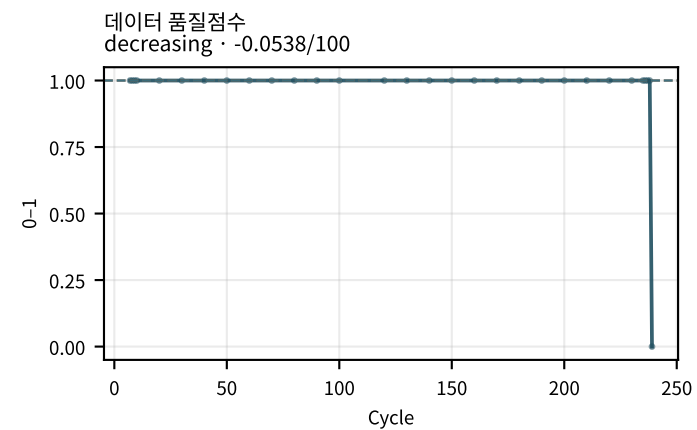
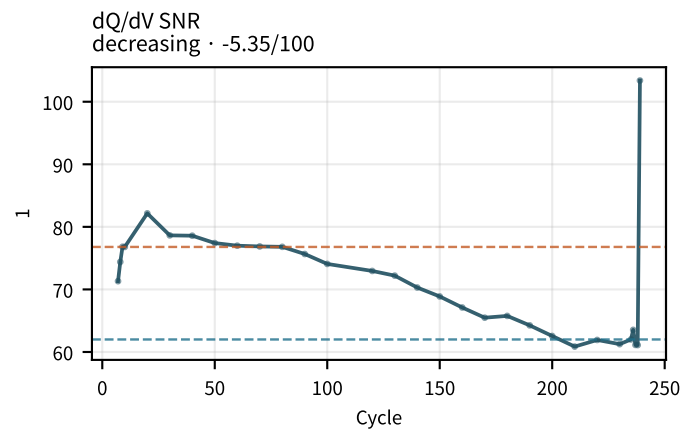
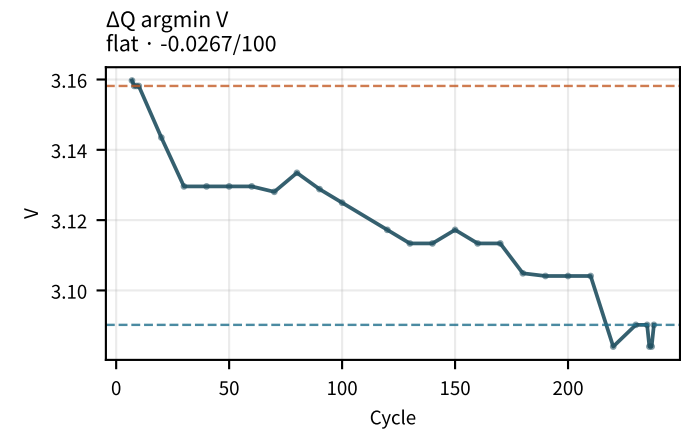
M02Ch111 · 곡선 fit · $\Delta Q(V)$



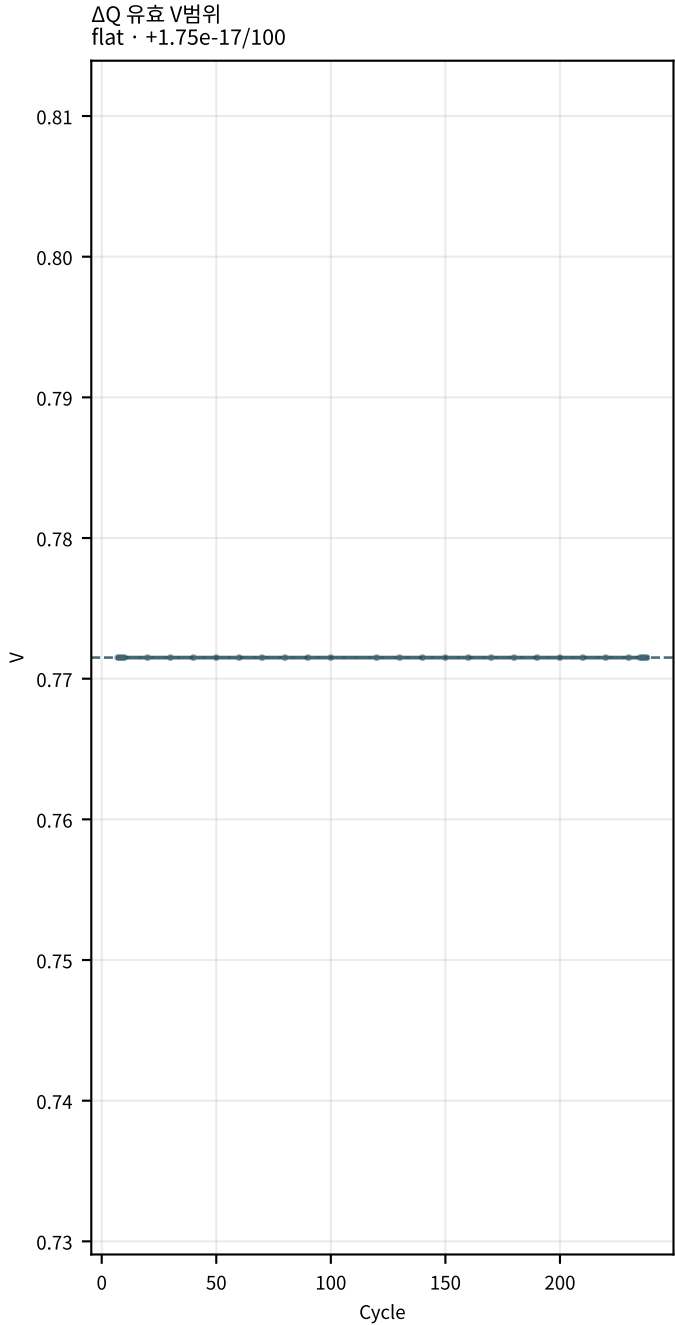
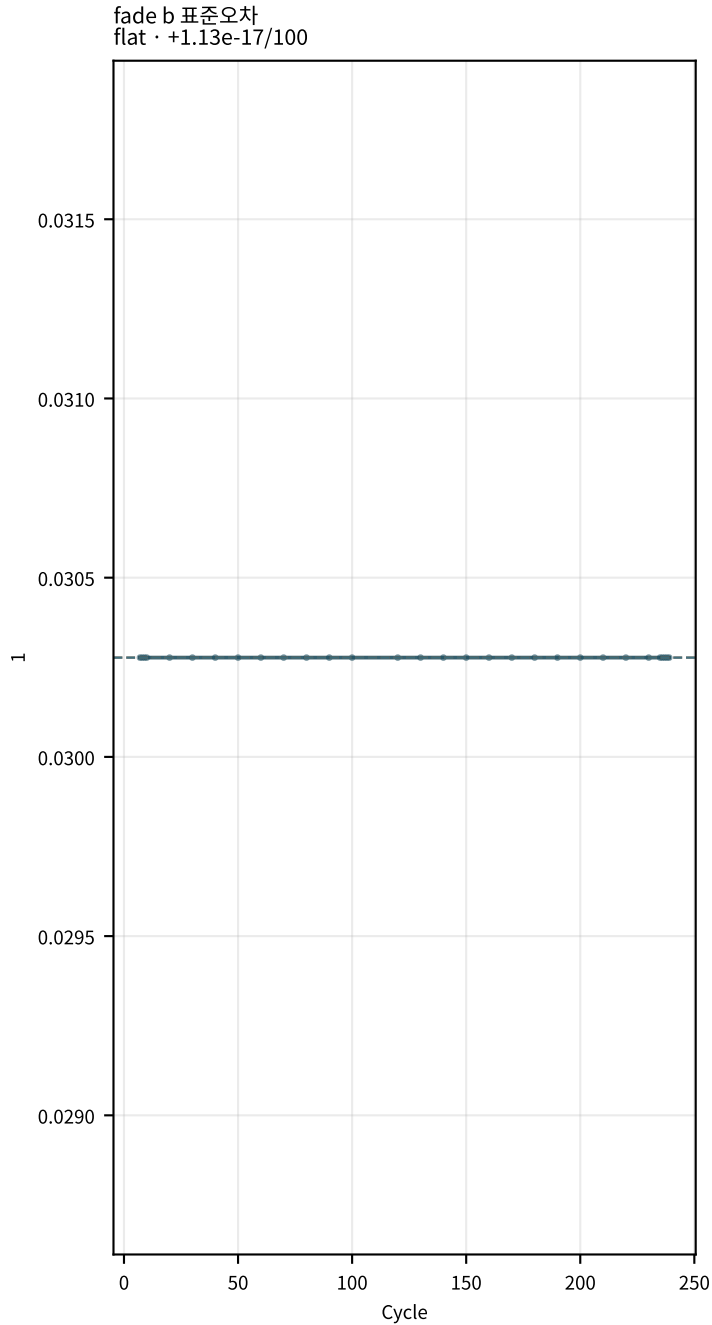
M02Ch111 · 곡선 fit · $\Delta Q(V)$



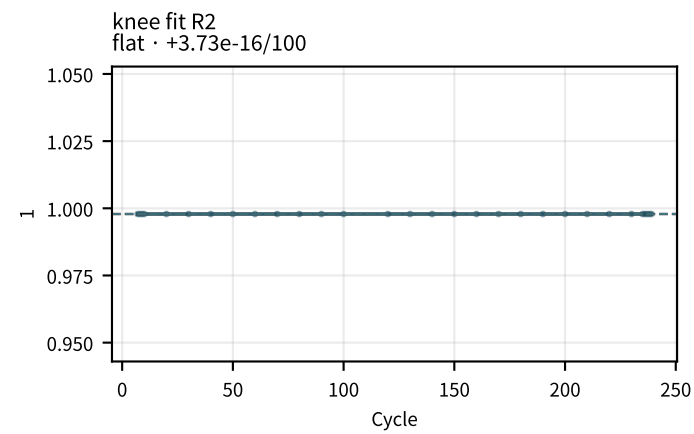
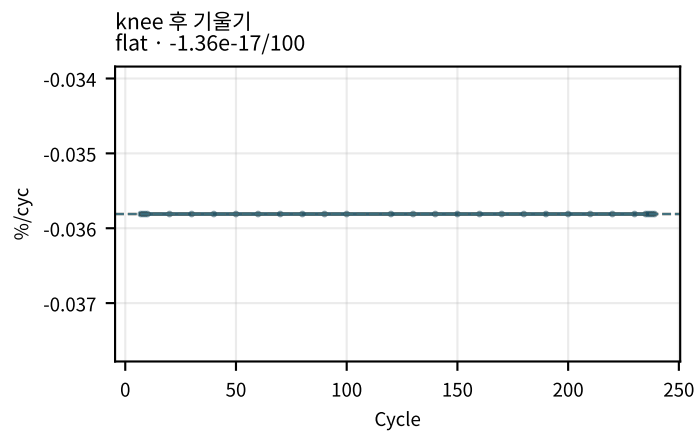
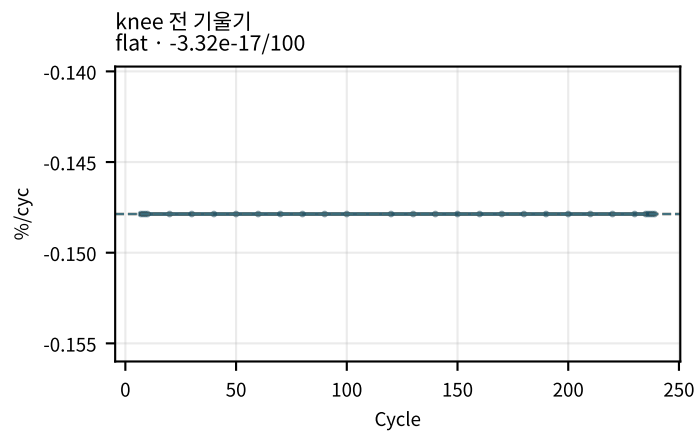
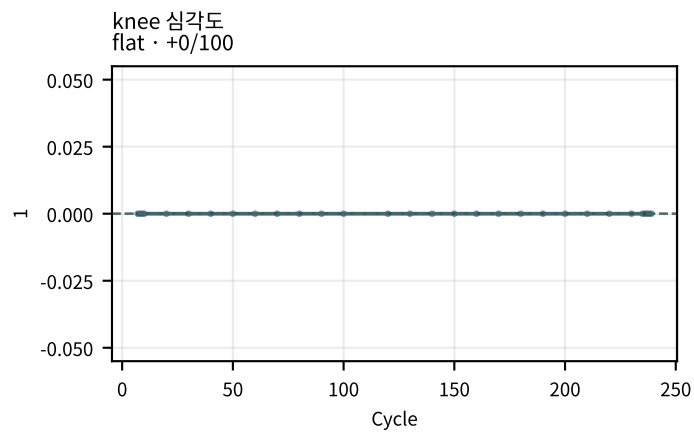
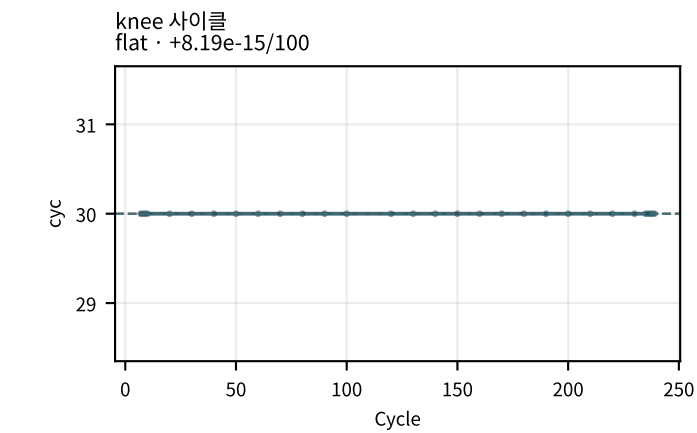
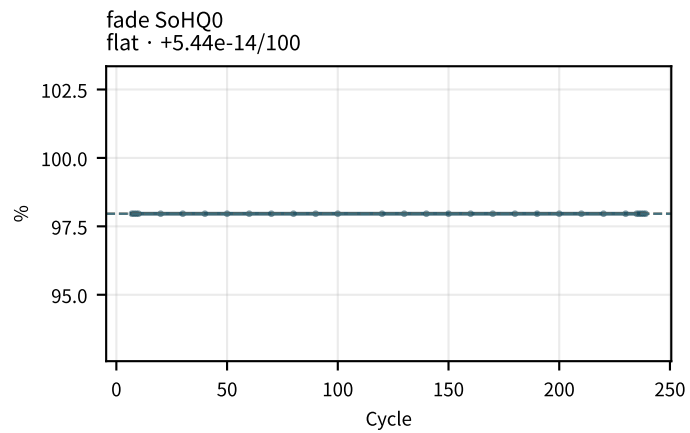
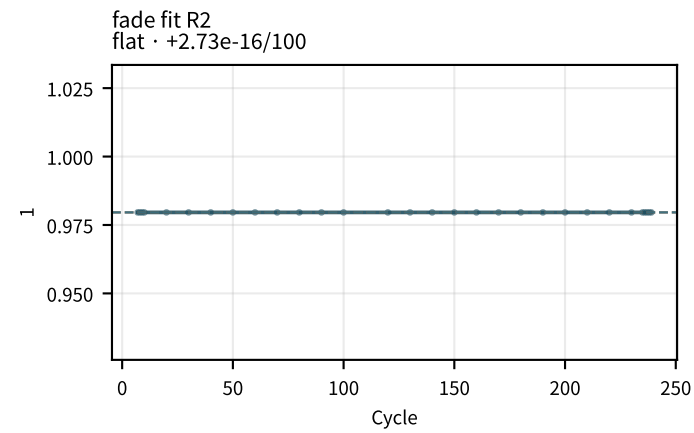
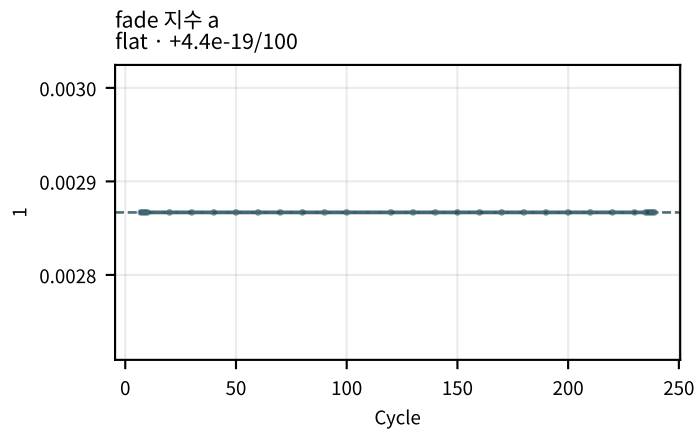
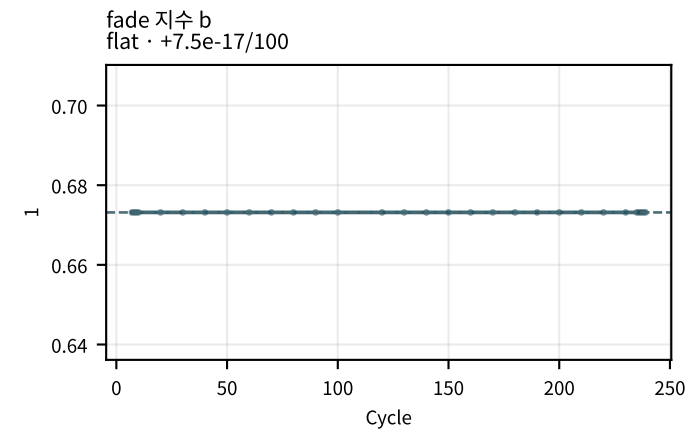
M02Ch111 · 곡선 fit · $\Delta Q(V)$



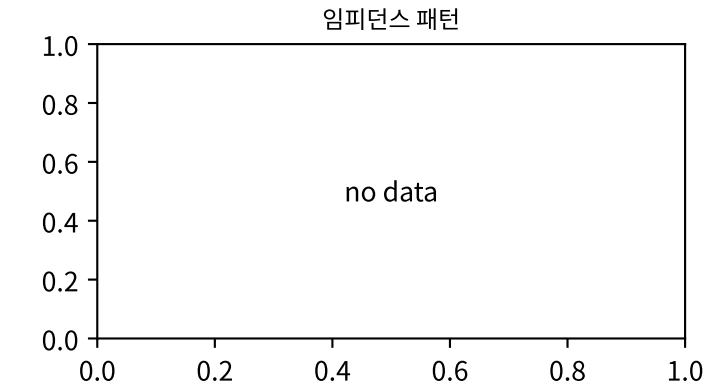
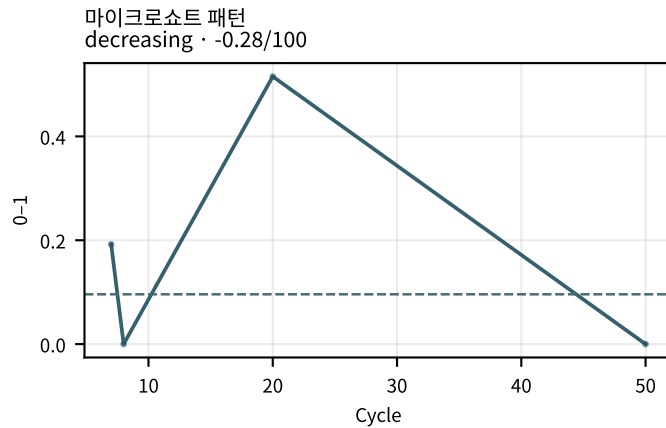
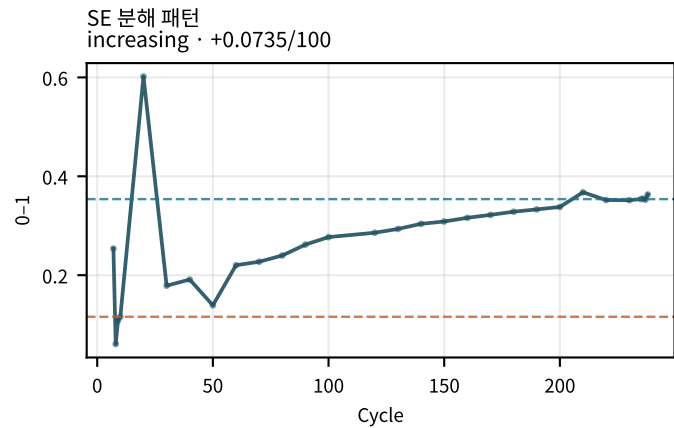
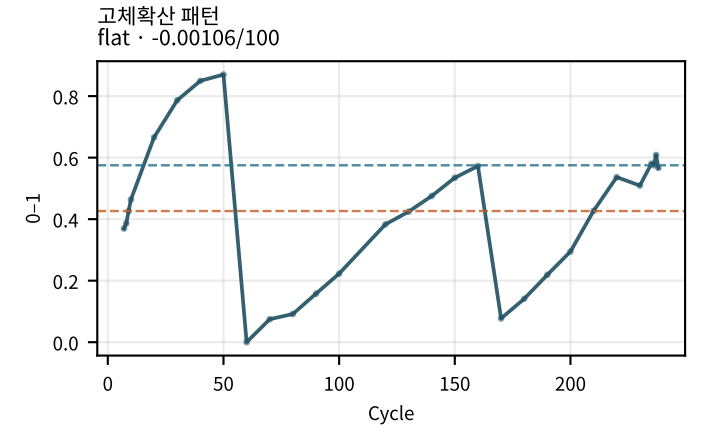
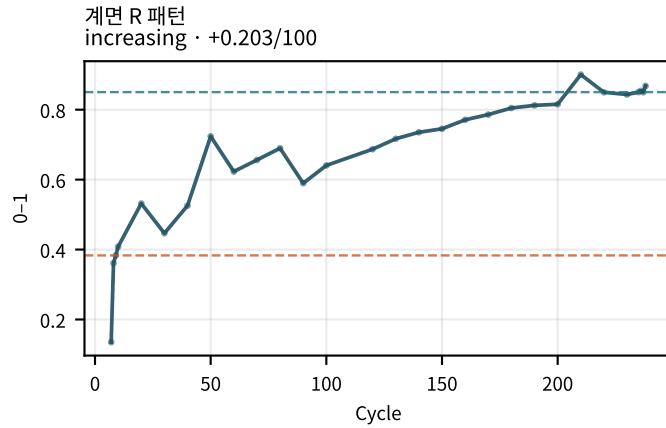
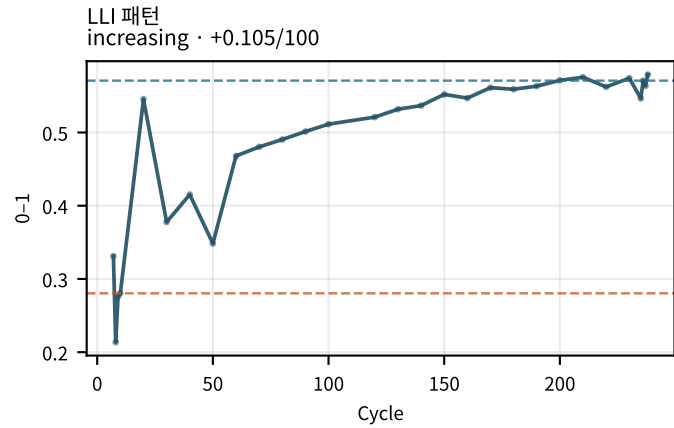
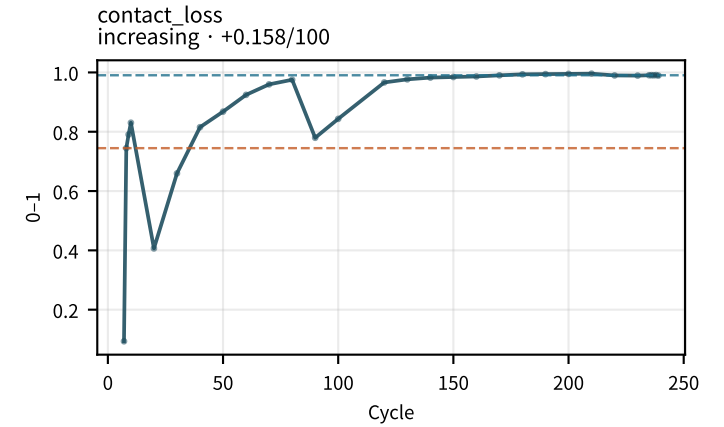
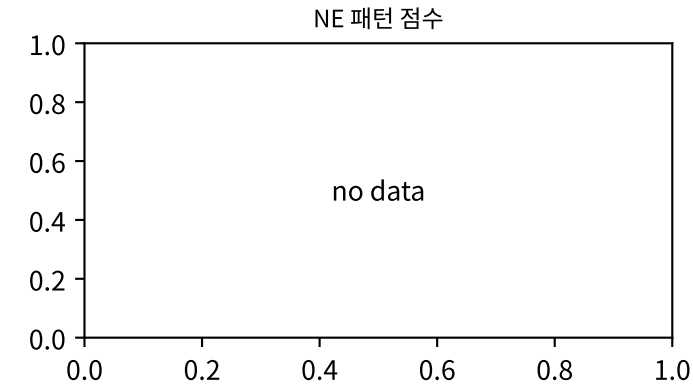
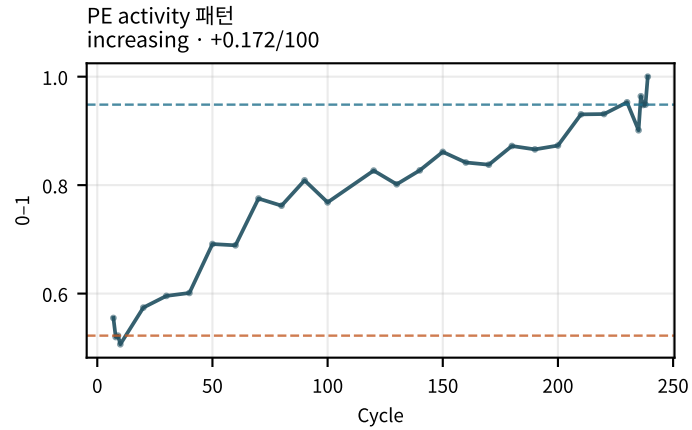
M02Ch111 · 곡선 fit · ΔQ(V)



M02Ch111 · fade · knee

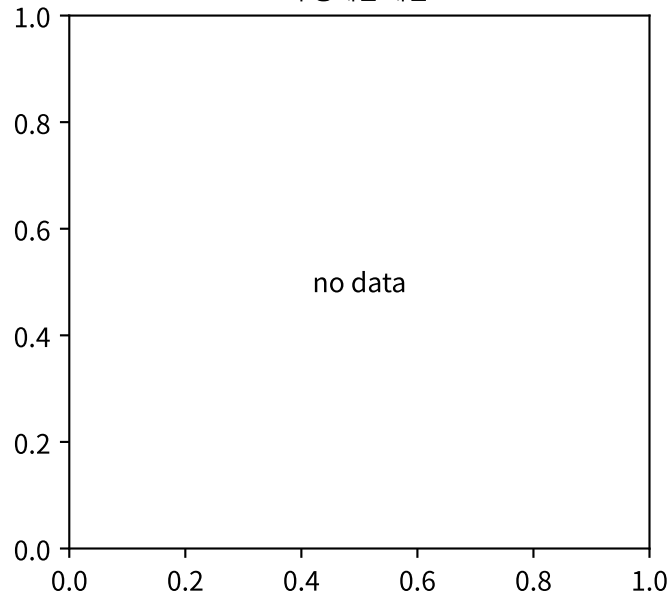


M02Ch111 · 열화 패턴 점수

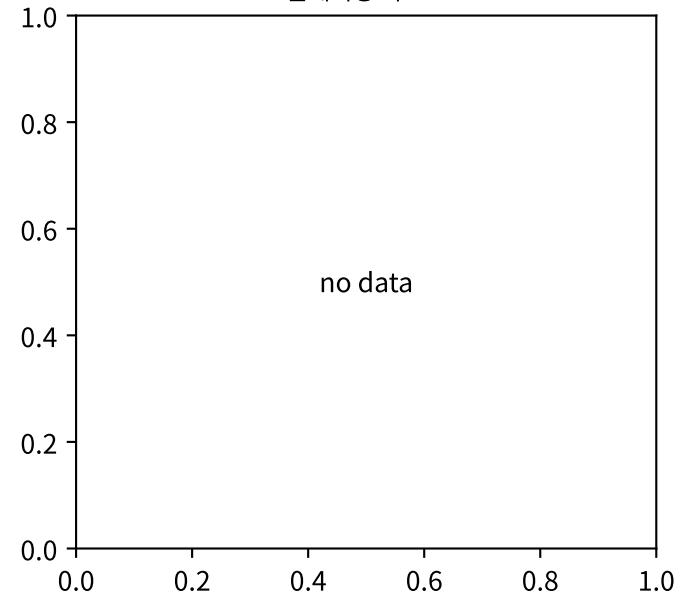


M02Ch111 · 열화 패턴 점수

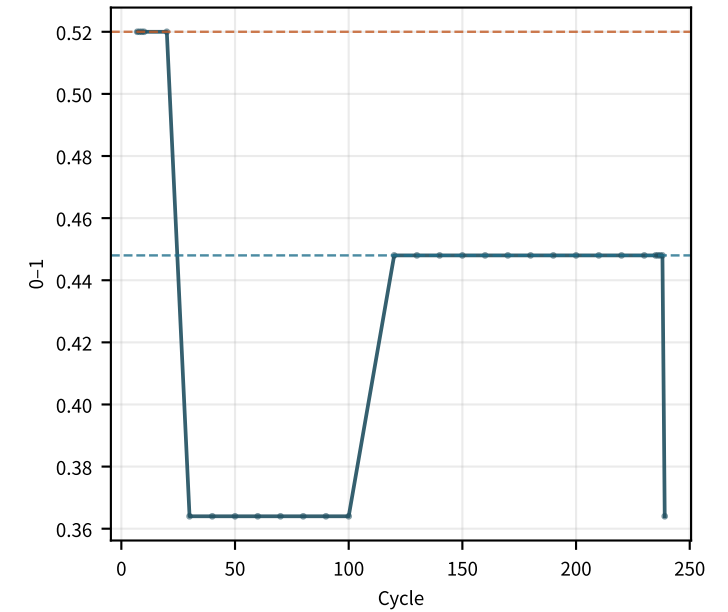
수송제한 패턴



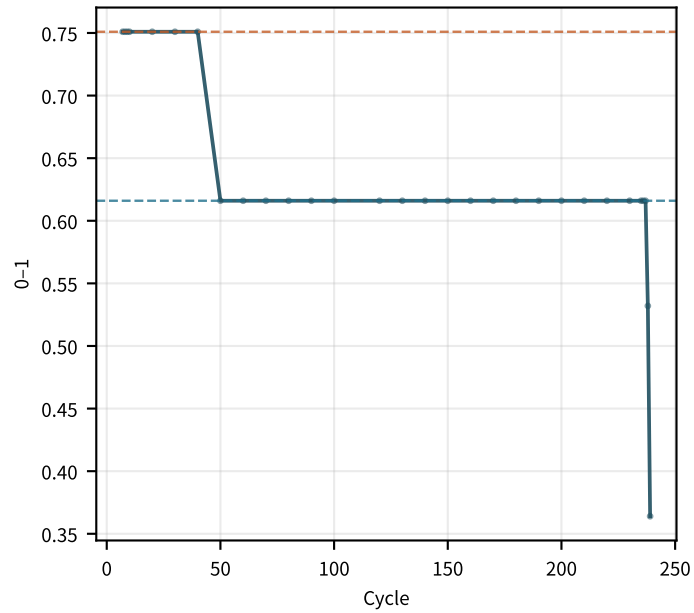
플레이팅 리스크



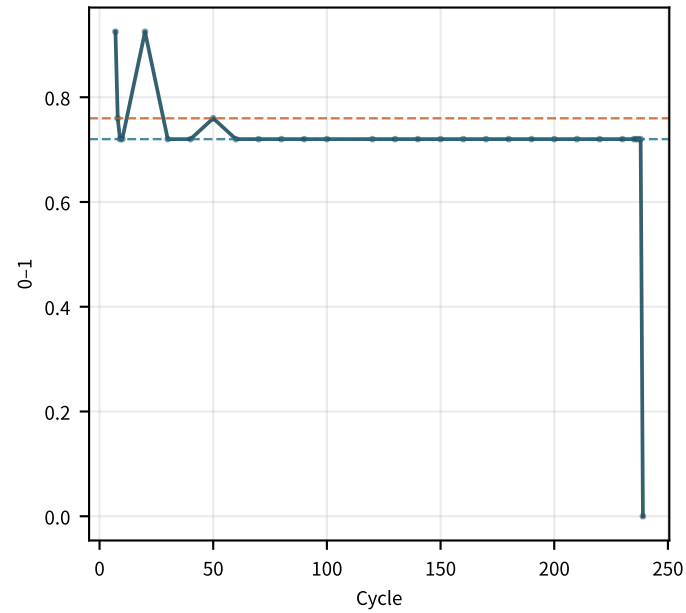
contact_loss 신뢰도
flat · -0.00436/100



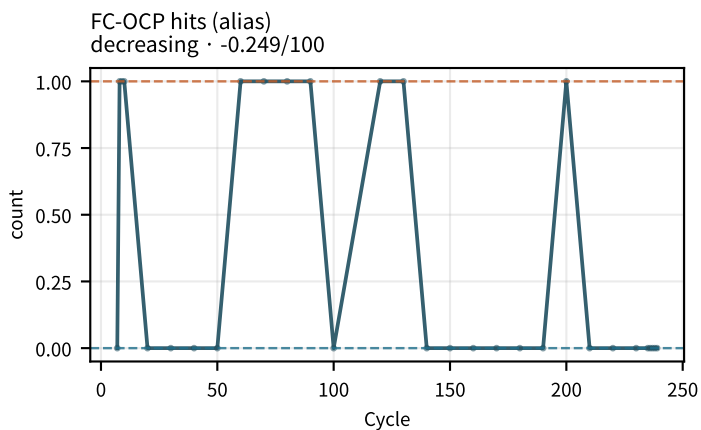
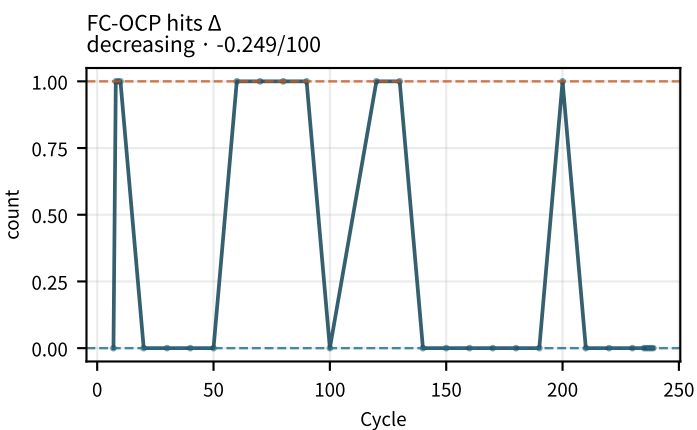
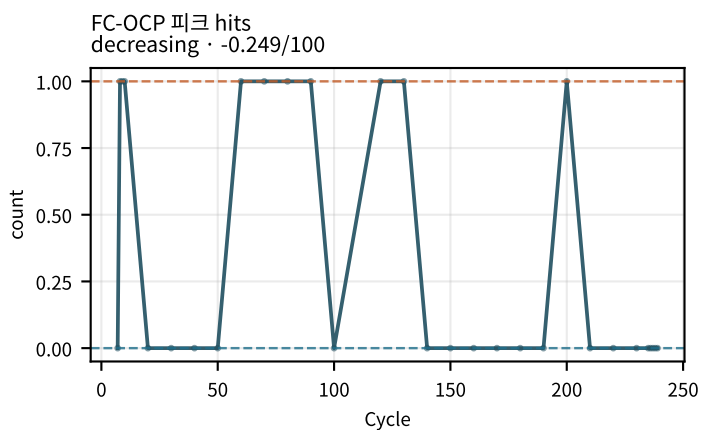
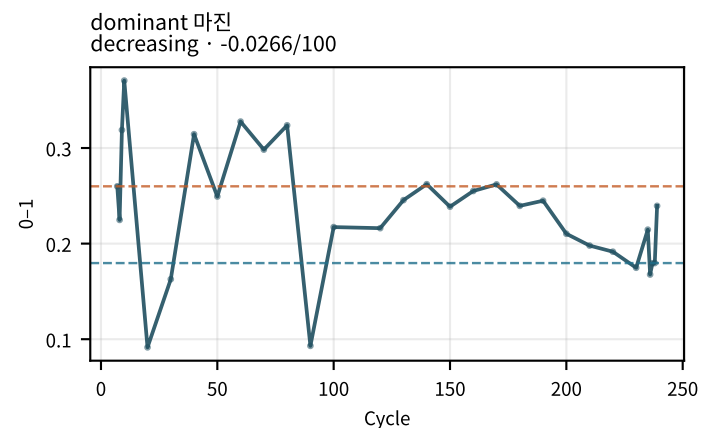
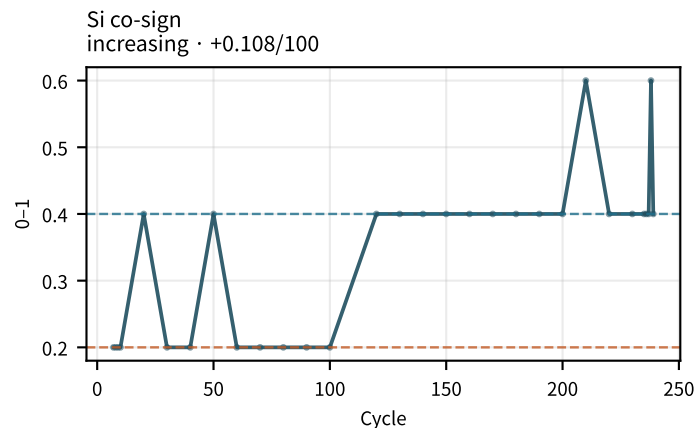
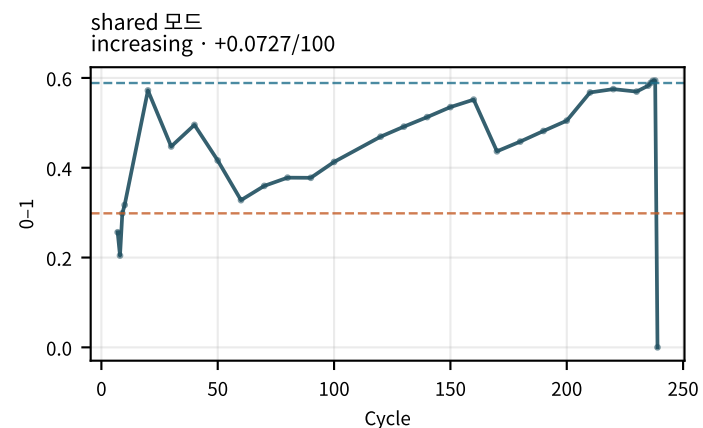
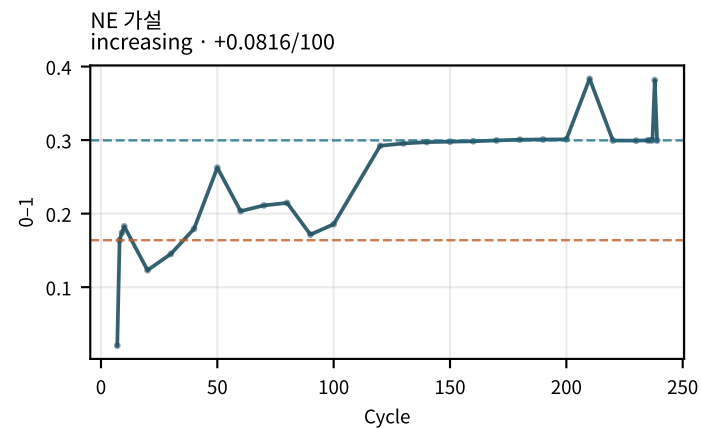
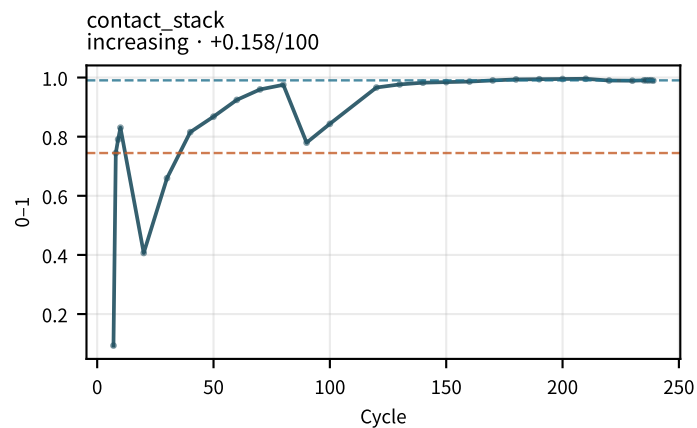
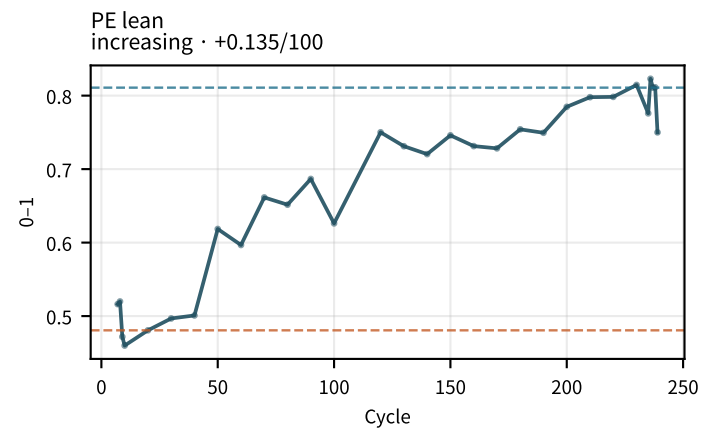
LAM_PE 신뢰도
decreasing · -0.0692/100



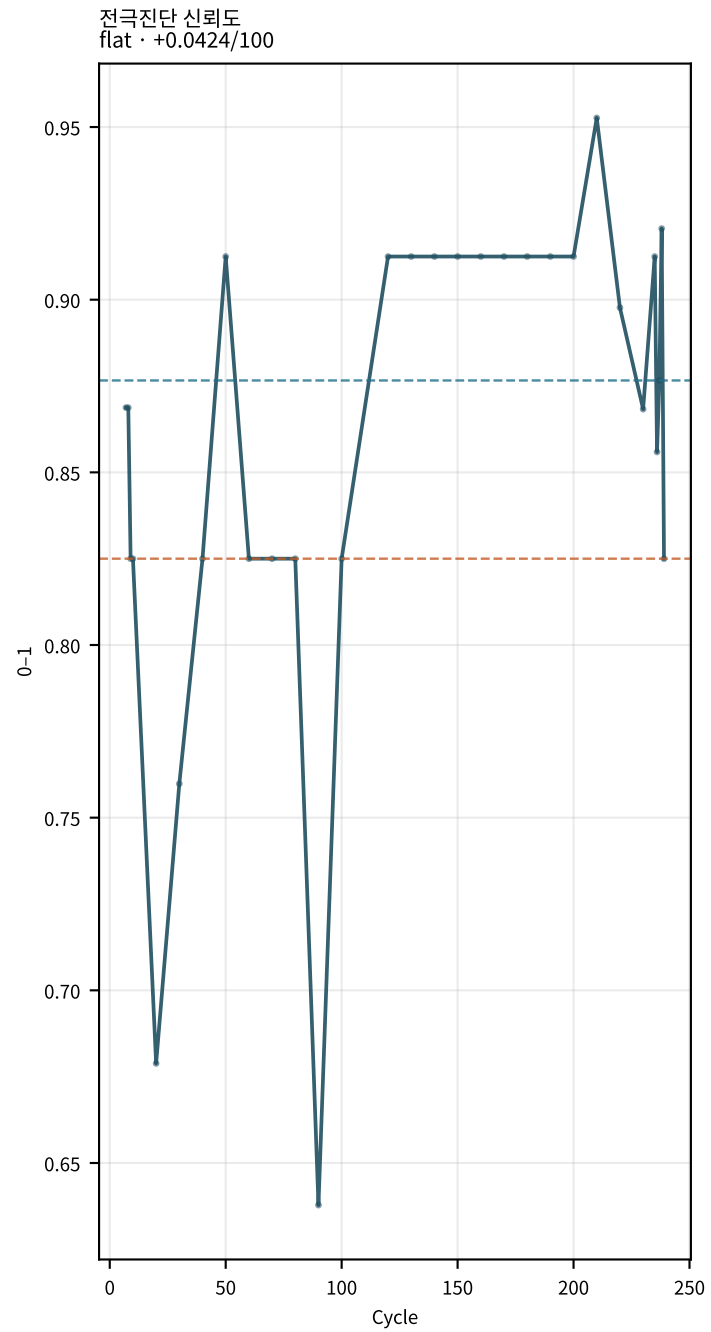
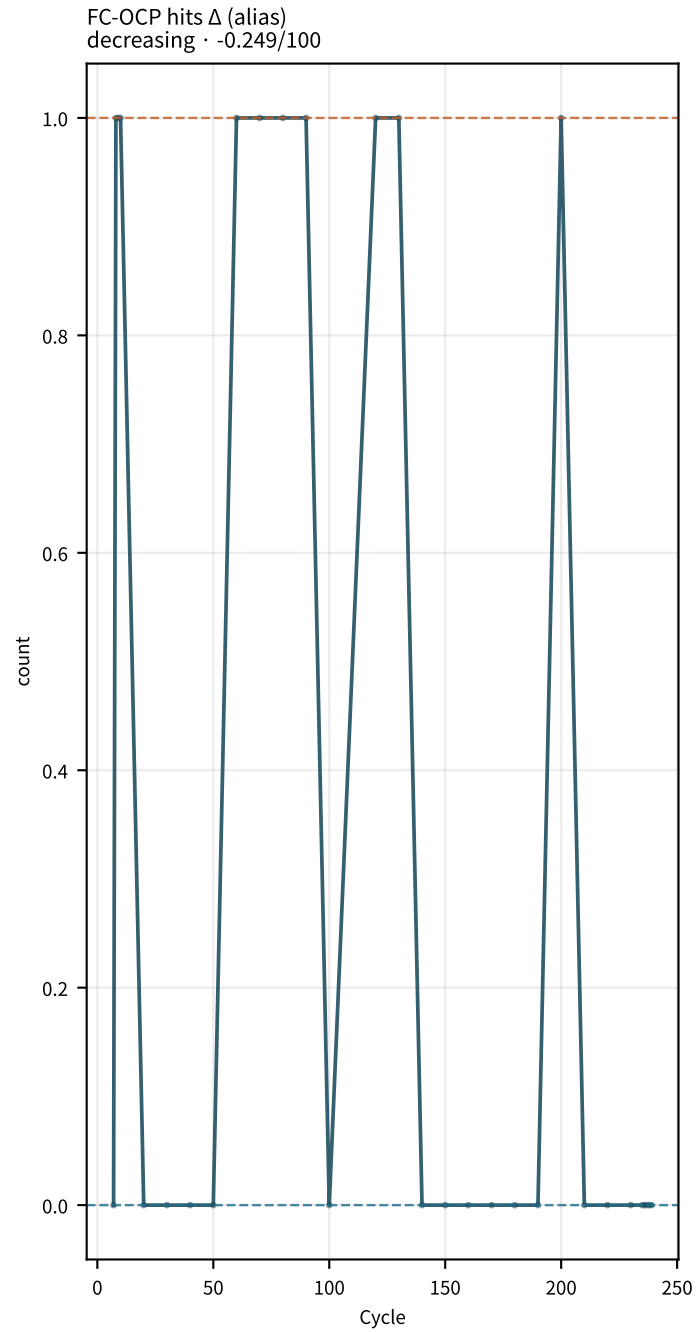
LLI 신뢰도
decreasing · -0.0657/100



M02Ch111 · 전극 lean 가설



M02Ch111 · 전극 lean 가설



지표 설명 · 트렌드 (M02Ch111)

충전 전압 컷오프 (chg_V_cutoff)

충전 종료 전압. | 계산: charge V max/cutoff.

트렌드: 충전 전압 컷오프: early=4.2 → late=4.2 ($\Delta=+5.18\text{e-}05$, $+3.568\text{e-}05\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전 전압 컷오프 (dchg_V_cutoff)

방전 종료 전압. | 계산: discharge V min/cutoff.

트렌드: 방전 전압 컷오프: early=2.5 → late=2.5 ($\Delta=-1.5\text{e-}05$, $+7.158\text{e-}06\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

충전 전류 컷오프 (chg_I_cutoff)

CV 종료 전류. | 계산: charge I cutoff.

트렌드: 충전 전류 컷오프: early=33.21 → late=34.45 ($\Delta=+1.242$, $+1.183\text{A}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

충전 평균 온도 (chg_temp_avg)

충전 구간 평균 온도. | 계산: mean Temp on charge.

트렌드: 충전 평균 온도: early=0 → late=0 ($\Delta=+0$, $+0\text{C}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전 평균 온도 (dchg_temp_avg)

방전 구간 평균 온도. | 계산: mean Temp on discharge.

트렌드: 방전 평균 온도: early=0 → late=0 ($\Delta=+0$, $+0\text{C}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

사이클 소요시간 (cycle_duration_h)

한 사이클 벽시계 시간. | 계산: total step time sum.

트렌드: 사이클 소요시간: early=5.086 → late=5.014 ($\Delta=-0.07194$, $-0.1598\text{h}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

용량 유지율 (SoHQ)

기준 대비 방전용량. | 계산: $Q_dchg/Q_base \times 100$.

트렌드: 용량 유지율: early=97.72 → late=86.75 ($\Delta=-10.97$, -4.347%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

방전 용량 (dchgCapa)

사이클 방전용량. | 계산: max discharge capacity.

트렌드: 방전 용량: early=67.59 → late=60 ($\Delta=-7.588$, -3.007Ah/100cyc) → flat · stable | aging 기대=decrease | vs=stable

충전 용량 (chgCapa)

사이클 충전용량. | 계산: max charge capacity.

트렌드: 충전 용량: early=66.47 → late=51.19 ($\Delta=-15.29$, -6.005Ah/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

CC 충전용량 (chgCCcapa)

CC 구간 용량. | 계산: CC capacity.

트렌드: CC 충전용량: early=65.57 → late=50.88 ($\Delta=-14.69$, -5.938Ah/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

CV 충전용량 (chgCVcapa)

CV 구간 용량. | 계산: signal/column CV Ah.

트렌드: CV 충전용량: early=0 → late=0 ($\Delta=+0$, -0.06824Ah/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

CC 용량비 (chgCapa_CCratio)

CC/(CC+CV). | 계산: chgCCcapa/chgCapa.

트렌드: CC 용량비: early=100 → late=100 ($\Delta=+0$, +0.048311/100cyc) → flat · stable | aging 기대=decrease | vs=stable

지표 설명 · 트렌드 (M02Ch111)

CC비 (정규화) (chgCapa_CCratio_norm)

기준 정규화 CC비. | 계산: $CCratio / baseline$.

트렌드: CC비 (정규화): early=95.23 → late=95.23 ($\Delta=+0$, +0.31261/100cyc) → flat · stable | aging 기대=decrease | vs=stable

CC비 Δ (delta_chgCapa_CCratio)

기준 대비 CC비 변화. | 계산: Δabs .

트렌드: CC비 Δ : early=0 → late=0 ($\Delta=+0$, +0.048311/100cyc) → flat · stable | aging 기대=decrease | vs=stable

CV 시간 (chgCVtime)

CV 지속시간. | 계산: CV step duration.

트렌드: CV 시간: early=0 → late=0 ($\Delta=+0$, -14.51s/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

CV 시정수 (τ_{CV})

CV 전류 감쇠 τ . | 계산: $I(t)$ exp fit.

트렌드: CV 시정수: early=843 → late=843 ($\Delta=+0$, +80.05s/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

CV Q @Tref ($Q_{CV_at_Tref}$)

온도 보정 CV 용량. | 계산: CV Q at ref T.

트렌드: CV Q @Tref: early=3.231 → late=3.231 ($\Delta=+0$, +0.5513Ah/100cyc) → increasing · context | aging 기대=either | vs=context

쿨롱 효율 (CE)

Q_{dchg}/Q_{chg} . | 계산: $dchg/chg*100$.

트렌드: 쿨롱 효율: early=101.9 → late=116.9 ($\Delta=+14.98$, +6.03%/100cyc) → increasing · opposite_aging | aging 기대=decrease | vs=opposite_aging

지표 설명 · 트렌드 (M02Ch111)

가역 CE (CE_rev)

가역 쿨롱 효율 proxy. | 계산: rev CE extract.

트렌드: 가역 CE: early=97.48 → late=85.31 (Δ =-12.17, -4.382%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

국소 CE(20) (CE_local_20)

최근 20사이클 국소 CE. | 계산: rolling CE.

트렌드: 국소 CE(20): early=498.2 → late=301.7 (Δ =-196.5, -63.24%/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

쿨롱 비효율 (CI)

100-CE. | 계산: 100-CE.

트렌드: 쿨롱 비효율: early=-1.924 → late=-16.9 (Δ =-14.98, -6.03%/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

CI /시간 (CI_per_hour)

시간당 쿨롱 손실. | 계산: CI / cycle_duration_h.

트렌드: CI /시간: early=-0.3778 → late=-3.372 (Δ =-2.994, -1.201%/h/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

전압 효율 (VE)

에너지 전압효율. | 계산: E_dchg/E_chg.

트렌드: 전압 효율: early=0.9062 → late=0.8627 (Δ =-0.04354, -0.018351/100cyc) → flat · stable | aging 기대=decrease | vs=stable

에너지 효율 (EE)

충방전 에너지비. | 계산: E_dchg/E_chg.

트렌드: 에너지 효율: early=0.9239 → late=1.009 (Δ =+0.08505, +0.033241/100cyc) → flat · stable | aging 기대=decrease | vs=stable

지표 설명 · 트렌드 (M02Ch111)

충전 에너지 (chg_E)

충전 Wh. | 계산: $\int VI \, dt$ charge.

트렌드: 충전 에너지: early=249.7 $\rightarrow$ late=194.6 (Δ =-55.06, -21.6Wh/100cyc) $\rightarrow$ decreasing · matches_aging | aging
기대=decrease | vs=matches_aging

방전 에너지 (dchg_E)

방전 Wh. | 계산: $\int VI \, dt$ discharge.

트렌드: 방전 에너지: early=230 $\rightarrow$ late=196.9 (Δ =-33.12, -13.35Wh/100cyc) $\rightarrow$ decreasing · matches_aging | aging
기대=decrease | vs=matches_aging

에너지 손실 (dE)

충전-방전 에너지. | 계산: chg_E-dchg_E.

트렌드: 에너지 손실: early=18.99 $\rightarrow$ late=-1.746 (Δ =-20.74, -8.288Wh/100cyc) $\rightarrow$ decreasing · opposite_aging | aging
기대=increase | vs=opposite_aging

완화 용량 (Q_relax)

휴지 회복 용량. | 계산: DCIR block ΔQ .

트렌드: 완화 용량: early=-0.308 $\rightarrow$ late=0.012 (Δ =+0.32, +0.175Ah/100cyc) $\rightarrow$ increasing · context | aging 기대=either |
vs=context

완화 용량 % (Q_relax_pct)

회복 용량 비율. | 계산: $Q_{\text{relax}}/Q \cdot 100$.

트렌드: 완화 용량 %: early=-0.4453 $\rightarrow$ late=0.01934 (Δ =+0.4647, +0.2544%/100cyc) $\rightarrow$ increasing · matches_aging | aging
기대=increase | vs=matches_aging

dSoHQ/dN (dSoHQ_dN)

용량 유지율 순간 기울기. | 계산: diff SoHQ.

트렌드: dSoHQ/dN: early=-0.1786 $\rightarrow$ late=-0.02735 (Δ =+0.1513, +0.1691%/cyc/100cyc) $\rightarrow$ increasing · opposite_aging |
aging 기대=decrease | vs=opposite_aging

지표 설명 · 트렌드 (M02Ch111)

d2SoHQ (d2SoHQ)

SoHQ 2차 미분. | 계산: diff dSoHQ.

트렌드: d2SoHQ: early=-0.001083 → late=-0.071 (Δ =-0.06992, -0.0469%/cyc2/100cyc) → decreasing · context | aging 기대=either | vs=context

충전후 휴지 초기 V (EoC_restV_init)

충전후 rest 시작 전압. | 계산: rest step 첫 샘플.

트렌드: 충전후 휴지 초기 V: early=4.2 → late=4.2 (Δ =+0.0001093, +3.172e-05V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 60s V (EoC_restV_60s)

충전후 rest 60초 전압. | 계산: rest $\approx$ 60 s 보간.

트렌드: 충전후 휴지 60s V: early=4.192 → late=4.181 (Δ =-0.01071, -0.004539V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 30분 V (EoC_restV_30m)

충전후 rest 30분 전압 (OCV에 가까움). | 계산: rest $\approx$ 1800 s 보간.

트렌드: 충전후 휴지 30분 V: early=4.179 → late=4.153 (Δ =-0.02615, -0.01123V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 종료 V (EoC_restV_end)

충전후 rest 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 충전후 휴지 종료 V: early=4.179 → late=4.153 (Δ =-0.02615, -0.01123V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 완화량 (EoC_restV_relax)

충전후 end-init 완화. | 계산: EoC_restV_end - EoC_restV_init.

트렌드: 충전후 휴지 완화량: early=-0.02061 → late=-0.04693 (Δ =-0.02632, -0.01126V/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

충전후 60s 완화량 (EoC_restV_relax_60s)

충전후 60s-init. | 계산: $\text{EoC_restV_60s} - \text{EoC_restV_init.}$

트렌드: 충전후 60s 완화량: early=-0.007592 → late=-0.01846 ($\Delta=-0.01086, -0.004571\text{V}/100\text{cyc}$) → decreasing · context | aging 기대=either | vs=context

충전후 휴지 완화 τ (EoC_restV_tau)

충전후 rest 완화 시정수 proxy. | 계산: rest V(t) 완화 fit τ .

트렌드: 충전후 휴지 완화 τ : early=534.7 → late=560.4 ($\Delta=+25.64, +14.47\text{s}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

충전후 60s V Δ vs기준 (delta_EoC_restV_60s)

기준 사이클 대비 EoC_restV_60s 이동. | 계산: $\text{EoC_restV_60s}(\text{cycle}) - \text{baseline.}$

트렌드: 충전후 60s V Δ vs기준: early=-0.0004723 → late=-0.01118 ($\Delta=-0.01071, -0.004539\text{V}/100\text{cyc}$) → decreasing · context | aging 기대=either | vs=context

충전후 30분 V Δ vs기준 (delta_EoC_restV_30m)

기준 사이클 대비 EoC_restV_30m 이동. | 계산: $\text{EoC_restV_30m}(\text{cycle}) - \text{baseline.}$

트렌드: 충전후 30분 V Δ vs기준: early=-0.000821 → late=-0.02698 ($\Delta=-0.02615, -0.01123\text{V}/100\text{cyc}$) → decreasing · context | aging 기대=either | vs=context

충전후 종료 V Δ vs기준 (delta_EoC_restV_end)

기준 사이클 대비 EoC_restV_end 이동. | 계산: $\text{EoC_restV_end}(\text{cycle}) - \text{baseline.}$

트렌드: 충전후 종료 V Δ vs기준: early=-0.000821 → late=-0.02698 ($\Delta=-0.02615, -0.01123\text{V}/100\text{cyc}$) → decreasing · context | aging 기대=either | vs=context

충전후 완화 τ Δ vs기준 (delta_EoC_restV_tau)

기준 사이클 대비 EoC_restV_tau 이동. | 계산: $\text{EoC_restV_tau}(\text{cycle}) - \text{baseline.}$

트렌드: 충전후 완화 τ Δ vs기준: early=-21.76 → late=3.882 ($\Delta=+25.64, +14.47\text{s}/100\text{cyc}$) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전후 휴지 초기 V (EoD_restV_init)

방전후 rest 시작 전압. | 계산: rest step 첫 샘플.

트렌드: 방전후 휴지 초기 V: early=2.5 $\rightarrow$ late=2.5 ($\Delta=-1.5e-05$, +7.158e-06V/100cyc) $\rightarrow$ flat · context | aging 기대=either
| vs=context

방전후 휴지 60s V (EoD_restV_60s)

방전후 rest 60초 전압. | 계산: rest $\approx$ 60 s 보간.

트렌드: 방전후 휴지 60s V: early=2.883 $\rightarrow$ late=2.951 ($\Delta=+0.06807$, +0.0323V/100cyc) $\rightarrow$ flat · context | aging 기대=either
| vs=context

방전후 휴지 30분 V (EoD_restV_30m)

방전후 rest 30분 전압 (OCV에 가까움). | 계산: rest $\approx$ 1800 s 보간.

트렌드: 방전후 휴지 30분 V: early=3.016 $\rightarrow$ late=3.074 ($\Delta=+0.05746$, +0.02446V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

방전후 휴지 종료 V (EoD_restV_end)

방전후 rest 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 방전후 휴지 종료 V: early=3.016 $\rightarrow$ late=3.074 ($\Delta=+0.05746$, +0.02446V/100cyc) $\rightarrow$ flat · context | aging 기대=either
| vs=context

방전후 휴지 완화량 (EoD_restV_relax)

방전후 end-init 완화. | 계산: EoD_restV_end – EoD_restV_init.

트렌드: 방전후 휴지 완화량: early=0.5163 $\rightarrow$ late=0.5736 ($\Delta=+0.05739$, +0.02446V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

방전후 60s 완화량 (EoD_restV_relax_60s)

방전후 60s-init. | 계산: EoD_restV_60s – EoD_restV_init.

트렌드: 방전후 60s 완화량: early=0.3826 $\rightarrow$ late=0.4507 ($\Delta=+0.06806$, +0.0323V/100cyc) $\rightarrow$ increasing · context | aging
기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전후 휴지 완화 τ (EoD_restV_tau)

방전후 rest 완화 시정수 proxy. | 계산: rest V(t) 완화 fit τ .

트렌드: 방전후 휴지 완화 τ : early=361.7 $\rightarrow$ late=324.2 (Δ =-37.57, -20.28s/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

방전후 60s V Δ vs기준 (delta_EoD_restV_60s)

기준 사이클 대비 EoD_restV_60s 이동. | 계산: EoD_restV_60s(cycle) - baseline.

트렌드: 방전후 60s V Δ vs기준: early=0.05328 $\rightarrow$ late=0.1214 (Δ =+0.06807, +0.0323V/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

방전후 30분 V Δ vs기준 (delta_EoD_restV_30m)

기준 사이클 대비 EoD_restV_30m 이동. | 계산: EoD_restV_30m(cycle) - baseline.

트렌드: 방전후 30분 V Δ vs기준: early=0.03952 $\rightarrow$ late=0.09698 (Δ =+0.05746, +0.02446V/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

방전후 종료 V Δ vs기준 (delta_EoD_restV_end)

기준 사이클 대비 EoD_restV_end 이동. | 계산: EoD_restV_end(cycle) - baseline.

트렌드: 방전후 종료 V Δ vs기준: early=0.03952 $\rightarrow$ late=0.09698 (Δ =+0.05746, +0.02446V/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

방전후 완화 τ Δ vs기준 (delta_EoD_restV_tau)

기준 사이클 대비 EoD_restV_tau 이동. | 계산: EoD_restV_tau(cycle) - baseline.

트렌드: 방전후 완화 τ Δ vs기준: early=3.59 $\rightarrow$ late=-33.98 (Δ =-37.57, -20.28s/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

EoC 방전 10s DCIR (EoC_dchgR_10s)

EoC 방전 시작 후 10s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V_0 - V(10s)|/|I| \cdot 1000$.

트렌드: EoC 방전 10s DCIR: early=0.1733 $\rightarrow$ late=0.2371 (Δ =+0.06378, +0.03014m Ω /100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch111)

EOC 방전 10s 증가% (EoC_dchgR_10s_inc)

기준 대비 EoC_dchgR_10s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoC 방전 10s 증가%: early=24.02 → late=69.66 (Δ =+45.64, +21.57%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 30s DCIR (EoC_dchgR_30s)

EOC 방전 시작 후 30s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(30s)|/|I| \cdot 1000$.

트렌드: EoC 방전 30s DCIR: early=0.4759 → late=0.6032 (Δ =+0.1273, +0.06098m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 30s 증가% (EoC_dchgR_30s_inc)

기준 대비 EoC_dchgR_30s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoC 방전 30s 증가%: early=18.59 → late=50.31 (Δ =+31.72, +15.2%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 60s DCIR (EoC_dchgR_60s)

EOC 방전 시작 후 60s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(60s)|/|I| \cdot 1000$.

트렌드: EoC 방전 60s DCIR: early=0.8405 → late=1.012 (Δ =+0.172, +0.08047m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 60s 증가% (EoC_dchgR_60s_inc)

기준 대비 EoC_dchgR_60s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoC 방전 60s 증가%: early=13.96 → late=37.28 (Δ =+23.32, +10.91%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOd 충전 10s DCIR (EoD_chgR_10s)

EOd 충전 시작 후 10s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(10s)|/|I| \cdot 1000$.

트렌드: EoD 충전 10s DCIR: early=0.2826 → late=0.3457 (Δ =+0.06308, +0.02728m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch111)

EoD 충전 10s 증가% (EoD_chgR_10s_inc)

기준 대비 EoD_chgR_10s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoD 충전 10s 증가%: early=-4.719 → late=16.55 (Δ =+21.26, +9.197%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD 충전 30s DCIR (EoD_chgR_30s)

EoD 충전 시작 후 30s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(30s)|/|I| \cdot 1000$.

트렌드: EoD 충전 30s DCIR: early=0.7679 → late=0.9127 (Δ =+0.1448, +0.05638m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD 충전 30s 증가% (EoD_chgR_30s_inc)

기준 대비 EoD_chgR_30s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoD 충전 30s 증가%: early=-8.423 → late=8.849 (Δ =+17.27, +6.724%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD 충전 60s DCIR (EoD_chgR_60s)

EoD 충전 시작 후 60s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(60s)|/|I| \cdot 1000$.

트렌드: EoD 충전 60s DCIR: early=1.374 → late=1.548 (Δ =+0.1747, +0.06353m Ω /100cyc) → flat · stable | aging 기대=increase | vs=stable

EoD 충전 60s 증가% (EoD_chgR_60s_inc)

기준 대비 EoD_chgR_60s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoD 충전 60s 증가%: early=-10.29 → late=1.118 (Δ =+11.41, +4.149%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoC R10/R60 (EoC_dchgR_10_60_ratio)

10s/60s 비. 초기 응답 비중. | 계산: EoC_dchgR_10s / EoC_dchgR_60s.

트렌드: EoC R10/R60: early=0.2062 → late=0.2345 (Δ =+0.02834, +0.013781/100cyc) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

EoD R10/R60 (EoD_chgR_10_60_ratio)

10s/60s 비. | 계산: EoD_chgR_10s / EoD_chgR_60s.

트렌드: EoD R10/R60: early=0.2065 → late=0.2233 (Δ =+0.01683, +0.0086811/100cyc) → flat · context | aging 기대=either | vs=context

EoC R10s @25C (EoC_dchgR_10s_T25)

온도 보정 10s DCIR. | 계산: Arrhenius-ish correct_r_to_25c.

트렌드: EoC R10s @25C: early=0.08282 → late=0.1133 (Δ =+0.03048, +0.0144m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD R10s @25C (EoD_chgR_10s_T25)

온도 보정 10s DCIR. | 계산: correct_r_to_25c.

트렌드: EoD R10s @25C: early=0.1351 → late=0.1652 (Δ =+0.03014, +0.01304m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoC R10s 성장/50cyc (EoC_dchgR_10s_growth_50)

롤링 기울기 (50사이클). | 계산: rolling slope of EoC_dchgR_10s.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R Ω (SOC80) (R_ohmic_soc80)

$\sqrt{t}$ 외삽 t→0 절편 (초기 비음 포함 proxy). | 계산: DCIR early $\sqrt{t}$ intercept.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct (SOC80) (R_ct_soc80)

중간 잔차 지수항. Cdl 미분리. | 계산: resid exp-sat fit.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

τ_{ct} (SOC80) (tau_ct_soc80)

Rct 시정수 (fit seed 2s). | 계산: curve_fit τ .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

A_diff (SOC80) (A_diff_soc80)

후반 $\sqrt{t}$ 확산 계수. | 계산: $t \in [10, 30]$ R vs $\sqrt{t}$ slope.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R30s 총 (SOC80) (R_30s_total_soc80)

펄스 30초 총 DCIR. | 계산: R($t \approx 30$ s).

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R Ω 분율 (SOC80) (R_ohmic_frac_soc80)

R Ω / R30s. | 계산: R_ohmic / R_30s_total.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct분율 (SOC80) (R_ct_frac_soc80)

Rct / R30s. | 계산: R_ct / R_30s_total.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Rdiff분율 (SOC80) (R_diff_frac_soc80)

$A\sqrt{30}$ / R30s. | 계산: A_diff*sqrt(30)/R30s.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

회복 τ_1 (SOC80) (R_recovery_tau1_soc80)

펄스 후 빠른 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 τ_2 (SOC80) (R_recovery_tau2_soc80)

펄스 후 느린 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} est (SOC80) (V_inf_est_soc80)

회복 fit 무한시간 전압. | 계산: recovery V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} rest (SOC80) (V_inf_rest_soc80)

rest 기반 V_{∞} . | 계산: rest asymptote.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 (SOC80) (self_discharge_rate_soc80)

휴지 중 전압 강하율. | 계산: dV/dt on long rest.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

DCIR fit R2 (SOC80) (dcir_fit_r2_soc80)

R(t) 3성분 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

DCIR fit RMSE (SOC80) (dcir_fit_rmse_soc80)

상대 RMSE. | 계산: rmse/meanR .

트렌드: 데이터 부족 | aging 기대=decrease | vs=n/a

펄스 전류 (SOC80) (pulse_current_A_soc80)

DCIR 펄스 || 중앙값. | 계산: median || on pulse.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$t \leq 1s$ 샘플수 (SOC80) (n_t_le_1s_soc80)

early 샘플 수 (R Ω 외삽 품질). | 계산: count $t \leq 1$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 amp비 (SOC80) (relax_amp_ratio_soc80)

느린/전체 회복 진폭비. | 계산: $|b_2|/(|b_1|+|b_2|)$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

완화 완성도 (SOC80) (relax_completeness_soc80)

rest 완화 충분성. | 계산: relax completeness.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 fit R2 (SOC80) (recovery_fit_r2_soc80)

two-exp 회복 fit 품질. | 계산: r^2 .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

DCIR fit 유효 (SOC80) (dcir_fit_valid_soc80)

rmse/r2/cond 게이트 통과. | 계산: dcir_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

DCIR fit 조건수 (SOC80) (dcir_fit_cond_soc80)

공분산 조건수 (축퇴 지표). | 계산: cond(pcov).

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 fit 유효 (SOC80) (sd_fit_valid_soc80)

self-discharge fit 게이트. | 계산: sd_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

펄스 포인트수 (SOC80) (n_points_soc80)

DCIR 펄스 샘플 수. | 계산: n_points.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R Ω (SOC50) (R_ohmic_soc50)

$\sqrt{t}$ 외삽 $t \rightarrow 0$ 절편 (초기 비음 포함 proxy). | 계산: DCIR early $\sqrt{t}$ intercept.

트렌드: R Ω (SOC50): early=1.092 $\rightarrow$ late=2.744 (Δ =+1.651, +0.7975m Ω /100cyc) $\rightarrow$ increasing · matches_aging | aging
기대=increase | vs=matches_aging

R_{ct} (SOC50) (R_ct_soc50)

중간 잔차 지수항. Cdl 미분리. | 계산: resid exp-sat fit.

트렌드: R_{ct} (SOC50): early=0.7379 $\rightarrow$ late=0.9016 (Δ =+0.1637, +0.06968m Ω /100cyc) $\rightarrow$ increasing · matches_aging |
aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch111)

τ_{ct} (SOC50) (tau_ct_soc50)

Rct 시정수 (fit seed 2s). | 계산: curve_fit τ .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

A_diff (SOC50) (A_diff_soc50)

후반 $\sqrt{t}$ 확산 계수. | 계산: $t \in [10, 30]$ R vs $\sqrt{t}$ slope.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R30s 총 (SOC50) (R_30s_total_soc50)

펄스 30초 총 DCIR. | 계산: R($t \approx 30s$).

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R Ω 분율 (SOC50) (R_ohmic_frac_soc50)

R Ω / R30s. | 계산: R_ohmic / R_30s_total.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct분율 (SOC50) (R_ct_frac_soc50)

Rct / R30s. | 계산: R_ct / R_30s_total.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Rdiff분율 (SOC50) (R_diff_frac_soc50)

A $\sqrt{30}$ / R30s. | 계산: A_diff*sqrt(30)/R30s.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

회복 τ_1 (SOC50) (R_recovery_tau1_soc50)

펄스 후 빠른 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 τ_2 (SOC50) (R_recovery_tau2_soc50)

펄스 후 느린 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} est (SOC50) (V_inf_est_soc50)

회복 fit 무한시간 전압. | 계산: recovery V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} rest (SOC50) (V_inf_rest_soc50)

rest 기반 V_{∞} . | 계산: rest asymptote.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 (SOC50) (self_discharge_rate_soc50)

휴지 중 전압 강하율. | 계산: dV/dt on long rest.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

DCIR fit R2 (SOC50) (dcir_fit_r2_soc50)

R(t) 3성분 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

DCIR fit RMSE (SOC50) (dcir_fit_rmse_soc50)

상대 RMSE. | 계산: rmse/meanR .

트렌드: 데이터 부족 | aging 기대=decrease | vs=n/a

펄스 전류 (SOC50) (pulse_current_A_soc50)

DCIR 펄스 || 중앙값. | 계산: median || on pulse.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$t \leq 1s$ 샘플수 (SOC50) (n_t_le_1s_soc50)

early 샘플 수 (R Ω 외삽 품질). | 계산: count $t \leq 1$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 amp비 (SOC50) (relax_amp_ratio_soc50)

느린/전체 회복 진폭비. | 계산: $|b_2|/(|b_1|+|b_2|)$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

완화 완성도 (SOC50) (relax_completeness_soc50)

rest 완화 충분성. | 계산: relax completeness.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 fit R2 (SOC50) (recovery_fit_r2_soc50)

two-exp 회복 fit 품질. | 계산: r^2 .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

DCIR fit 유효 (SOC50) (dcir_fit_valid_soc50)

rmse/r2/cond 게이트 통과. | 계산: dcir_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

DCIR fit 조건수 (SOC50) (dcir_fit_cond_soc50)

공분산 조건수 (축퇴 지표). | 계산: cond(pcov).

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 fit 유효 (SOC50) (sd_fit_valid_soc50)

self-discharge fit 게이트. | 계산: sd_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

펄스 포인트수 (SOC50) (n_points_soc50)

DCIR 펄스 샘플 수. | 계산: n_points.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$R\Omega$ (SOC20) (R_ohmic_soc20)

$\sqrt{t}$ 외삽 $t \rightarrow 0$ 절편 (초기 비음 포함 proxy). | 계산: DCIR early $\sqrt{t}$ intercept.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R_{ct} (SOC20) (R_ct_soc20)

중간 잔차 지수항. Cdl 미분리. | 계산: resid exp-sat fit.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

τ_{ct} (SOC20) (τ_{ct_soc20})

Rct 시정수 (fit seed 2s). | 계산: $\text{curve_fit } \tau$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

A_diff (SOC20) (A_{diff_soc20})

후반 $\sqrt{t}$ 확산 계수. | 계산: $t \in [10, 30]$ R vs $\sqrt{t}$ slope.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R30s 총 (SOC20) ($R_{30s_total_soc20}$)

펄스 30초 총 DCIR. | 계산: $R(t \approx 30s)$.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

RΩ분율 (SOC20) ($R_{ohmic_frac_soc20}$)

R_{Ω} / R_{30s} . | 계산: $R_{ohmic} / R_{30s_total}$.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct분율 (SOC20) ($R_{ct_frac_soc20}$)

R_{ct} / R_{30s} . | 계산: R_{ct} / R_{30s_total} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Rdiff분율 (SOC20) ($R_{diff_frac_soc20}$)

$A\sqrt{30} / R_{30s}$. | 계산: $A_{diff} \cdot \sqrt{30} / R_{30s}$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

회복 τ_1 (SOC20) (R_recovery_tau1_soc20)

펄스 후 빠른 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 τ_2 (SOC20) (R_recovery_tau2_soc20)

펄스 후 느린 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} est (SOC20) (V_inf_est_soc20)

회복 fit 무한시간 전압. | 계산: recovery V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} rest (SOC20) (V_inf_rest_soc20)

rest 기반 V_{∞} . | 계산: rest asymptote.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 (SOC20) (self_discharge_rate_soc20)

휴지 중 전압 강하율. | 계산: dV/dt on long rest.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

DCIR fit R2 (SOC20) (dcir_fit_r2_soc20)

R(t) 3성분 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

DCIR fit RMSE (SOC20) (dcir_fit_rmse_soc20)

상대 RMSE. | 계산: rmse/meanR.

트렌드: 데이터 부족 | aging 기대=decrease | vs=n/a

펄스 전류 (SOC20) (pulse_current_A_soc20)

DCIR 펄스 || 중앙값. | 계산: median || on pulse.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$t \leq 1s$ 샘플수 (SOC20) (n_t_le_1s_soc20)

early 샘플 수 (R Ω 외삽 품질). | 계산: count $t \leq 1$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 amp비 (SOC20) (relax_amp_ratio_soc20)

느린/전체 회복 진폭비. | 계산: $|b_2|/(|b_1|+|b_2|)$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

완화 완성도 (SOC20) (relax_completeness_soc20)

rest 완화 충분성. | 계산: relax completeness.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 fit R2 (SOC20) (recovery_fit_r2_soc20)

two-exp 회복 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

DCIR fit 유효 (SOC20) (dcir_fit_valid_soc20)

rmse/r2/cond 게이트 통과. | 계산: dcir_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

DCIR fit 조건수 (SOC20) (dcir_fit_cond_soc20)

공분산 조건수 (축퇴 지표). | 계산: cond(pcov).

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 fit 유효 (SOC20) (sd_fit_valid_soc20)

self-discharge fit 게이트. | 계산: sd_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

펄스 포인트수 (SOC20) (n_points_soc20)

DCIR 펄스 샘플 수. | 계산: n_points.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

기계/화학 비 (mech_vs_chem_ratio)

R Ω /R_{ct} 상대 비중. | 계산: R_ohmic_soc50 / R_ct_soc50.

트렌드: 기계/화학 비: early=1.48 → late=3.043 (Δ =+1.563, +0.79681/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

R Ω 성장/100cyc (R_ohmic_growth_100)

기준 대비 R Ω 성장률 (레벨과 별개). | 계산: (R-R0)/((N-N0)/100).

트렌드: R Ω 성장/100cyc: early=1.104 → late=0.7864 (Δ =-0.3172, -0.3213m Ω /100cyc/100cyc) → decreasing · context |
aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

Rct 성장/100cyc (R_ct_growth_100)

기준 대비 Rct 성장률. | 계산: $(Rct-Rct0)/((N-N0)/100)$.

트렌드: Rct 성장/100cyc: early=0.05901 → late=0.07797 (Δ =+0.01896, +0.01921mΩ/100cyc/100cyc) → increasing · context
| aging 기대=either | vs=context

R30s 20/50 (R_ratio_20_50)

SOC20 vs 50 총저항 비. | 계산: R30s_20 / R30s_50.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R30s 80/50 (R_ratio_80_50)

SOC80 vs 50 총저항 비. | 계산: R30s_80 / R30s_50.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R-SOC 기울기 (R_SOC_slope)

SOC20/50/80 R30s 선형 기울기. | 계산: linreg R vs SOC.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R-SOC 곡률 (R_SOC_curvature)

3점 이차 계수. | 계산: polyfit deg2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

RΩ SOC50 (ff) (R_ohmic_soc50_ff)

DCIR 블록 forward-fill 값. | 계산: block stamp + ffill.

트렌드: RΩ SOC50 (ff): early=1.092 → late=2.744 (Δ =+1.651, +0.7975mΩ/100cyc) → increasing · matches_aging |
aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch111)

Rct SOC50 (ff) (R_ct_soc50_ff)

DCIR 블록 forward-fill 값. | 계산: block stamp + ffill.

트렌드: Rct SOC50 (ff): early=0.7379 → late=0.9016 (Δ =+0.1637, +0.06968m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전 평균 V (chg_V_avg)

충전 평균 전압. | 계산: mean V charge.

트렌드: 충전 평균 V: early=3.774 → late=3.824 (Δ =+0.05024, +0.02046V/100cyc) → flat · context | aging 기대=either | vs=context

방전 평균 V (dchg_V_avg)

방전 평균 전압. | 계산: mean V discharge.

트렌드: 방전 평균 V: early=3.413 → late=3.291 (Δ =-0.1219, -0.05321V/100cyc) → flat · context | aging 기대=either | vs=context

충전 평균V Δ (delta_chg_V_avg)

기준 대비. | 계산: delta.

트렌드: 충전 평균V Δ : early=0.03413 → late=0.08437 (Δ =+0.05024, +0.02046V/100cyc) → increasing · context | aging 기대=either | vs=context

방전 평균V Δ (delta_dchg_V_avg)

기준 대비. | 계산: delta.

트렌드: 방전 평균V Δ : early=-0.02634 → late=-0.1482 (Δ =-0.1219, -0.05321V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전 IR drop proxy (chg_ir_drop_proxy)

충전 초반 IR 강하 proxy. | 계산: early Δ V.

트렌드: 충전 IR drop proxy: early=0.04014 → late=0.04014 (Δ =-6.9e-06, +2.066e-05V/100cyc) → flat · stable | aging 기대=increase | vs=stable

지표 설명 · 트렌드 (M02Ch111)

방전 IR drop proxy (dchg_ir_drop_proxy)

방전 초반 IR 강하 proxy. | 계산: early ΔV .

트렌드: 방전 IR drop proxy: early=0.04009 → late=0.04015 (Δ =+5.57e-05, +0.0004129V/100cyc) → flat · stable | aging 기대=increase | vs=stable

히스테리시스 면적 (hyst_area)

전체 총방전 히스테리시스. | 계산: $\oint (V_{chg} - V_{dchg})dQ$.

트렌드: 히스테리시스 면적: early=0.5537 → late=0.5996 (Δ =+0.04594, +0.02149V/100cyc) → flat · stable | aging 기대=increase | vs=stable

저SOC 히스테리시스 (hyst_area_low)

저SOC 밴드. Si chemo-mech. | 계산: band integral.

트렌드: 저SOC 히스테리시스: early=0.08677 → late=0.0439 (Δ =-0.04287, -0.01806V/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

중SOC 히스테리시스 (hyst_area_mid)

중SOC 밴드. | 계산: band integral.

트렌드: 중SOC 히스테리시스: early=0.2494 → late=0.319 (Δ =+0.06957, +0.03057V/100cyc) → increasing · context | aging 기대=either | vs=context

고SOC 히스테리시스 (hyst_area_high)

고SOC 밴드. PE 보조. | 계산: band integral.

트렌드: 고SOC 히스테리시스: early=0.2143 → late=0.2348 (Δ =+0.02048, +0.009044V/100cyc) → flat · stable | aging 기대=increase | vs=stable

히스테리시스 저SOC분율 (hyst_frac_low)

low/total. | 계산: hyst_low/hyst.

트렌드: 히스테리시스 저SOC분율: early=0.157 → late=0.07325 (Δ =-0.08377, -0.035291/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

히스테리시스 고SOC분율 (hyst_frac_high)

high/total. | 계산: hyst_high/hyst.

트렌드: 히스테리시스 고SOC분율: early=0.3874 → late=0.3919 (Δ =+0.004497, +0.0011271/100cyc) → flat · context | aging
기대=either | vs=context

최대 히스테리시스 dV (hyst_max_dV)

최대 총방전 전압차. | 계산: max|Vchg-Vdchg|.

트렌드: 최대 히스테리시스 dV: early=1.543 → late=1.602 (Δ =+0.05813, +0.02742V/100cyc) → flat · stable | aging
기대=increase | vs=stable

max dV 저SOC (hyst_max_dV_low)

저SOC 최대 dV. | 계산: band max.

트렌드: max dV 저SOC: early=0.7193 → late=0.4784 (Δ =-0.2409, -0.1008V/100cyc) → decreasing · opposite_aging |
aging 기대=increase | vs=opposite_aging

max dV 중SOC (hyst_max_dV_mid)

중SOC 최대 dV. | 계산: band max.

트렌드: max dV 중SOC: early=0.9535 → late=1.088 (Δ =+0.1344, +0.05858V/100cyc) → increasing · context | aging
기대=either | vs=context

max dV 고SOC (hyst_max_dV_high)

고SOC 최대 dV. | 계산: band max.

트렌드: max dV 고SOC: early=1.543 → late=1.602 (Δ =+0.05813, +0.02742V/100cyc) → flat · context | aging
기대=either | vs=context

히스테리시스 Δ (delta_hyst_area)

기준 대비 면적. | 계산: delta.

트렌드: 히스테리시스 Δ : early=0.004923 → late=0.05087 (Δ =+0.04594, +0.02149V/100cyc) → increasing · matches_aging |
aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch111)

max dV Δ (delta_hyst_max_dV)

기준 대비. | 계산: delta.

트렌드: max dV Δ: early=0.008159 → late=0.06629 (Δ=+0.05813, +0.02742V/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전 플래토 V (chg_plateau_V)

충전 플래토 전압. | 계산: plateau detect.

트렌드: 충전 플래토 V: early=3.686 → late=3.774 (Δ=+0.08743, +0.01714V/100cyc) → flat · context | aging 기대=either | vs=context

충전 플래토 폭 (chg_plateau_width)

충전 플래토 Q 폭. | 계산: plateau width.

트렌드: 충전 플래토 폭: early=10.34 → late=9.889 (Δ=-0.4511, +0.4667Q-units/100cyc) → flat · context | aging 기대=either | vs=context

방전 플래토 V (dchg_plateau_V)

방전 플래토 전압. | 계산: plateau detect.

트렌드: 방전 플래토 V: early=3.19 → late=3.116 (Δ=-0.07325, -0.03213V/100cyc) → flat · context | aging 기대=either | vs=context

방전 플래토 폭 (dchg_plateau_width)

방전 플래토 Q 폭. | 계산: plateau width.

트렌드: 방전 플래토 폭: early=17.42 → late=14.33 (Δ=-3.09, -1.201Q-units/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 플래토 ΔV (delta_dchg_plateau_V)

기준 대비 이동. | 계산: delta plateau V.

트렌드: 방전 플래토 ΔV: early=-0.03324 → late=-0.1065 (Δ=-0.07325, -0.03213V/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전 컷오프 마진 (dchg_V_cutoff_margin)

컷오프까지 여유. | 계산: Vmin-margin.

트렌드: 방전 컷오프 마진: early=0.4223 → late=0.2915 (Δ =-0.1308, -0.05775V/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

컷오프 마진 Δ (delta_dchg_V_cutoff_margin)

기준 대비. | 계산: delta.

트렌드: 컷오프 마진 Δ : early=-0.01926 → late=-0.1501 (Δ =-0.1308, -0.05775V/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 형상 DTW (dchg_shape_DTW)

기준 곡선 DTW 거리. | 계산: DTW vs baseline.

트렌드: 방전 형상 DTW: early=0.003666 → late=0.01155 (Δ =+0.007883, +0.0036571/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

DTW Δ (delta_dchg_shape_DTW)

기준 대비 DTW. | 계산: delta.

트렌드: DTW Δ : early=0.002511 → late=0.01039 (Δ =+0.007883, +0.0036571/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전 dQ/dV 피크1 V (chg_dQdV_peak1_V)

충전 IC 1번 피크 전압. | 계산: SG + find_peaks.

트렌드: 충전 dQ/dV 피크1 V: early=3.464 → late=3.761 (Δ =+0.2962, +0.05425V/100cyc) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크1 높이 (chg_dQdV_peak1)

충전 IC 1번 피크 높이. | 계산: peak prominence/height.

트렌드: 충전 dQ/dV 피크1 높이: early=80.48 → late=81.33 (Δ =+0.8509, -3433Ah/V/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전 dQ/dV 피크1 V (dchg_dQdV_peak1_V)

방전 IC 1번 피크 전압. | 계산: SG + find_peaks.

트렌드: 방전 dQ/dV 피크1 V: early=3.169 → late=3.116 (Δ =-0.05271, -0.02131V/100cyc) → flat · context | aging 기대=either | vs=context

방전 dQ/dV 피크1 높이 (dchg_dQdV_peak1)

방전 IC 1번 피크 높이. | 계산: peak height.

트렌드: 방전 dQ/dV 피크1 높이: early=-71.29 → late=-56.51 (Δ =+14.78, +6.118Ah/V/100cyc) → increasing · context | aging 기대=either | vs=context

충전 dV/dQ 피크1 Q (chg_dVdQ_peak1_Q)

충전 dV/dQ 1번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 충전 dV/dQ 피크1 Q: early=55.15 → late=47.41 (Δ =-7.742, -3.048Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

충전 dV/dQ 피크1 높이 (chg_dVdQ_peak1)

충전 dV/dQ 1번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 충전 dV/dQ 피크1 높이: early=0.01582 → late=0.01802 (Δ =+0.002205, +0.0009162V/Ah/100cyc) → increasing · context | aging 기대=either | vs=context

방전 dV/dQ 피크1 Q (dchg_dVdQ_peak1_Q)

방전 dV/dQ 1번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 방전 dV/dQ 피크1 Q: early=11.59 → late=25.05 (Δ =+13.46, +8.147Ah/100cyc) → increasing · context | aging 기대=either | vs=context

방전 dV/dQ 피크1 높이 (dchg_dVdQ_peak1)

방전 dV/dQ 1번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 방전 dV/dQ 피크1 높이: early=-0.02213 → late=-0.0227 (Δ =-0.0005672, -0.0002853V/Ah/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

충전 dQ/dV 피크2 V (chg_dQdV_peak2_V)

충전 IC 2번 피크 전압. | 계산: SG + find_peaks.

트렌드: 충전 dQ/dV 피크2 V: early=3.698 → late=3.981 (Δ =+0.2825, +0.08312V/100cyc) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크2 높이 (chg_dQdV_peak2)

충전 IC 2번 피크 높이. | 계산: peak prominence/height.

트렌드: 충전 dQ/dV 피크2 높이: early=89.44 → late=75.44 (Δ =-14, -4.197Ah/V/100cyc) → flat · context | aging 기대=either | vs=context

방전 dQ/dV 피크2 V (dchg_dQdV_peak2_V)

방전 IC 2번 피크 전압. | 계산: SG + find_peaks.

트렌드: 방전 dQ/dV 피크2 V: early=3.645 → late=3.629 (Δ =-0.01606, -0.01394V/100cyc) → flat · context | aging 기대=either | vs=context

방전 dQ/dV 피크2 높이 (dchg_dQdV_peak2)

방전 IC 2번 피크 높이. | 계산: peak height.

트렌드: 방전 dQ/dV 피크2 높이: early=-53.28 → late=-50.82 (Δ =+2.457, +1.283Ah/V/100cyc) → flat · context | aging 기대=either | vs=context

충전 dV/dQ 피크2 Q (chg_dVdQ_peak2_Q)

충전 dV/dQ 2번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 충전 dV/dQ 피크2 Q: early=65.54 → late=65.54 (Δ =+0, n/a) → flat · context | aging 기대=either | vs=context

충전 dV/dQ 피크2 높이 (chg_dVdQ_peak2)

충전 dV/dQ 2번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 충전 dV/dQ 피크2 높이: early=0.01257 → late=0.01257 (Δ =+0, n/a) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전 dV/dQ 피크2 Q (dchg_dVdQ_peak2_Q)

방전 dV/dQ 2번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 방전 dV/dQ 피크2 Q: early=30.03 → late=29.09 (Δ =-0.9408, -1.651Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

방전 dV/dQ 피크2 높이 (dchg_dVdQ_peak2)

방전 dV/dQ 2번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 방전 dV/dQ 피크2 높이: early=-0.02153 → late=-0.02176 (Δ =-0.0002347, -0.0005918V/Ah/100cyc) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크3 V (chg_dQdV_peak3_V)

충전 IC 3번 피크 전압. | 계산: SG + find_peaks.

트렌드: 충전 dQ/dV 피크3 V: early=3.953 → late=3.997 (Δ =+0.04471, -0.01215V/100cyc) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크3 높이 (chg_dQdV_peak3)

충전 IC 3번 피크 높이. | 계산: peak prominence/height.

트렌드: 충전 dQ/dV 피크3 높이: early=82.18 → late=78.36 (Δ =-3.826, +7.334Ah/V/100cyc) → increasing · context | aging 기대=either | vs=context

방전 dQ/dV 피크3 V (dchg_dQdV_peak3_V)

방전 IC 3번 피크 전압. | 계산: SG + find_peaks.

트렌드: 방전 dQ/dV 피크3 V: early=3.981 → late=3.855 (Δ =-0.1263, -0.09386V/100cyc) → flat · context | aging 기대=either | vs=context

방전 dQ/dV 피크3 높이 (dchg_dQdV_peak3)

방전 IC 3번 피크 높이. | 계산: peak height.

트렌드: 방전 dQ/dV 피크3 높이: early=-59.57 → late=-45.25 (Δ =+14.32, +6.625Ah/V/100cyc) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

충전 dV/dQ 피크3 Q (chg_dVdQ_peak3_Q)

충전 dV/dQ 3번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크3 높이 (chg_dVdQ_peak3)

충전 dV/dQ 3번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크3 Q (dchg_dVdQ_peak3_Q)

방전 dV/dQ 3번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크3 높이 (dchg_dVdQ_peak3)

방전 dV/dQ 3번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크4 V (chg_dQdV_peak4_V)

충전 IC 4번 피크 전압. | 계산: SG + find_peaks.

트렌드: 충전 dQ/dV 피크4 V: early=4.161 → late=4.161 ($\Delta=+0$, -0.05226V/100cyc) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크4 높이 (chg_dQdV_peak4)

충전 IC 4번 피크 높이. | 계산: peak prominence/height.

트렌드: 충전 dQ/dV 피크4 높이: early=96.7 → late=96.7 ($\Delta=+0$, +40.62Ah/V/100cyc) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전 dQ/dV 피크4 V (dchg_dQdV_peak4_V)

방전 IC 4번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크4 높이 (dchg_dQdV_peak4)

방전 IC 4번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크4 Q (chg_dVdQ_peak4_Q)

충전 dV/dQ 4번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크4 높이 (chg_dVdQ_peak4)

충전 dV/dQ 4번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크4 Q (dchg_dVdQ_peak4_Q)

방전 dV/dQ 4번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크4 높이 (dchg_dVdQ_peak4)

방전 dV/dQ 4번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

충전 dQ/dV 피크5 V (chg_dQdV_peak5_V)

충전 IC 5번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크5 높이 (chg_dQdV_peak5)

충전 IC 5번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크5 V (dchg_dQdV_peak5_V)

방전 IC 5번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크5 높이 (dchg_dQdV_peak5)

방전 IC 5번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크5 Q (chg_dVdQ_peak5_Q)

충전 dV/dQ 5번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크5 높이 (chg_dVdQ_peak5)

충전 dV/dQ 5번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

방전 dV/dQ 피크5 Q (dchg_dVdQ_peak5_Q)

방전 dV/dQ 5번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크5 높이 (dchg_dVdQ_peak5)

방전 dV/dQ 5번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크6 V (chg_dQdV_peak6_V)

충전 IC 6번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크6 높이 (chg_dQdV_peak6)

충전 IC 6번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크6 V (dchg_dQdV_peak6_V)

방전 IC 6번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크6 높이 (dchg_dQdV_peak6)

방전 IC 6번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

충전 dV/dQ 피크6 Q (chg_dVdQ_peak6_Q)

충전 dV/dQ 6번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크6 높이 (chg_dVdQ_peak6)

충전 dV/dQ 6번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크6 Q (dchg_dVdQ_peak6_Q)

방전 dV/dQ 6번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크6 높이 (dchg_dVdQ_peak6)

방전 dV/dQ 6번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크7 V (chg_dQdV_peak7_V)

충전 IC 7번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크7 높이 (chg_dQdV_peak7)

충전 IC 7번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

방전 dQ/dV 피크7 V (dchg_dQdV_peak7_V)

방전 IC 7번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크7 높이 (dchg_dQdV_peak7)

방전 IC 7번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크7 Q (chg_dVdQ_peak7_Q)

충전 dV/dQ 7번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크7 높이 (chg_dVdQ_peak7)

충전 dV/dQ 7번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크7 Q (dchg_dVdQ_peak7_Q)

방전 dV/dQ 7번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크7 높이 (dchg_dVdQ_peak7)

방전 dV/dQ 7번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

충전 dQ/dV 피크8 V (chg_dQdV_peak8_V)

충전 IC 8번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크8 높이 (chg_dQdV_peak8)

충전 IC 8번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크8 V (dchg_dQdV_peak8_V)

방전 IC 8번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크8 높이 (dchg_dQdV_peak8)

방전 IC 8번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크8 Q (chg_dVdQ_peak8_Q)

충전 dV/dQ 8번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크8 높이 (chg_dVdQ_peak8)

충전 dV/dQ 8번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

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방전 dV/dQ 피크8 Q (dchg_dVdQ_peak8_Q)

방전 dV/dQ 8번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크8 높이 (dchg_dVdQ_peak8)

방전 dV/dQ 8번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 피크1 ΔV (delta_chg_dQdV_peak1_V)

기준 대비 충전 dQ/dV 피크1 이동. | 계산: delta_chg_dQdV_peak1_V

트렌드: 충전 피크1 ΔV: early=0.04817 → late=0.3443 (Δ=+0.2962, +0.05425V/100cyc) → increasing · context | aging 기대=either | vs=context

방전 피크1 ΔV (delta_dchg_dQdV_peak1_V)

기준 대비 방전 dQ/dV 피크1 이동. | 계산: delta_dchg_dQdV_peak1_V

트렌드: 방전 피크1 ΔV: early=-0.03239 → late=-0.0851 (Δ=-0.05271, -0.02131V/100cyc) → decreasing · context | aging 기대=either | vs=context

방전 dV/dQ @SOC0 (dchg_dVdQ_SOC0)

저SOC cliff dV/dQ. | 계산: dchg_dVdQ_SOC0

트렌드: 방전 dV/dQ @SOC0: early=0.1078 → late=0.06981 (Δ=-0.03795, -0.01832V/Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

방전 dV/dQ @SOC5 (dchg_dVdQ_SOC5)

SOC≈5% dV/dQ. | 계산: dchg_dVdQ_SOC5

트렌드: 방전 dV/dQ @SOC5: early=0.04914 → late=0.04273 (Δ=-0.00641, -0.002951V/Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전 dV/dQ @SOC10 (dchg_dVdQ_SOC10)

SOC $\approx$ 10% dV/dQ. | 계산: dchg_dVdQ_SOC10

트렌드: 방전 dV/dQ @SOC10: early=0.02913 $\rightarrow$ late=0.03017 (Δ =+0.001044, +0.0002802V/Ah/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 dV/dQ @mid (dchg_dVdQ_SOCmid)

중SOC dV/dQ. | 계산: dchg_dVdQ_SOCmid

트렌드: 방전 dV/dQ @mid: early=0.0196 $\rightarrow$ late=0.02076 (Δ =+0.001167, +0.0004514V/Ah/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 cliff Q (dchg_dVdQ_SOC0_Q)

SOC0 dV/dQ 위치 Q. | 계산: dchg_dVdQ_SOC0_Q

트렌드: 방전 cliff Q: early=67.39 $\rightarrow$ late=59.8 (Δ =-7.583, -2.982Ah/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 cliff 폭 (dchg_dVdQ_SOC0_cliff_width)

저SOC cliff 폭. | 계산: dchg_dVdQ_SOC0_cliff_width

트렌드: 방전 cliff 폭: early=4.997 $\rightarrow$ late=3.478 (Δ =-1.519, -0.6294Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

cliff/mid 비 (dchg_dVdQ_SOC0_to_mid_ratio)

SOC0/mid dV/dQ 비. | 계산: dchg_dVdQ_SOC0_to_mid_ratio

트렌드: cliff/mid 비: early=5.499 $\rightarrow$ late=3.369 (Δ =-2.13, -1.0051/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

충전 dV/dQ @100 (chg_dVdQ_SOC100)

만충 부근 dV/dQ. | 계산: chg_dVdQ_SOC100

트렌드: 충전 dV/dQ @100: early=0.009986 $\rightarrow$ late=0.01344 (Δ =+0.003457, +0.001421V/Ah/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

dV/dQ SOC0 Δ (delta_dchg_dVdQ_SOC0)

기준 대비. | 계산: delta_dchg_dVdQ_SOC0

트렌드: dV/dQ SOC0 Δ : early=0.001093 $\rightarrow$ late=-0.03686 (Δ =-0.03795, -0.01832V/Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

dV/dQ SOC5 Δ (delta_dchg_dVdQ_SOC5)

기준 대비. | 계산: delta_dchg_dVdQ_SOC5

트렌드: dV/dQ SOC5 Δ : early=-0.002466 $\rightarrow$ late=-0.008876 (Δ =-0.00641, -0.002951V/Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

dV/dQ SOC10 Δ (delta_dchg_dVdQ_SOC10)

기준 대비. | 계산: delta_dchg_dVdQ_SOC10

트렌드: dV/dQ SOC10 Δ : early=-0.001896 $\rightarrow$ late=-0.0008519 (Δ =+0.001044, +0.0002802V/Ah/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

dV/dQ mid Δ (delta_dchg_dVdQ_SOCmid)

기준 대비. | 계산: delta_dchg_dVdQ_SOCmid

트렌드: dV/dQ mid Δ : early=0.0002808 $\rightarrow$ late=0.001447 (Δ =+0.001167, +0.0004514V/Ah/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

dV/dQ 100 Δ (delta_chg_dVdQ_SOC100)

기준 대비. | 계산: delta_chg_dVdQ_SOC100

트렌드: dV/dQ 100 Δ : early=-0.004091 $\rightarrow$ late=-0.0006338 (Δ =+0.003457, +0.001421V/Ah/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

cliff 폭 Δ (delta_dchg_dVdQ_SOC0_cliff_width)

기준 대비. | 계산: delta_dchg_dVdQ_SOC0_cliff_width

트렌드: cliff 폭 Δ : early=-0.5255 $\rightarrow$ late=-2.044 (Δ =-1.519, -0.6294Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

cliff/mid Δ (delta_dchg_dVdQ_SOC0_to_mid_ratio)

기준 대비. | 계산: delta_dchg_dVdQ_SOC0_to_mid_ratio

트렌드: cliff/mid Δ : early=-0.02339 $\rightarrow$ late=-2.154 (Δ =-2.13, -1.0051/100cyc) $\rightarrow$ decreasing · context | aging
기대=either | vs=context

충전 IC 면적합 (chg_dQdV_area_sum)

dQ/dV 피크 면적 합. | 계산: chg_dQdV_area_sum

트렌드: 충전 IC 면적합: early=66.15 $\rightarrow$ late=50.88 (Δ =-15.27, -5.683Ah/100cyc) $\rightarrow$ decreasing · context | aging
기대=either | vs=context

방전 IC 면적합 (dchg_dQdV_area_sum)

dQ/dV 피크 면적 합. | 계산: dchg_dQdV_area_sum

트렌드: 방전 IC 면적합: early=67.39 $\rightarrow$ late=59.8 (Δ =-7.583, -2.982Ah/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

RCF (RCF)

Q_0.5C / Q_C/3. | 계산: routine/RPT Q.

트렌드: RCF: early=0.9772 $\rightarrow$ late=0.9672 (Δ =-0.009985, +0.0016941/100cyc) $\rightarrow$ flat · stable | aging
기대=decrease | vs=stable

RCF 기울기/100 (RCF_slope_100)

RCF 변화율. | 계산: first-last slope.

트렌드: RCF 기울기/100: early=-0.01156 $\rightarrow$ late=-0.01156 (Δ =+0, +3.54e-181/100cyc/100cyc) $\rightarrow$ flat · stable | aging
기대=decrease | vs=stable

PER (PER)

$\eta/(\Delta I \cdot R)$. | 계산: eta_SOC50/(dl*R).

트렌드: 데이터 부족 | aging
기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

η @SOC20 (eta_SOC20)

저SOC 과전위. | 계산: C/3 vs 0.5C.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η @SOC50 (eta_SOC50)

중SOC 과전위. | 계산: C/3 vs 0.5C.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η @SOC80 (eta_SOC80)

고SOC 과전위. | 계산: C/3 vs 0.5C.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η max (eta_max)

최대 과전위. | 계산: max η (SOC).

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η mean (eta_mean)

평균 과전위. | 계산: mean η .

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η argmax SOC (eta_argmax_SOC)

η 최대 SOC. | 계산: argmax.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

η 저SOC 기울기 (eta_slope_lowSOC)

저SOC η 기울기. | 계산: slope low band.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff scale (Reff_scale)

유효 R 스케일. | 계산: shape fit scale.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Reff fit R2 (Reff_shape_fit_r2)

Reff 형상 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff 잔차 SOC20 (Reff_resid_soc20)

Reff 모델 잔차. | 계산: resid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff 잔차 SOC50 (Reff_resid_soc50)

Reff 모델 잔차. | 계산: resid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff 잔차 SOC80 (Reff_resid_soc80)

Reff 모델 잔차. | 계산: resid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

l_{∞} 정규화 (l_inf_norm)

CV 잔류전류 정규화. | 계산: $l_{\infty} / l_{\text{ref}}$.

트렌드: l_{∞} 정규화: early=5.405e-17 → late=5.405e-17 ($\Delta=+0$, +0.12631/100cyc) → increasing · context | aging 기대=either | vs=context

펄스 1s 샘플수 (pulse_sample_count_1s)

$t \leq 1s$ 샘플 수. | 계산: quality.

트렌드: 펄스 1s 샘플수: early=5 → late=7 ($\Delta=+2$, +0.8611count/100cyc) → increasing · context | aging 기대=either | vs=context

펄스 전류 안정도 (pulse_current_stability)

$\text{std}(I)/|I|$. | 계산: quality.

트렌드: 펄스 전류 안정도: early=0.02607 → late=0.04566 ($\Delta=+0.01959$, +0.0083421/100cyc) → increasing · opposite_aging | aging 기대=decrease | vs=opposite_aging

rest 충분성 (rest_sufficiency)

휴지 길이/품질. | 계산: quality.

트렌드: rest 충분성: early=3 → late=3 ($\Delta=+0$, -0.16151/100cyc) → decreasing · context | aging 기대=either | vs=context

레그 완전성 (leg_completeness)

충방전 레그 완전성. | 계산: quality.

트렌드: 레그 완전성: early=0.9998 → late=0.9999 ($\Delta=+8.035e-05$, -0.018141/100cyc) → flat · context | aging 기대=either | vs=context

완화 완성도 max (relax_completeness_max)

SOC별 최대 완화 완성도. | 계산: max.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

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샘플/mV (samples_per_mV)

전압 해상도 샘플밀도. | 계산: dqdv quality.

트렌드: 샘플/mV: early=0.3618 → late=0.3506 (Δ =-0.01121, -0.011761/mV/100cyc) → flat · context | aging 기대=either | vs=context

OCV V_{∞} SOC80 (ocv_V_inf_soc80)

고SOC 무한시간 OCV. | 계산: recovery/rest V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV V_{∞} SOC50 (ocv_V_inf_soc50)

중SOC OCV. | 계산: V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV V_{∞} SOC20 (ocv_V_inf_soc20)

저SOC OCV. | 계산: V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV spread 20–80 (ocv_spread_20_80)

SOC20–80 OCV 폭. | 계산: V_{80} - V_{20} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV spread 50–80 (ocv_spread_50_80)

SOC50–80 폭. | 계산: V_{80} - V_{50} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

OCV spread 20–50 (ocv_spread_20_50)

SOC20–50 폭. | 계산: V50-V20.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV 평행이동 (ocv_parallel_shift)

OCV 곡선 평행 시프트. | 계산: block OCV shift.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV 폭 압축 (ocv_spread_compression)

spread 축소. | 계산: Δ spread.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV80 Δ (delta_ocv_V_inf_soc80)

기준 대비. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV50 Δ (delta_ocv_V_inf_soc50)

기준 대비. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV20 Δ (delta_ocv_V_inf_soc20)

기준 대비. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

OCV spread Δ (delta_ocv_spread_20_80)

기준 대비 폭 변화. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

LAM 곡선 proxy (LAM_curve_proxy)

V-Q scale 축소 proxy (절대 LAM% 아님). | 계산: (1-s)*100.

트렌드: LAM 곡선 proxy: early=-1.959 → late=-12.31 (Δ =-10.36, -4.262%/100cyc) → decreasing · context | aging 기대=either | vs=context

LLI 곡선 proxy (LLI_curve_proxy)

Q 오프셋 proxy (절대 LLI% 아님). | 계산: offset/Qmax*100.

트렌드: LLI 곡선 proxy: early=-0.2132 → late=1.26 (Δ =+1.473, +0.4021%/100cyc) → increasing · context | aging 기대=either | vs=context

R 곡선 proxy (R_curve_proxy)

곡선 fit dR proxy. | 계산: fit_dR.

트렌드: R 곡선 proxy: early=0.9586 → late=4.291 (Δ =+3.333, +1.524m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

방전 fit scale (dchg_fit_scale)

기준 대비 scale s. | 계산: 3-param fit.

트렌드: 방전 fit scale: early=1.02 → late=1.123 (Δ =+0.1036, +0.042621/100cyc) → flat · stable | aging 기대=decrease | vs=stable

방전 fit offset (dchg_fit_offset)

Q 오프셋. | 계산: 3-param fit.

트렌드: 방전 fit offset: early=-0.1474 → late=0.8715 (Δ =+1.019, +0.2781Ah/100cyc) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

방전 fit dR (dchg_fit_dR)

저항 항. | 계산: 3-param fit.

트렌드: 방전 fit dR: early=0.9586 → late=4.291 (Δ =+3.333, +1.524m Ω /100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

fit 잔차 RMS (dchg_fit_residual_rms)

잔차 rms. | 계산: RMS(resid).

트렌드: fit 잔차 RMS: early=4.072 → late=20.02 (Δ =+15.95, +6.417mV/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

fit 잔차 max (dchg_fit_residual_max)

잔차 최대. | 계산: max|resid|.

트렌드: fit 잔차 max: early=34.04 → late=203.3 (Δ =+169.3, +62.24mV/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

잔차 argmax SOC (dchg_fit_residual_argmax_SOC)

잔차 최대 SOC (방전 DOD→SOC 변환). | 계산: argmax residual.

트렌드: 잔차 argmax SOC: early=100 → late=100 (Δ =+0, +0.01526%/100cyc) → flat · context | aging 기대=either |
vs=context

fit R2 (dchg_fit_r2)

곡선 fit 품질. | 계산: r2.

트렌드: fit R2: early=0.9999 → late=0.9977 (Δ =-0.002247, -0.00096861/100cyc) → flat · context | aging 기대=either |
vs=context

fit corr(s,o) (dchg_fit_corr_s_o)

scale-offset 상관 (축퇴 지표). | 계산: corr.

트렌드: fit corr(s,o): early=0.5388 → late=0.7869 (Δ =+0.2481, +0.087661/100cyc) → increasing · context | aging
기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

잔차 argmax DOD (dchg_fit_residual_argmax_DOD)

잔차 최대 DOD. | 계산: argmax DOD .

트렌드: 잔차 argmax DOD : early=0 $\rightarrow$ late=0 ($\Delta=+0$, -0.01526%/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

$\Delta Q(V)$ min (dQV_min)

전압빈 ΔQ 최소. | 계산: histogram.

트렌드: $\Delta Q(V)$ min: early=-3.286 $\rightarrow$ late=-15.66 ($\Delta=-12.38$, -5.241Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

$\Delta Q(V)$ mean (dQV_mean)

ΔQ 평균. | 계산: mean.

트렌드: $\Delta Q(V)$ mean: early=-2.433 $\rightarrow$ late=-12.03 ($\Delta=-9.599$, -4.07Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

$\Delta Q(V)$ var (dQV_var)

ΔQ 분산. | 계산: var.

트렌드: $\Delta Q(V)$ var: early=0.2287 $\rightarrow$ late=4.005 ($\Delta=+3.776$, +1.662Ah²/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

$\Delta Q(V)$ log-var (dQV_log_var)

log10 분산. | 계산: log10(var).

트렌드: $\Delta Q(V)$ log-var: early=-0.6408 $\rightarrow$ late=0.6026 ($\Delta=+1.243$, +0.48451/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

$\Delta Q(V)$ skew (dQV_skew)

왜도. | 계산: skew.

트렌드: $\Delta Q(V)$ skew: early=-0.3955 $\rightarrow$ late=-0.762 ($\Delta=-0.3665$, -0.11451/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

$\Delta Q(V)$ kurtosis (dQV_kurtosis)

첨도. | 계산: kurtosis.

트렌드: $\Delta Q(V)$ kurtosis: early=-1.093 → late=-0.9439 (Δ =+0.149, +0.073451/100cyc) → increasing · context | aging 기대=either | vs=context

ΔQ argmin V (dQV_argmin_V)

ΔQ 최소 전압. | 계산: argmin.

트렌드: ΔQ argmin V: early=3.158 → late=3.09 (Δ =-0.06796, -0.02668V/100cyc) → flat · context | aging 기대=either | vs=context

dQ/dV SNR (dqdv_snr)

IC 신호대잡음. | 계산: snr estimate.

트렌드: dQ/dV SNR: early=76.79 → late=62 (Δ =-14.79, -5.3481/100cyc) → decreasing · context | aging 기대=either | vs=context

데이터 품질점수 (quality_score)

추출 품질 종합. | 계산: quality gates.

트렌드: 데이터 품질점수: early=1 → late=1 (Δ =+0, -0.053850-1/100cyc) → decreasing · context | aging 기대=either | vs=context

전압 노이즈 σ (v_noise_sigma)

전압 노이즈 추정. | 계산: noise sigma.

트렌드: 전압 노이즈 σ : early=0.02213 → late=0.02742 (Δ =+0.005281, +0.00179V/100cyc) → increasing · context | aging 기대=either | vs=context

$\Delta Q(V)$ 기준 사이클 (dQV_ref_cycle)

ΔQ 비교 기준 사이클. | 계산: ref cycle id.

트렌드: $\Delta Q(V)$ 기준 사이클: early=3 → late=3 (Δ =+0, -4.883e-16cyc/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

온도 가용 (temperature_available)

Temp 컬럼 유효 여부. | 계산: bool→float.

트렌드: 온도 가용: early=0 → late=0 ($\Delta=+0$, +00/1/100cyc) → flat · context | aging 기대=either | vs=context

Q_relax 유의 (Q_relax_significant)

완화 용량 유의 플래그. | 계산: threshold flag.

트렌드: Q_relax 유의: early=1 → late=0 ($\Delta=-1$, -0.53580/1/100cyc) → decreasing · context | aging 기대=either | vs=context

fit 축퇴 플래그 (dchg_fit_degenerate_flag)

scale bound 포화 등. | 계산: degenerate flag.

트렌드: fit 축퇴 플래그: early=0 → late=0 ($\Delta=+0$, +00/1/100cyc) → flat · context | aging 기대=either | vs=context

η 유효 (eta_valid)

과전위 계산 유효 플래그. | 계산: eta_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

fade b 표준오차 (fade_exponent_b_se)

fade 지수 표준오차. | 계산: fit se.

트렌드: fade b 표준오차: early=0.03028 → late=0.03028 ($\Delta=+0$, +1.126e-171/100cyc) → flat · context | aging 기대=either | vs=context

ΔQ 유효 V범위 (dQV_valid_V_range)

ΔQ 집계 전압폭. | 계산: Vmax-Vmin used.

트렌드: ΔQ 유효 V범위: early=0.7715 → late=0.7715 ($\Delta=+0$, +1.747e-17V/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

fade 지수 b (fade_exponent_b)

SoHQ power-law 지수. | 계산: SoHQ fit.

트렌드: fade 지수 b: early=0.6732 → late=0.6732 ($\Delta=+0$, +7.504e-171/100cyc) → flat · context | aging 기대=either | vs=context

fade 지수 a (fade_exponent_a)

power-law 계수. | 계산: SoHQ fit.

트렌드: fade 지수 a: early=0.002867 → late=0.002867 ($\Delta=+0$, +4.396e-191/100cyc) → flat · context | aging 기대=either | vs=context

fade fit R2 (fade_fit_r2)

fade 적합도. | 계산: r2.

트렌드: fade fit R2: early=0.9796 → late=0.9796 ($\Delta=+0$, +2.735e-161/100cyc) → flat · context | aging 기대=either | vs=context

fade SoHQ0 (fade_sohq0)

fit 초기 SoHQ. | 계산: intercept.

트렌드: fade SoHQ0: early=97.96 → late=97.96 ($\Delta=+0$, +5.437e-14%/100cyc) → flat · context | aging 기대=either | vs=context

knee 사이클 (knee_cycle_bw)

bilinear knee 위치. | 계산: broken-stick SoHQ.

트렌드: knee 사이클: early=30 → late=30 ($\Delta=+0$, +8.192e-15cyc/100cyc) → flat · context | aging 기대=either | vs=context

knee 심각도 (knee_severity)

전후 기울기 차이. | 계산: slope_after-before.

트렌드: knee 심각도: early=0 → late=0 ($\Delta=+0$, +01/100cyc) → flat · stable | aging 기대=increase | vs=stable

지표 설명 · 트렌드 (M02Ch111)

knee 전 기울기 (knee_slope_before)

knee 이전 fade 기울기. | 계산: bilinear.

트렌드: knee 전 기울기: early=-0.1479 → late=-0.1479 ($\Delta=+0$, -3.32e-17%/cyc/100cyc) → flat · stable | aging
기대=decrease | vs=stable

knee 후 기울기 (knee_slope_after)

knee 이후 fade 기울기. | 계산: bilinear.

트렌드: knee 후 기울기: early=-0.03581 → late=-0.03581 ($\Delta=+0$, -1.356e-17%/cyc/100cyc) → flat · stable | aging
기대=decrease | vs=stable

knee fit R2 (knee_fit_r2)

knee 적합도. | 계산: r2.

트렌드: knee fit R2: early=0.9979 → late=0.9979 ($\Delta=+0$, +3.735e-161/100cyc) → flat · context | aging
기대=either | vs=context

PE activity 패턴 (LAM_PE_pattern_score)

NCM activity/isolation (절대 LAM% 아님). | 계산: mode_weights LAM_PE.

트렌드: PE activity 패턴: early=0.5224 → late=0.9484 ($\Delta=+0.426$, +0.17170-1/100cyc) → increasing · matches_aging |
aging 기대=increase | vs=matches_aging

NE 패턴 점수 (LAM_NE_pattern_score)

NE 관련 패턴 (Si-on-Gr에선 보조). | 계산: mode_weights LAM_NE.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

contact_loss (contact_loss_score)

옴/스택/접촉 증거 합. | 계산: RQ growth 등 가중합.

트렌드: contact_loss: early=0.7447 → late=0.9905 ($\Delta=+0.2458$, +0.15760-1/100cyc) → increasing · matches_aging |
aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch111)

LLI 패턴 (LLI_pattern_score)

CE·slippage·offset 기반. | 계산: mode_weights LLI.

트렌드: LLI 패턴: early=0.2804 → late=0.5708 (Δ =+0.2904, +0.10470-1/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

계면 R 패턴 (interface_R_score)

Rct·VE 등 계면저항. | 계산: mode_weights interface_R.

트렌드: 계면 R 패턴: early=0.3832 → late=0.8502 (Δ =+0.467, +0.20280-1/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

고체확산 패턴 (solid_diffusion_score)

A_diff·PER·RCF. | 계산: mode_weights solid_diffusion.

트렌드: 고체확산 패턴: early=0.4264 → late=0.5751 (Δ =+0.1488, -0.0010570-1/100cyc) → flat · stable | aging
기대=increase | vs=stable

SE 분해 패턴 (SE_decomposition_score)

CE ↓ · Rct ↑ 등 SE 분해 가설. | 계산: mode_weights SE_decomposition.

트렌드: SE 분해 패턴: early=0.1159 → late=0.3538 (Δ =+0.238, +0.073470-1/100cyc) → increasing · matches_aging | aging
aging 기대=increase | vs=matches_aging

마이크로쇼트 패턴 (microshort_score)

자기방전·CE 기반 soft-short 가설. | 계산: mode_weights microshort.

트렌드: 마이크로쇼트 패턴: early=0.09603 → late=0.09603 (Δ =+0, -0.28020-1/100cyc) → decreasing · opposite_aging | aging
기대=increase | vs=opposite_aging

임피던스 패턴 (impedance_pattern_score)

총 임피던스 성장 패턴. | 계산: mode score.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch111)

수송제한 패턴 (transport_limitation_score)

rate/수송 제한 패턴. | 계산: mode score.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

플레이팅 리스크 (plating_risk_score)

Li plating 위험 패턴. | 계산: mode score.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

contact_loss 신뢰도 (contact_loss_confidence)

패턴 점수 신뢰도. | 계산: evidence coverage.

트렌드: contact_loss 신뢰도: early=0.52 → late=0.448 ($\Delta=-0.072$, -0.0043570–1/100cyc) → flat · context | aging
기대=either | vs=context

LAM_PE 신뢰도 (LAM_PE_confidence)

패턴 신뢰도. | 계산: confidence.

트렌드: LAM_PE 신뢰도: early=0.751 → late=0.616 ($\Delta=-0.135$, -0.069180–1/100cyc) → decreasing · context | aging
기대=either | vs=context

LLI 신뢰도 (LLI_confidence)

패턴 신뢰도. | 계산: confidence.

트렌드: LLI 신뢰도: early=0.7598 → late=0.72 ($\Delta=-0.03978$, -0.065670–1/100cyc) → decreasing · context | aging
기대=either | vs=context

PE lean (PE_side_score)

$0.75 \cdot \text{LAM_PE} + \text{feature} + \text{FC-OCP } \Delta \text{hits}$. | 계산: electrode_side v1.3.

트렌드: PE lean: early=0.4806 → late=0.8109 ($\Delta=+0.3302$, +0.13490–1/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch111)

contact_stack (contact_stack_score)

$\approx$ contact_loss (R-centric). | 계산: clip(contact_loss).

트렌드: contact_stack: early=0.7447 $\rightarrow$ late=0.9905 (Δ =+0.2458, +0.15760-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

NE 가설 (NE_side_score)

contact $\times$ Si co-sign. | 계산: electrode_side.

트렌드: NE 가설: early=0.1638 $\rightarrow$ late=0.2996 (Δ =+0.1358, +0.08160-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

shared 모드 (shared_side_score)

LLI/interface 등 공유 모드 평균. | 계산: shared modes mean.

트렌드: shared 모드: early=0.2985 $\rightarrow$ late=0.5887 (Δ =+0.2902, +0.072660-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

Si co-sign (si_cosign)

저SOC hyst·Q_relax·mech/chem·CV 동시 신호. | 계산: SI_NE_COSIGN boost.

트렌드: Si co-sign: early=0.2 $\rightarrow$ late=0.4 (Δ =+0.2, +0.10750-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

dominant 마진 (dominance_margin)

1위-2위 점수차. | 계산: top-second.

트렌드: dominant 마진: early=0.2599 $\rightarrow$ late=0.1796 (Δ =-0.0803, -0.026560-1/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

FC-OCP 피크 hits (pe_peak_hits)

충전 dQ/dV $\leftrightarrow$ 합성 FC-OCP 매칭 수. | 계산: unique nearest $\pm$ 60mV.

트렌드: FC-OCP 피크 hits: early=1 $\rightarrow$ late=0 (Δ =-1, -0.2485count/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch111)

FC-OCP hits Δ (pe_peak_hits_delta)

기준 대비 hits 증가. | 계산: hits-hits0.

트렌드: FC-OCP hits Δ : early=1 $\rightarrow$ late=0 ($\Delta=-1$, $-0.2485\text{count}/100\text{cyc}$) $\rightarrow$ decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

FC-OCP hits (alias) (fc_ocp_hits)

pe_peak_hits 별칭. | 계산: same as pe_peak_hits.

트렌드: FC-OCP hits (alias): early=1 $\rightarrow$ late=0 ($\Delta=-1$, $-0.2485\text{count}/100\text{cyc}$) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

FC-OCP hits Δ (alias) (fc_ocp_hits_delta)

pe_peak_hits_delta 별칭. | 계산: same as pe_peak_hits_delta.

트렌드: FC-OCP hits Δ (alias): early=1 $\rightarrow$ late=0 ($\Delta=-1$, $-0.2485\text{count}/100\text{cyc}$) $\rightarrow$ decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

전극진단 신뢰도 (electrode_confidence)

coverage·분리·OCP 가용성. | 계산: $0.35\text{cov}+0.35\text{sep}+0.30\text{ocp}$.

트렌드: 전극진단 신뢰도: early=0.825 $\rightarrow$ late=0.8766 ($\Delta=+0.05161$, $+0.042380-1/100\text{cyc}$) $\rightarrow$ flat · context | aging 기대=either | vs=context